Real Analysis

Lecture notes: Background and Chapters 4–7

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Background: Results Used from Chapters 1–3

How to use this

Chapters 4–7 rely on standard facts about Lebesgue measure, the Lebesgue integral and L^p spaces that were developed in Chapters 1–3. They are collected here, in the order in which they arise, and are cited in the later chapters as *Fact B.1*, *Fact B.2*, ... Throughout, $(X, \mathcal{S}, \mu)$ is a measure space, m is Lebesgue measure on $\mathbb{R}$ with σ-algebra $\mathcal{M}$ of Lebesgue measurable sets, and $1 \le p \le \infty$.

Measures

Fact B.1 (translation invariance)

For $E \in \mathcal{M}$ and $x \in \mathbb{R}$, $E + x \in \mathcal{M}$ and $m(E + x) = m(E)$; consequently $\int f(y + x)\, dm(y) = \int f(y)\, dm(y)$ for integrable f.

Fact B.2 (completeness of Lebesgue measure)

$(\mathbb{R}, \mathcal{M}, m)$ is complete: every subset of a set of Lebesgue measure zero is Lebesgue measurable (and has measure zero).

Fact B.3 (regularity of Lebesgue measurable sets)

For every $S \in \mathcal{M}$ there are an F_σ-set A and a G_δ-set B with $A \subset S \subset B$ and $m(B \setminus A) = 0$.

Fact B.4 (counting measure)

On any set X with $\mathcal{S} = 2^X$, the counting measure $c(E) = \#E$ (the number of elements, possibly ∞) is a measure. On $X = \mathbb{N}$, $\int f\, dc = \sum_n f(n)$ for $f \ge 0$.

Measurable functions and the integral

Fact B.5 (approximation by simple functions)

If $f : X \to [0, \infty]$ is measurable, there is an increasing sequence of measurable simple functions $0 \le s_1 \le s_2 \le \cdots$ with $s_n \to f$ pointwise.

Fact B.6 (series of characteristic functions)

Every measurable $f : X \to [0, \infty]$ can be written as $f = \sum_{k=1}^{\infty} c_k \chi_{E_k}$ with constants $c_k \ge 0$ and measurable sets E_k.

Fact B.7 (measures defined by densities)

If $h : X \to [0, \infty]$ is measurable, then $\lambda(E) = \int_E h\, d\mu$ $(E \in \mathcal{S})$ defines a measure on $\mathcal{S}$.

Fact B.8 (functions with vanishing integrals)

(a) If $h \geq 0$ and $\int_X h\,d\mu = 0$ then $h = 0$ almost everywhere.
(b) If h is integrable and $\int_E h\,d\mu = 0$ for every $E \in \mathcal{S}$, then $h = 0$ almost everywhere.

Fact B.9 (triangle inequality for integrals)

For integrable f, $|\int_X f\,d\mu| \leq \int_X |f|\,d\mu$.

Fact B.10 (linearity of the integral)

If f, g are integrable and $a, b \in \mathbb{C}$, then $af + bg$ is integrable and $\int (af + bg)\,d\mu = a\int f\,d\mu + b\int g\,d\mu$.

Fact B.11 (differentiation under the integral sign)

Let $f(x,t)$ be integrable in x for each t in an open interval I, differentiable in t, and suppose $|\partial_t f(x,t)| \leq g(x)$ for an integrable g independent of t. Then $F(t) = \int f(x,t)\,d\mu(x)$ is differentiable on I with $F'(t) = \int \partial_t f(x,t)\,d\mu(x)$.

Fact B.12 (periodisation)

For $f \in L^1(\mathbb{R})$ the series $F(t) = \sum_{n\in\mathbb{Z}} f(t - 2n\pi)$ converges absolutely for almost every t, is 2π-periodic, and $\int_{-\pi}^{\pi} \sum_n |f(t-2n\pi)|\,dt = \int_{\mathbb{R}} |f|\,dm$ (term-by-term integration, Fact B.25).

Fact B.13 (Riemann integral via step functions)

For bounded $f : [a,b] \to \mathbb{R}$ and a partition $P = \{a = x_0 < x_1 < \cdots < x_n = b\}$ put $M_i = \sup_{[x_{i-1},x_i]} f$, $m_i = \inf_{[x_{i-1},x_i]} f$ and define the step functions

$$U_P = \sum_{i=1}^{n} M_i \chi_{(x_{i-1},x_i]}, \qquad L_P = \sum_{i=1}^{n} m_i \chi_{(x_{i-1},x_i]}.$$

f is Riemann integrable iff $\inf_P \int U_P = \sup_P \int L_P$, and then this common value is $\int_a^b f$. Improper Riemann integrals are defined as limits of such integrals; Lebesgue integrals need no such limiting procedure.

Differentiation

Fact B.14 (derivative of a monotone function)

If $\alpha : [a,b] \to \mathbb{R}$ is nondecreasing, then α' exists almost everywhere, $\alpha' \geq 0$ is integrable, and $\int_{[a,b]} \alpha'\,dm \leq \alpha(b) - \alpha(a)$.

Fact B.15 (absolute continuity of the integral)

If $f \in L^1(\mathbb{R})$, then for every $\epsilon > 0$ there is $\delta > 0$ such that $m(E) < \delta$ implies $\int_E |f|\,dm < \epsilon$.

Fact B.16 (fundamental theorem for absolutely continuous functions)

If $\alpha : [a,b] \to \mathbb{R}$ is absolutely continuous, then α' exists almost everywhere, is integrable, and $\alpha(x) - \alpha(a) = \int_{[a,x]} \alpha'\, dm$ for all $x \in [a,b]$.

L^p spaces

Fact B.17 (completeness of L^p)

For $1 \le p \le \infty$, $L^p(\mu)$ is a Banach space.

Fact B.18 (a.e. convergent subsequence)

If $\|f_n - f\|_p \to 0$ for some $1 \le p < \infty$, then a subsequence $f_{n_k} \to f$ almost everywhere.

Fact B.19 (translates)

For a function f on $\mathbb{R}$ and $s \in \mathbb{R}$, the translate is $f_s(x) = f(x - s)$.

Fact B.20 (density of simple functions)

For $1 \le p < \infty$, the measurable simple functions s with $\mu(\{s \neq 0\}) < \infty$ are dense in $L^p(\mu)$; moreover every $f \in L^p(\mu)$ is the L^p-limit of simple functions s_n with $|s_n| \le |f|$.

Fact B.21 (density of continuous functions on an interval)

For $1 \le p < \infty$, the continuous functions on $[a,b]$ are dense in $L^p[a,b]$.

Fact B.22 (density of $C_c(\mathbb{R})$)

For $1 \le p < \infty$, the continuous functions with compact support are dense in $L^p(\mathbb{R})$.

Fact B.23 (continuity of translation)

For $f \in L^p(\mathbb{R})$ and $1 \le p < \infty$, $\|f_s - f\|_p \to 0$ as $s \to 0$ (with f_s as in Fact B.19).

Further properties of measures, functions and sums

Fact B.24 (continuity of measures)

Let $E_1, E_2, \dots \in \mathcal{S}$.
(a) Countable additivity: $\mu(\bigsqcup_n E_n) = \sum_n \mu(E_n)$ for disjoint E_n.
(b) Continuity from below: if $E_1 \subset E_2 \subset \cdots$, then $\mu(\bigcup_n E_n) = \lim_n \mu(E_n)$.
(c) Continuity from above: if $E_1 \supset E_2 \supset \cdots$ and $\mu(E_1) < \infty$, then $\mu(\bigcap_n E_n) = \lim_n \mu(E_n)$.

Fact B.25 (term-by-term integration)

For measurable $f_n : X \to [0,\infty]$, $\int_X \sum_{n=1}^{\infty} f_n \, d\mu = \sum_{n=1}^{\infty} \int_X f_n \, d\mu$ (a consequence of the monotone convergence theorem).

Fact B.26 (operations preserving measurability)

Let f, g be measurable (real or complex valued, with respect to a fixed σ-algebra).
(a) $f+g$ and fg are measurable.
(b) Pointwise sup, inf, lim sup, lim inf and limits of sequences of measurable functions are measurable; in particular a pointwise convergent series of measurable functions is measurable.
(c) $|f|$, $\operatorname{Re} f$, $\operatorname{Im} f$ are measurable.
(d) If $g : \mathbb{R} \to \mathbb{R}$ is continuous, then $g \circ f$ is measurable.

Fact B.27 (averages in a closed set)

Let $\mu(X) < \infty$, $g \in L^1(\mu)$ and S a closed subset of $\mathbb{C}$. If $\frac{1}{\mu(E)} \int_E g \, d\mu \in S$ for every $E \in \mathcal{S}$ with $\mu(E) > 0$, then $g(x) \in S$ for almost every x. (For σ-finite μ, apply this on each set of finite measure.)

Fact B.28 (sums over arbitrary index sets)

For $a_\alpha \geq 0$ ($\alpha \in A$, A any set), $\sum_{\alpha \in A} a_\alpha := \sup\{\sum_{\alpha \in F} a_\alpha : F \subset A \text{ finite}\}$. Hence if $\sum_{\alpha \in F} a_\alpha \leq M$ for every finite $F \subset A$, then $\sum_{\alpha \in A} a_\alpha \leq M$; and if the sum is finite, $a_\alpha \neq 0$ for at most countably many α.

Fact B.29 (inclusions between L^p spaces)

If $\mu(X) < \infty$ and $1 \leq p < q \leq \infty$, then $L^q(\mu) \subset L^p(\mu)$ and $\|f\|_p \leq \mu(X)^{1/p - 1/q} \|f\|_q$. In particular, on $\mathbb{T}$ with total mass 1, $L^2(\mathbb{T}) \subset L^1(\mathbb{T})$ and $\|f\|_1 \leq \|f\|_2$.

Chapter 4. Double Integrals

Overview

Results from Chapters 1–3 are cited as *Fact B.k*; they are stated in the separate *Background* notes. When dealing with functions of several variables one constantly needs to interchange the order of integration. To do this with the Lebesgue integral we must first have a measure on the plane. This chapter defines the **product** of two general measures, uses it to define double integrals, and determines when a double integral equals the iterated integrals (Tonelli, Fubini). Double integrals are essential for defining the **convolution**, which appears constantly in Fourier analysis.

4.1 Product Measures

Motivation. Double integrals over $\mathbb{R}^2$ should be computable as iterated integrals, but with the Lebesgue integral this needs a measure on $X \times Y$ built from measures on X and Y. The construction must handle sets far more complicated than rectangles, which is why the monotone class argument appears.

4.1.1 Setting up the problem

Let $(X, \mathcal{S}, \mu)$ and $(Y, \mathcal{T}, \nu)$ be measure spaces. To put a measure on $X \times Y$ we first need a σ-algebra, which should certainly contain the **measurable rectangles**

$$\mathcal{R} = \{S \times T \subset X \times Y : \ S \in \mathcal{S}, \ T \in \mathcal{T}\}.$$

Once a σ-algebra is fixed, the way to define the measure is clear. For $E \subset X \times Y$, $x \in X$, $y \in Y$ define the **sections**

$$E_x = \{y \in Y : (x, y) \in E\}, \qquad E^y = \{x \in X : (x, y) \in E\}.$$

If $E_x \in \mathcal{T}$ and $E^y \in \mathcal{S}$ for all x, y, we can define

$$\varphi : x \mapsto \nu(E_x) = \int_Y \chi_E(x, y)\, d\nu(y), \qquad \psi : y \mapsto \mu(E^y) = \int_X \chi_E(x, y)\, d\mu(x). \tag{1}$$

If $\varphi : X \to [0, \infty]$ and $\psi : Y \to [0, \infty]$ are measurable and

$$\int_X \varphi \, d\mu = \int_Y \psi \, d\nu, \tag{2}$$

it is natural to define this common value to be the measure of E. The conclusion will be: if $\mathcal{S} \times \mathcal{T}$ denotes the smallest σ-algebra containing $\mathcal{R}$, every $E \in \mathcal{S} \times \mathcal{T}$ has all these properties.

4.1.2 The smallest σ-algebra

Let $\mathcal{U}$ be any collection of subsets of a set Z and let Ω be the family of all σ-algebras containing $\mathcal{U}$; $\Omega \neq \emptyset$ since the power set belongs to it. Put

$$\mathcal{W} = \bigcap \{\mathcal{M} : \mathcal{M} \in \Omega\}.$$

Clearly $\emptyset \in \mathcal{W}$. If $A \in \mathcal{W}$ then $A \in \mathcal{M}$ for every $\mathcal{M} \in \Omega$, hence $A^c \in \mathcal{M}$ for every $\mathcal{M}$, so $A^c \in \mathcal{W}$. Likewise $A_n \in \mathcal{W}$ $(n = 1, 2, \dots)$ implies $\bigcup_n A_n \in \mathcal{W}$. Thus $\mathcal{W}$ is a σ-algebra, and if $\mathcal{M}$ is any σ-algebra containing $\mathcal{U}$ then $\mathcal{W} \subset \mathcal{M}$: $\mathcal{W}$ is the **smallest σ-algebra containing $\mathcal{U}$**.

4.1.3 Sections of sets in $\mathcal{S} \times \mathcal{T}$

To show $E_x \in \mathcal{T}$ for every $E \in \mathcal{S} \times \mathcal{T}$ and $x \in X$, let

$$\mathcal{A} = \{E \in \mathcal{S} \times \mathcal{T} : \ E_x \in \mathcal{T} \ (x \in X)\}.$$

Clearly $\mathcal{R} \subset \mathcal{A}$, so it suffices to show $\mathcal{A}$ is a σ-algebra (then $\mathcal{A} \supset \mathcal{S} \times \mathcal{T}$). $\emptyset \in \mathcal{A}$. If $E \in \mathcal{A}$ then $(E^c)_x = (E_x)^c$, so $E^c \in \mathcal{A}$. If $E_n \in \mathcal{A}$ then $(\bigcup_n E_n)_x = \bigcup_n (E_n)_x$, so $\bigcup_n E_n \in \mathcal{A}$. Treating E^y the same way gives

$$E \in \mathcal{S} \times \mathcal{T} \implies E_x \in \mathcal{T}, \ E^y \in \mathcal{S} \qquad (x \in X, \ y \in Y). \tag{3}$$

Exercise 4.1.1

In the argument above, show that $(E^c)_x = (E_x)^c$ and $(\bigcup_n E_n)_x = \bigcup_n (E_n)_x$.

Solution [Supplement]. $y \in (E^c)_x \iff (x,y) \in E^c \iff (x,y) \notin E \iff y \notin E_x \iff y \in (E_x)^c$. $y \in (\bigcup_n E_n)_x \iff (x,y) \in E_n$ for some $n \iff y \in (E_n)_x$ for some $n \iff y \in \bigcup_n (E_n)_x$. □

4.1.4 The monotone class argument

From now on assume X and Y can be partitioned into sets of finite measure:

$$X = \bigsqcup_{n=1}^{\infty} X_n, \qquad Y = \bigsqcup_{m=1}^{\infty} Y_m, \qquad \mu(X_n), \ \nu(Y_m) < \infty.$$

To show that every $E \in \mathcal{S} \times \mathcal{T}$ satisfies (2), consider

$$\mathcal{B} = \{E \in \mathcal{S} \times \mathcal{T} : \ \text{the functions } \varphi, \psi \text{ of (1) are measurable and (2) holds}\},$$
$$\mathcal{C} = \{E \in \mathcal{S} \times \mathcal{T} : \ E \cap (X_n \times Y_m) \in \mathcal{B} \ \ (m, n = 1, 2, \dots)\}.$$

Lemma 4.1.1

For the collections $\mathcal{B}$ and $\mathcal{C}$:

(a) $\mathcal{B}$ is closed under unions of increasing sequences.

(b) $\mathcal{B}$ is closed under countable disjoint unions.

(c) $\mathcal{C}$ is closed under unions of increasing sequences and intersections of decreasing sequences.

(d) If $E_1, \dots, E_N \in \mathcal{R}$ are pairwise disjoint, then $\bigcup_{n=1}^{N} E_n \in \mathcal{C}$.

Proof. (a) Let $\langle E_i \rangle$ be an increasing sequence in $\mathcal{B}$ with union E, and let φ_i, ψ_i and φ, ψ be the corresponding functions in (1). Since $\langle (E_i)_x \rangle$ increases to E_x, Fact B.24(b) (continuity from below) gives $\varphi_i \nearrow \varphi$. Each φ_i is measurable, so φ is measurable; the same holds for ψ_i, ψ. By the monotone convergence theorem,

$$\int_X \varphi \, d\mu = \lim_{i \to \infty} \int_X \varphi_i \, d\mu = \lim_{i \to \infty} \int_Y \psi_i \, d\nu = \int_Y \psi \, d\nu.$$

(b) If $E, F \in \mathcal{B}$ are disjoint, the functions in (1) for $E \sqcup F$ are sums of those for E and F, so $E \sqcup F \in \mathcal{B}$; by induction $\mathcal{B}$ is closed under finite disjoint unions. A countable disjoint union is the increasing union of the finite partial unions, so (a) applies.

(c) If $\langle E_i \rangle$ is an increasing sequence in $\mathcal{C}$, then for each m, n the sequence

$$\langle E_i \cap (X_n \times Y_m) : i = 1, 2, \dots \rangle \tag{4}$$

is an increasing sequence in $\mathcal{B}$, so by (a) $\bigcup_i E_i \in \mathcal{C}$. If $\langle E_i \rangle$ is decreasing in $\mathcal{C}$, consider the intersection of the sequence (4). Proceed as in (a), but use Fact B.24(c) (continuity from above, valid because $\nu(Y_m), \mu(X_n) < \infty$) and the Lebesgue (dominated) convergence theorem (the functions are bounded by $\nu(Y_m)\chi_{X_n}$, which is integrable). Hence the intersection lies in $\mathcal{B}$, proving (c).

(d) Clearly $\mathcal{R} \subset \mathcal{C}$: for $E = S \times T$, $\varphi = \nu(T \cap Y_m)\chi_{S \cap X_n}$ and $\psi = \mu(S \cap X_n)\chi_{T \cap Y_m}$ are measurable with equal integrals $\mu(S \cap X_n)\nu(T \cap Y_m)$. Then (b) gives (d). ■

Now let $\mathcal{D}$ be the smallest collection containing $\mathcal{R}$ and satisfying (c) and (d) of Lemma 4.1.1. Then

$$\mathcal{D} \subset \mathcal{C} \subset \mathcal{S} \times \mathcal{T}, \tag{5}$$

and we will show that $\mathcal{D}$ is a σ-algebra. Let $\mathcal{E}$ be the collection of sets expressible as finite disjoint unions of measurable rectangles. Then $\mathcal{D}$ is the smallest collection containing $\mathcal{E}$ and closed under unions of increasing sequences and intersections of decreasing sequences (a *monotone class*).

Exercise 4.1.2

Show that for any collection of sets there is a smallest collection containing it that is closed under unions of increasing sequences and intersections of decreasing sequences.

Solution [Supplement]**.** Let $\mathcal{U}$ be a collection of subsets of Z and Ω the family of all monotone classes containing $\mathcal{U}$ ($\Omega \neq \emptyset$: the power set of Z is one). The intersection $\mathcal{W} = \bigcap \Omega$ contains $\mathcal{U}$. If $\langle A_i \rangle$ is an increasing (decreasing) sequence in $\mathcal{W}$, then it is such a sequence in every $\mathcal{M} \in \Omega$, so $\bigcup A_i$ ($\bigcap A_i$) lies in every $\mathcal{M}$, hence in $\mathcal{W}$. So $\mathcal{W}$ is a monotone class, and it is contained in every monotone class containing $\mathcal{U}$. □

A simple set computation shows that $\mathcal{E}$ is closed under differences and unions:

$$E, F \in \mathcal{E} \implies E \setminus F \in \mathcal{E},\ F \setminus E \in \mathcal{E},\ E \cup F \in \mathcal{E}. \tag{6}$$

Exercise 4.1.3

Prove (6).

Solution [Supplement]**.** (i) The intersection of two rectangles is a rectangle: $(S \times T) \cap (S' \times T') = (S \cap S') \times (T \cap T')$. Hence if $E = \bigsqcup_i R_i$, $F = \bigsqcup_j R'_j$ are in $\mathcal{E}$, then $E \cap F = \bigsqcup_{i,j}(R_i \cap R'_j) \in \mathcal{E}$; so $\mathcal{E}$ is closed under finite intersections.

(ii) The complement of a rectangle is a disjoint union of two rectangles: $(S \times T)^c = (S^c \times Y) \sqcup (S \times T^c)$. So $(\bigsqcup_i R_i)^c = \bigcap_i R_i^c$ is a finite intersection of members of $\mathcal{E}$, hence in $\mathcal{E}$ by (i).

(iii) Therefore $E \setminus F = E \cap F^c \in \mathcal{E}$, likewise $F \setminus E \in \mathcal{E}$, and $E \cup F = (E \setminus F) \sqcup F \in \mathcal{E}$, the union being disjoint and each piece a finite disjoint union of rectangles. □

For an arbitrary $E \subset X \times Y$ define

$$\Omega(E) = \{F \subset X \times Y :\ E \setminus F \in \mathcal{D},\ F \setminus E \in \mathcal{D},\ E \cup F \in \mathcal{D}\}.$$

Then $\Omega(E)$ also satisfies Lemma 4.1.1(c).

Exercise 4.1.4

Show that $\Omega(E)$ satisfies (c) of Lemma 4.1.1.

Solution [Supplement]. Let $F_i \in \Omega(E)$ increase to F. Then $\langle E \setminus F_i\rangle$ decreases to $E \setminus F$, $\langle F_i \setminus E\rangle$ increases to $F \setminus E$, and $\langle E \cup F_i\rangle$ increases to $E \cup F$; all terms lie in $\mathcal{D}$, which is a monotone class, so the three limits lie in $\mathcal{D}$ and $F \in \Omega(E)$. If F_i decreases to F, then $E \setminus F_i \nearrow E \setminus F$, $F_i \setminus E \searrow F \setminus E$, $E \cup F_i \searrow E \cup F$, and the same argument applies. □

If $E, F \in \mathcal{E}$, then by (6) $F \in \Omega(E)$, so $\mathcal{E} \subset \Omega(E)$ and therefore $\mathcal{D} \subset \Omega(E)$, i.e.

$$E \in \mathcal{E} \implies \mathcal{D} \subset \Omega(E).$$

Hence if $E \in \mathcal{D}$ and $F \in \mathcal{E}$, then $E \in \Omega(F)$, which is equivalent to $F \in \Omega(E)$ (the definition is symmetric). So $\mathcal{E} \subset \Omega(E)$ and thus $\mathcal{D} \subset \Omega(E)$. In other words

$$E \in \mathcal{D} \implies \mathcal{D} \subset \Omega(E),$$

which says exactly that $\mathcal{D}$ is closed under differences and finite unions. Since $X \times Y \in \mathcal{D}$, $\mathcal{D}$ is closed under complements. If $E_n \in \mathcal{D}$, put $F_n = E_1 \cup \cdots \cup E_n$; then $\langle F_n\rangle$ is increasing in $\mathcal{D}$, so $\bigcup_n E_n = \bigcup_n F_n \in \mathcal{D}$. Thus $\mathcal{D}$ is a σ-algebra. Since $\mathcal{S} \times \mathcal{T}$ is the smallest σ-algebra containing $\mathcal{E}$, (5) gives

$$\mathcal{D} = \mathcal{C} = \mathcal{S} \times \mathcal{T}. \tag{7}$$

Consequently, if $E \in \mathcal{S} \times \mathcal{T} = \mathcal{C}$, then $E \cap (X_n \times Y_m) \in \mathcal{B}$ for all m, n; these sets are pairwise disjoint with union E, so $E \in \mathcal{B}$ by Lemma 4.1.1(b). We have shown $\mathcal{S} \times \mathcal{T} \subset \mathcal{B}$.

Definition (σ-finite)

A measure space $(X, \mathcal{S}, \mu)$ is **σ-finite** if X is a countable union of sets of finite measure.

Theorem 4.1.2

Let $(X, \mathcal{S}, \mu)$, $(Y, \mathcal{T}, \nu)$ be measure spaces and let $\mathcal{S} \times \mathcal{T}$ be the smallest σ-algebra containing the measurable rectangles. If $E \in \mathcal{S} \times \mathcal{T}$, then $E_x \in \mathcal{T}$ and $E^y \in \mathcal{S}$ for all $x \in X$, $y \in Y$. If $(X, \mathcal{S}, \mu)$ and $(Y, \mathcal{T}, \nu)$ are σ-finite, then the functions φ, ψ of (1) are measurable and (2) holds.

(The proof is the discussion above.) Writing the common value of (2) as $(\mu \times \nu)(E)$,

$$\begin{aligned}(\mu \times \nu)(E) &= \int_X \Big[\int_Y \chi_E(x,y)\, d\nu(y)\Big] d\mu(x) = \int_X \nu(E_x)\, d\mu(x) \\ &= \int_Y \Big[\int_X \chi_E(x,y)\, d\mu(x)\Big] d\nu(y) = \int_Y \mu(E^y)\, d\nu(y).\end{aligned} \tag{8}$$

$\mu \times \nu$ is a measure: if $E_n \in \mathcal{S} \times \mathcal{T}$ are disjoint with $E = \bigsqcup_n E_n$, then $E_x = \bigsqcup_n (E_n)_x$, and by Fact B.25 (term-by-term integration of nonnegative series),

$$\sum_{n=1}^{\infty} (\mu \times \nu)(E_n) = \sum_{n=1}^{\infty} \int_X \nu((E_n)_x)\, d\mu(x) = \int_X \sum_{n=1}^{\infty} \nu((E_n)_x)\, d\mu(x) = \int_X \nu(E_x)\, d\mu(x) = (\mu \times \nu)(E).$$

The measure space $(X \times Y, \mathcal{S} \times \mathcal{T}, \mu \times \nu)$ is the **product measure** of the σ-finite measure spaces $(X, \mathcal{S}, \mu)$ and $(Y, \mathcal{T}, \nu)$.

Exercise 4.1.5

Show that $(\mu \times \nu)(S \times T) = \mu(S)\nu(T)$ for every $S \in \mathcal{S}$, $T \in \mathcal{T}$.

Solution [Supplement]. $(S \times T)_x = T$ if $x \in S$ and $= \emptyset$ otherwise, so $\nu((S \times T)_x) = \nu(T)\chi_S(x)$. By (8), $(\mu \times \nu)(S \times T) = \int_X \nu(T)\chi_S\, d\mu = \nu(T)\mu(S)$ (with $0 \cdot \infty = 0$). □

Exercise 4.1.6

If $(X, \mathcal{S}, \mu)$ and $(Y, \mathcal{T}, \nu)$ are σ-finite, show that $(X \times Y, \mathcal{S} \times \mathcal{T}, \mu \times \nu)$ is σ-finite.

Solution [Supplement]. With $X = \bigsqcup_n X_n$, $Y = \bigsqcup_m Y_m$ as above, $X \times Y = \bigsqcup_{m,n}(X_n \times Y_m)$ is a countable union, and by Exercise 4.1.5 $(\mu \times \nu)(X_n \times Y_m) = \mu(X_n)\nu(Y_m) < \infty$. □

4.1.5 The product of Lebesgue measures; Borel sets

Consider the product of Lebesgue measure $(\mathbb{R}, \mathcal{M}, m)$ with itself. Given an open set $U \subset \mathbb{R}^2$, the collection

$$\{(a, b) \times (c, d) \subset U : \ a, b, c, d \text{ rational}\}$$

is countable and its union is U, so $U \in \mathcal{M} \times \mathcal{M}$: every open set is in $\mathcal{M} \times \mathcal{M}$. Let $\mathcal{B}_n$ be the smallest σ-algebra containing the open sets of $\mathbb{R}^n$. Then

$$\mathcal{B}_2 \subset \mathcal{M} \times \mathcal{M}.$$

Members of $\mathcal{B}_n$ are called **Borel sets** of $\mathbb{R}^n$; closed sets, G_δ-sets and F_σ-sets are typical examples.

However, $\mathcal{M} \times \mathcal{M}$ has a defect. Take $E \in \mathcal{M}$ with $m(E) = 0$, $E \neq \emptyset$, and $F \notin \mathcal{M}$, $F \subset \mathbb{R}$. By (3), $E \times F \notin \mathcal{M} \times \mathcal{M}$ (its sections $(E \times F)^y = E$ for $y \in F$, but $(E \times F)_x = F \notin \mathcal{M}$ for $x \in E$). Yet $E \times F \subset E \times \mathbb{R}$ and $(m \times m)(E \times \mathbb{R}) = 0$. So a subset of a null set can fail to be measurable.

Definition (complete measure)

A measure space $(X, \mathcal{S}, \mu)$ is **complete** if

$$E \subset F, \ F \in \mathcal{S}, \ \mu(F) = 0 \implies E \in \mathcal{S}.$$

Thus $(\mathbb{R}^2, \mathcal{M} \times \mathcal{M}, m \times m)$ is not complete, while $(\mathbb{R}, \mathcal{M}, m)$ is complete by Fact B.2.

Proposition 4.1.3 (completion)

For a measure space $(X, \mathcal{S}, \mu)$ let

$$\mathcal{S}^* = \{E \subset X : \text{ there exist } A, B \in \mathcal{S} \text{ with } A \subset E \subset B, \ \mu(B \setminus A) = 0\},$$

and for $E \in \mathcal{S}^*$ put $\mu^*(E) = \mu(A)$. Then $(X, \mathcal{S}^*, \mu^*)$ is a complete measure space.

Proof. *Well-definedness.* Suppose $A_i \subset E \subset B_i$, $\mu(B_i \setminus A_i) = 0$ for $i = 1, 2$. Then

$$A_2 \setminus A_1 \subset E \setminus A_1 \subset B_1 \setminus A_1,$$

so $\mu(A_2 \setminus A_1) = 0$; likewise $\mu(A_1 \setminus A_2) = 0$, hence $\mu(A_1) = \mu(A_2)$.

$\mathcal{S}^*$ is a σ-algebra. Clearly $\emptyset \in \mathcal{S}^*$. If $E \in \mathcal{S}^*$ with $A \subset E \subset B$, $\mu(B \setminus A) = 0$, then $B^c \subset E^c \subset A^c$ and $A^c \setminus B^c = B \setminus A$, so $E^c \in \mathcal{S}^*$. Finally let $A_n \subset E_n \subset B_n$ with $\mu(B_n \setminus A_n) = 0$, and put $A = \bigcup_n A_n$, $E = \bigcup_n E_n$, $B = \bigcup_n B_n$. Then $A \subset E \subset B$ and

$$B \setminus A = \bigcup_n (B_n \setminus A) \subset \bigcup_n (B_n \setminus A_n),$$

so $\mu(B \setminus A) = 0$ and $E \in \mathcal{S}^*$.

μ^* is a measure. If the E_n are disjoint, so are the A_n, and

$$\mu^*(E) = \mu(A) = \sum_{n=1}^{\infty} \mu(A_n) = \sum_{n=1}^{\infty} \mu^*(E_n).$$

Completeness. Let $E \subset F$, $F \in \mathcal{S}^*$, $\mu^*(F) = 0$. Choose $A, B \in \mathcal{S}$ with $A \subset F \subset B$ and $\mu(A) = \mu(B) = 0$. Then $\emptyset \subset E \subset B$ and $\mu(B) = 0$, so $E \in \mathcal{S}^*$. ■

Exercise 4.1.7

If $(X, \mathcal{S}, \mu)$ is complete, show that $\mathcal{S}^* = \mathcal{S}$.

Solution [Supplement]**.** $\mathcal{S} \subset \mathcal{S}^*$ always (take $A = B = E$). Conversely, let $E \in \mathcal{S}^*$ with $A \subset E \subset B$, $\mu(B \setminus A) = 0$. Then $E \setminus A \subset B \setminus A$, a null set in $\mathcal{S}$, so $E \setminus A \in \mathcal{S}$ by completeness, and $E = A \cup (E \setminus A) \in \mathcal{S}$. □

4.1.6 Lebesgue measure on $\mathbb{R}^2$

Compare the two measures $(\mathbb{R}^2, \mathcal{B}_2, m \times m)$ and $(\mathbb{R}^2, \mathcal{M} \times \mathcal{M}, m \times m)$, i.e. $\mathcal{B}_2^*$ and $(\mathcal{M} \times \mathcal{M})^*$. Since $\mathcal{B}_2 \subset \mathcal{M} \times \mathcal{M}$, clearly $\mathcal{B}_2^* \subset (\mathcal{M} \times \mathcal{M})^*$.

If $S \in \mathcal{M}$, Fact B.3 gives an F_σ-set A and a G_δ-set B with $A \subset S \subset B$, $m(B \setminus A) = 0$. Then $A \times \mathbb{R}$ and $B \times \mathbb{R}$ are F_σ- and G_δ-sets of $\mathbb{R}^2$, so they lie in $\mathcal{B}_2$, and

$$(m \times m)[(B \times \mathbb{R}) \setminus (A \times \mathbb{R})] = (m \times m)[(B \setminus A) \times \mathbb{R}] = 0,$$

so $S \times \mathbb{R} \in \mathcal{B}_2^*$. Similarly $T \in \mathcal{M}$ implies $\mathbb{R} \times T \in \mathcal{B}_2^*$, so every measurable rectangle $S \times T = (S \times \mathbb{R}) \cap (\mathbb{R} \times T)$ belongs to $\mathcal{B}_2^*$. Hence $\mathcal{M} \times \mathcal{M} \subset \mathcal{B}_2^*$, and since $(\mathcal{M} \times \mathcal{M})^* \subset \mathcal{B}_2^{**} = \mathcal{B}_2^*$,

$$(\mathcal{M} \times \mathcal{M})^* = \mathcal{B}_2^*.$$

Write $\mathcal{M}_2$ for this σ-algebra and m_2 for the measure $(m \times m)^*$ on it. The complete measure space $(\mathbb{R}^2, \mathcal{M}_2, m_2)$ is the **Lebesgue measure on** $\mathbb{R}^2$. (If one only wants Lebesgue measure on $\mathbb{R}^n$ from scratch, the same method used for $\mathbb{R}^1$ applies.)

Exercise 4.1.8

Use induction to define Lebesgue measure on $\mathbb{R}^n$.

Solution [Supplement]**.** Suppose the complete measure space $(\mathbb{R}^{n-1}, \mathcal{M}_{n-1}, m_{n-1})$ has been defined ($m_1 = m$); it is σ-finite since $\mathbb{R}^{n-1} = \bigcup_k [-k, k]^{n-1}$. Identify $\mathbb{R}^n = \mathbb{R}^{n-1} \times \mathbb{R}$, form the product measure $(\mathbb{R}^n, \mathcal{M}_{n-1} \times \mathcal{M}, m_{n-1} \times m)$ (Theorem 4.1.2), and define

$$\mathcal{M}_n = (\mathcal{M}_{n-1} \times \mathcal{M})^*, \qquad m_n = (m_{n-1} \times m)^*$$

(Proposition 4.1.3). Exactly as above one checks $\mathcal{B}_n \subset \mathcal{M}_{n-1} \times \mathcal{M}$ (open sets are countable unions of rational boxes) and $\mathcal{M}_n = \mathcal{B}_n^*$, using inductively that each $S \in \mathcal{M}_{n-1}$ is squeezed between an F_σ- and a G_δ-set of $\mathbb{R}^{n-1}$ differing by a null set. By Exercise 4.1.5 and induction, m_n of a box $\prod_i [a_i, b_i]$ is $\prod_i (b_i - a_i)$. □

4.2 Fubini's Theorem

Motivation. The point of product measures is to justify swapping the order of integration. Tonelli's theorem allows this for nonnegative functions without conditions, Fubini's for integrable ones; the classical counterexamples show that some hypothesis is necessary.

Equation (8) says that the order of integration can be interchanged for characteristic functions of sets in $\mathcal{S} \times \mathcal{T}$. Since the integral is built from integrals of characteristic functions, we expect the same for nonnegative measurable functions and for integrable functions. For a function f on $X \times Y$ define

$$f_x : y \mapsto f(x, y), \qquad f^y : x \mapsto f(x, y).$$

Proposition 4.2.1

If $f : X \times Y \to \mathbb{R}^*$ is $\mathcal{S} \times \mathcal{T}$-measurable, then for every $x \in X$ and $y \in Y$, f_x is $\mathcal{T}$-measurable and f^y is $\mathcal{S}$-measurable.

Proof. For $\alpha \in \mathbb{R}$ put $E = \{(x, y) \in X \times Y : f(x, y) > \alpha\}$. Then $E_x = \{y \in Y : f_x(y) > \alpha\}$. Since $E \in \mathcal{S} \times \mathcal{T}$, (3) gives $E_x \in \mathcal{T}$, so f_x is measurable. Similarly f^y is measurable. ■

From now on, "measurable" on $X \times Y$ means measurable with respect to $\mathcal{S} \times \mathcal{T}$.

Exercise 4.2.1

If $f : \mathbb{R}^2 \to \mathbb{R}$ is measurable with respect to Lebesgue measure on $\mathbb{R}^2$, is $f_x : \mathbb{R} \to \mathbb{R}$ always Lebesgue measurable for every $x \in \mathbb{R}$?

Solution [Supplement]**.** No. Let $N \subset \mathbb{R}$ be a non-measurable set and $f = \chi_{\{0\} \times N}$. Since $\{0\} \times N \subset \{0\} \times \mathbb{R}$ and $m_2(\{0\} \times \mathbb{R}) = m(\{0\})m(\mathbb{R}) = 0$, completeness gives $\{0\} \times N \in \mathcal{M}_2$, so f is $\mathcal{M}_2$-measurable. But $f_0 = \chi_N$ is not Lebesgue measurable. (What is true is that f_x is measurable for *almost every* x; see Theorem 4.2.6.) □

Theorem 4.2.2 (Tonelli)

Let $(X, \mathcal{S}, \mu)$ and $(Y, \mathcal{T}, \nu)$ be σ-finite. For every measurable $f : X \times Y \to [0, \infty]$:

(a) the functions $x \mapsto \int_Y f_x \, d\nu$ and $y \mapsto \int_X f^y \, d\mu$ are measurable;

(b)
$$\int_X \Big[\int_Y f_x \, d\nu\Big] d\mu = \int_{X \times Y} f \, d(\mu \times \nu) = \int_Y \Big[\int_X f^y \, d\mu\Big] d\nu.$$

Proof. By Fact B.5 there is an increasing sequence of simple functions $\langle s_n \rangle$ with $s_n \nearrow f$ pointwise. Put
$$\varphi(x) = \int_Y f_x \, d\nu, \qquad \varphi_n(x) = \int_Y (s_n)_x \, d\nu.$$
By Theorem 4.1.2, characteristic functions of measurable sets satisfy (a) and (b), and by linearity so do simple functions. Since $s_n \nearrow f$, we have $(s_n)_x \nearrow f_x$, and the monotone convergence theorem for ν gives $\varphi_n \nearrow \varphi$. Hence φ is measurable. Applying the monotone convergence theorem for $\mu \times \nu$ and for μ,
$$\int_{X \times Y} f \, d(\mu \times \nu) = \lim_{n \to \infty} \int_{X \times Y} s_n \, d(\mu \times \nu) = \lim_{n \to \infty} \int_X \varphi_n \, d\mu = \int_X \varphi \, d\mu.$$
The other half is proved in exactly the same way. ■

Exercise 4.2.2

Derive Fact B.25 from Tonelli's theorem.

Solution [Supplement]**.** Formula Fact B.25 is term-by-term integration: for measurable $f_n : X \to [0, \infty]$, $\int_X \sum_n f_n \, d\mu = \sum_n \int_X f_n \, d\mu$. Let $Y = \mathbb{N}$ with the counting measure c (on the power set), which is σ-finite, and define $f(x, n) = f_n(x)$. Every subset of $\mathbb{N}$ is measurable, and $\{f > \alpha\} = \bigcup_n \{f_n > \alpha\} \times \{n\}$, so f is $\mathcal{S} \times \mathcal{P}(\mathbb{N})$-measurable. Since $\int_{\mathbb{N}} g \, dc = \sum_n g(n)$, Tonelli gives
$$\int_X \sum_n f_n(x) \, d\mu(x) = \int_{X \times \mathbb{N}} f \, d(\mu \times c) = \sum_n \int_X f_n \, d\mu.$$
□

The same holds for integrable functions; we first find a sufficient condition for f to be integrable.

Theorem 4.2.3 (Fubini)

Let $(X,\mathcal{S},\mu)$ and $(Y,\mathcal{T},\nu)$ be σ-finite and let $f: X\times Y\to\mathbb{C}$ be measurable with

$$\int_X \Big[\int_Y |f(x,y)|\,d\nu(y)\Big]d\mu(x)<\infty.$$

Then:

(a) $f\in L^1(\mu\times\nu)$; moreover $f_x\in L^1(\nu)$ for almost every $x\in X$ and $f^y\in L^1(\mu)$ for almost every $y\in Y$;

(b) the functions $x\mapsto\int_Y f_x\,d\nu$ and $y\mapsto\int_X f^y\,d\mu$, defined almost everywhere, are integrable;

(c)
$$\int_X\Big[\int_Y f_x\,d\nu\Big]d\mu=\int_{X\times Y} f\,d(\mu\times\nu)=\int_Y\Big[\int_X f^y\,d\mu\Big]d\nu.$$

Proof. By Tonelli, $\int_{X\times Y}|f|=\int_X[\int_Y|f(x,y)|\,d\nu(y)]d\mu(x)<\infty$, so $f\in L^1(\mu\times\nu)$. Writing $f=u+iv$, it suffices to treat $f: X\times Y\to\mathbb{R}$. Put

$$\varphi(x)=\int_Y f_x\,d\nu,\qquad \varphi_1(x)=\int_Y (f_+)_x\,d\nu,\qquad \varphi_2(x)=\int_Y (f_-)_x\,d\nu.$$

By Tonelli,

$$\int_X\varphi_1\,d\mu=\int_{X\times Y} f_+\,d(\mu\times\nu)<\infty,$$

so $\varphi_1\in L^1(\mu)$, and likewise $\varphi_2\in L^1(\mu)$. Therefore the set

$$E=\{x\in X:\varphi_1(x)<\infty,\ \varphi_2(x)<\infty\}$$

satisfies $\mu(E^c)=0$. For each $x\in E$, $f_x=(f_+)_x-(f_-)_x$ with both parts ν-integrable, so $f_x\in L^1(\nu)$; thus $f_x\in L^1(\nu)$ for almost every x. Also, for $x\in E$, $\varphi(x)=\varphi_1(x)-\varphi_2(x)$, and since $\varphi_1,\varphi_2\in L^1(\mu)$, $\varphi\in L^1(\mu)$. Finally, applying Tonelli to f_+ and f_-,

$$\int_X\varphi\,d\mu=\int_X\varphi_1\,d\mu-\int_X\varphi_2\,d\mu=\int_{X\times Y} f_+\,d(\mu\times\nu)-\int_{X\times Y} f_-\,d(\mu\times\nu)=\int_{X\times Y} f\,d(\mu\times\nu).$$

This proves half of the theorem; the other half is identical. ∎

Exercise 4.2.3

In Fubini's theorem let Y be the set of natural numbers with the measure of Fact B.4 (counting measure). What conclusion do you get? What if X is also taken to be $\mathbb{N}$ in the same way?

Solution [Supplement]**.** With $Y=\mathbb{N}$ and counting measure c, a function on $X\times\mathbb{N}$ is a sequence $f_n(x)=f(x,n)$ of measurable functions, and $\int_{\mathbb{N}} g\,dc=\sum_n g(n)$. Fubini says: if $\sum_n\int_X|f_n|\,d\mu<\infty$ (equivalently $\int_X\sum_n|f_n|\,d\mu<\infty$, by Tonelli), then $\sum_n f_n(x)$ converges absolutely for a.e. x, defines an integrable function, and

$$\int_X\sum_{n=1}^{\infty} f_n\,d\mu=\sum_{n=1}^{\infty}\int_X f_n\,d\mu.$$

If also $X=\mathbb{N}$ with counting measure, $f(m,n)=a_{mn}$ is a double sequence and we get: if $\sum_m\sum_n|a_{mn}|<\infty$, then

$$\sum_{m=1}^{\infty}\sum_{n=1}^{\infty}a_{mn}=\sum_{n=1}^{\infty}\sum_{m=1}^{\infty}a_{mn},$$

both sides converging absolutely; Tonelli gives the same equality for $a_{mn}\ge 0$ without any condition. □

Exercise 4.2.4

On $\mathbb{N}\times\mathbb{N}$ consider

$$f(m,n)=\begin{cases}2-2^{-m}, & m=n,\\ -2+2^{-m}, & m=n+1,\\ 0, & \text{otherwise.}\end{cases}$$

Compute the two iterated sums and see how this relates to the hypotheses of Tonelli's and Fubini's theorems. Do the same for

$$f(x,y)=\frac{x^2-y^2}{(x^2+y^2)^2},\qquad x,y\in[1,\infty).$$

Solution [Supplement]. (i) Fix n and sum over m: only $m=n$ and $m=n+1$ contribute, giving $(2-2^{-n})+(-2+2^{-n-1})=-2^{-n-1}$. Hence $\sum_n\sum_m f(m,n)=-\sum_{n\ge1}2^{-n-1}=-\frac12$. Fix m and sum over n: for $m=1$ only $n=1$ contributes, giving $2-\frac12=\frac32$; for $m\ge2$, $n=m$ and $n=m-1$ give $(2-2^{-m})+(-2+2^{-m})=0$. Hence $\sum_m\sum_n f(m,n)=\frac32$. The iterated sums differ. This does not contradict the theorems: f takes both signs, so Tonelli does not apply, and $\sum\sum|f(m,n)|\ge\sum_n(2-2^{-n})=\infty$, so the hypothesis of Fubini fails.

(ii) Since $\dfrac{\partial}{\partial y}\dfrac{y}{x^2+y^2}=\dfrac{x^2-y^2}{(x^2+y^2)^2}$,

$$\int_1^\infty f(x,y)\,dy=\Big[\frac{y}{x^2+y^2}\Big]_{y=1}^{\infty}=-\frac{1}{1+x^2},\qquad \int_1^\infty\!\!\int_1^\infty f\,dy\,dx=-\int_1^\infty\frac{dx}{1+x^2}=-\frac{\pi}{4}.$$

Since $f(y,x)=-f(x,y)$, the other order gives $+\frac{\pi}{4}$. Again f changes sign and $\iint|f|=\infty$ (otherwise Fubini would force equality), so neither theorem applies. □

4.2.1 Borel functions

A function on $\mathbb{R}^n$ is a **Borel function** if it is measurable with respect to $\mathcal{B}_n$; plain "measurable" means measurable with respect to Lebesgue measure $(\mathcal{M}_n,m_n)$. Borel functions are of course measurable. The following extends Fact B.26(d).

Proposition 4.2.4

If $f:X\to\mathbb{R}$ is measurable with respect to $(X,\mathcal{S},\mu)$ and $g:\mathbb{R}\to\mathbb{R}$ is a Borel function, then $g\circ f:X\to\mathbb{R}$ is measurable.

Proof. For each $\alpha\in\mathbb{R}$, $g^{-1}(\alpha,\infty)\in\mathcal{B}_1$, so it suffices to show $f^{-1}(E)\in\mathcal{S}$ for every $E\in\mathcal{B}_1$. Let

$$\mathcal{A}=\{E\in\mathcal{B}_1:\ f^{-1}(E)\in\mathcal{S}\}.$$

$\mathcal{A}$ is a σ-algebra of subsets of $\mathbb{R}$ (Exercise 4.2.5), and every open set lies in $\mathcal{A}$ (an open set is a countable union of open intervals, and $f^{-1}(a,b)=\{f>a\}\cap\{f<b\}\in\mathcal{S}$). Hence $\mathcal{A}=\mathcal{B}_1$. ■

Exercise 4.2.5

In the proof of Proposition 4.2.4, show that $\mathcal{A}$ is a σ-algebra.

Solution [Supplement]. $f^{-1}(\emptyset)=\emptyset\in\mathcal{S}$. If $E\in\mathcal{A}$ then $f^{-1}(E^c)=X\setminus f^{-1}(E)\in\mathcal{S}$. If $E_n\in\mathcal{A}$ then $f^{-1}(\bigcup_n E_n)=\bigcup_n f^{-1}(E_n)\in\mathcal{S}$. Since $\mathcal{B}_1$ is itself a σ-algebra, these sets also lie in $\mathcal{B}_1$. □

To apply Tonelli or Fubini to a function of two variables we must know it is $\mathcal{M}\times\mathcal{M}$-measurable, which is tedious to verify directly. Every Borel function on $\mathbb{R}^2$ is $\mathcal{M}\times\mathcal{M}$-measurable (as $\mathcal{B}_2\subset$

$\mathcal{M} \times \mathcal{M}$), so the next proposition is useful; it also yields Tonelli–Fubini for arbitrary measurable functions of two variables.

Proposition 4.2.5

For every measurable function f on $\mathbb{R}^n$ there is a Borel function g with $f = g$ almost everywhere.

Proof. It suffices to treat $f \geq 0$. By Fact B.6 write $f(x) = \sum_{k=1}^{\infty} c_k \chi_{E_k}(x)$ with $c_k \geq 0$ and $E_k \in \mathcal{M}_n$. For each k there are Borel sets $A_k \subset E_k \subset B_k$ with $m_n(B_k \setminus A_k) = 0$. Put

$$g(x) = \sum_{k=1}^{\infty} c_k \chi_{A_k}(x), \qquad x \in \mathbb{R}^n.$$

By Fact B.26, g is a Borel function. Moreover

$$\{x : f(x) \neq g(x)\} \subset \bigcup_{k=1}^{\infty} (E_k \setminus A_k) \subset \bigcup_{k=1}^{\infty} (B_k \setminus A_k),$$

a null set, so $f = g$ almost everywhere. ∎

Theorem 4.2.6 (Tonelli–Fubini on $\mathbb{R}^2$)

Let f be a measurable function on $\mathbb{R}^2$.

(a) For almost every x and almost every y, f_x and f^y are measurable.

(b) If $0 \leq f \leq \infty$, then

$$\int_{\mathbb{R}} \int_{\mathbb{R}} f(x,y)\, dx\, dy = \int_{\mathbb{R}\times\mathbb{R}} f\, dm_2 = \int_{\mathbb{R}} \int_{\mathbb{R}} f(x,y)\, dy\, dx. \tag{9}$$

(c) If $f : \mathbb{R}^2 \to \mathbb{C}$ and $\int_{\mathbb{R}} \int_{\mathbb{R}} |f(x,y)|\, dx\, dy < \infty$, then $f \in L^1(m_2)$; moreover $f_x, f^y \in L^1(m_1)$ for almost every x, y, and (9) holds.

Proof. By Proposition 4.2.5, $f = g + h$ where g is a Borel function and h is measurable with $h = 0$ almost everywhere. As discussed above, g is $\mathcal{M} \times \mathcal{M}$-measurable, so g satisfies everything in the theorem by Proposition 4.2.1 and Theorems 4.2.2, 4.2.3 (note $\int g\, dm_2 = \int g\, d(m \times m)$ since m_2 extends $m \times m$). It therefore suffices to prove the statements for h. Let

$$E = \{(x,y) \in \mathbb{R}^2 : h(x,y) \neq 0\}.$$

Then $E \in \mathcal{M}_2$ and $m_2(E) = 0$, so there is $F \in \mathcal{M} \times \mathcal{M}$ with $E \subset F$ and $(m \times m)(F) = 0$. By (8), $\int_{\mathbb{R}} m(F_x)\, dx = (m \times m)(F) = 0$, so

$$A = \{x \in \mathbb{R} : m(F_x) > 0\}$$

has $m(A) = 0$. Fix $x \in \mathbb{R} \setminus A$. Then $m(F_x) = 0$ and $E_x \subset F_x$, so $m(E_x) = 0$ (by completeness of m). If $y \notin E_x$ then $h_x(y) = h(x,y) = 0$; so h_x vanishes outside a null set, hence is measurable and $h_x = 0$ a.e. Thus for almost every x, h_x is measurable and $h_x(y) = 0$ for almost every y. The same holds for h^y, proving (a);

(b) and (c) for h follow at once since all integrals of h, h_x, h^y (for a.e. x, y) vanish. ∎

Exercise 4.2.6

Complete the proof of Theorem 4.2.6.

Solution [Supplement]**.** Write $f = g + h$ as in the proof. For (b), let $f \geq 0$; we may take $g \geq 0$ (replace g by g_+, still Borel, and h by $f - g_+$, still zero a.e.). For $x \notin A$, $f_x = g_x$ a.e., so $\int f_x\,dy = \int g_x\,dy$ for a.e. x, hence $\int\!\int f\,dy\,dx = \int\!\int g\,dy\,dx$. Also $\int f\,dm_2 = \int g\,dm_2 = \int g\,d(m \times m)$ because $f = g$ m_2-a.e. and g is $\mathcal{M} \times \mathcal{M}$-measurable with $m_2 = m \times m$ there. Tonelli for g gives $\int\!\int g\,dy\,dx = \int g\,d(m \times m) = \int\!\int g\,dx\,dy$, and the $dx\,dy$ side for f is handled by the symmetric argument with h^y. This is (9). For (c), $|f|$ is measurable and nonnegative, so by (b) $\int |f|\,dm_2 = \int\!\int |f|\,dx\,dy < \infty$, i.e. $f \in L^1(m_2)$; then $g \in L^1(m \times m)$ and Fubini applies to g. Since $f_x = g_x$ a.e. for a.e. x (and $f^y = g^y$ a.e. for a.e. y), the integrability of the sections and equality (9) transfer from g to f. □

Example 1 (the Dirichlet integral)

Apply Fubini to $f(x,t) = e^{-xt}\sin x$ on $[0,A] \times [0,\infty)$. Since

$$\frac{1}{x} = \int_0^\infty e^{-xt}\,dt, \qquad x > 0,$$

we have

$$\int_0^1\!\int_0^\infty |e^{-xt}\sin x|\,dt\,dx \leq \int_0^1 x\int_0^\infty e^{-xt}\,dt\,dx \leq \int_0^1 dx = 1,$$
$$\int_1^A\!\int_0^\infty |e^{-xt}\sin x|\,dt\,dx \leq \int_1^A\!\int_0^\infty e^{-xt}\,dt\,dx = \int_1^A \frac{1}{x}\,dx < \infty,$$

so Fubini applies (using $|\sin x| \leq x$ on $[0,1]$). Therefore

$$\int_0^A \frac{\sin x}{x}\,dx = \int_0^A\!\int_0^\infty e^{-xt}\sin x\,dt\,dx = \int_0^\infty\!\int_0^A e^{-xt}\sin x\,dx\,dt$$
$$= \int_0^\infty \Big[\frac{e^{-At}}{1+t^2}(-t\sin A - \cos A) + \frac{1}{1+t^2}\Big]dt.$$

Since

$$\int_0^\infty \Big|\frac{e^{-tA}}{1+t^2}(-t\sin A - \cos A)\Big|dt \leq \int_0^\infty 2e^{-tA}\,dt = \frac{2}{A}$$

(using $|t\sin A + \cos A| \leq 1 + t$ and $\frac{1+t}{1+t^2} \leq 2$), we conclude

$$\lim_{A\to\infty}\int_0^A \frac{\sin x}{x}\,dx = \int_0^\infty \frac{dt}{1+t^2} = \frac{\pi}{2}.$$

4.3 Convolution of Integrable Functions

Motivation. Convolution $f * g(x) = \int f(x-y)g(y)\,dy$ averages f with weights given by g. It smooths functions, turns into multiplication under the Fourier transform, and, with approximate identities, approximates any L^p function by nice ones. Fubini's theorem is what makes it well defined.

Among the Banach spaces $L^p(X,\mu)$ $(1 \leq p \leq \infty)$, $L^1(X,\mu)$ plays a special role: if X carries an operation and μ has a property like Fact B.1 (translation invariance), one can define a product on $L^1(X,\mu)$. Here we treat only Lebesgue integration on the real line.

4.3.1 Definition

For measurable $f, g : \mathbb{R} \to \mathbb{C}$ and $x \in \mathbb{R}$ consider

$$\int_{\mathbb{R}} f(y)g(x-y)\,dy.$$

By Proposition 4.2.5 we may assume f, g are Borel functions without changing this integral. The maps $(x,y) \mapsto y$ and $(x,y) \mapsto x-y$ are continuous, so by Proposition 4.2.4 the functions $(x,y) \mapsto f(y)$ and $(x,y) \mapsto g(x-y)$ are Borel, and by Fact B.26(a) their product

$$(x,y) \mapsto f(y)g(x-y) \tag{10}$$

is also a Borel function. (When f, g are merely measurable, (10) is still a measurable function of two variables.) Now

$$\int_{\mathbb{R}} \Big[\int_{\mathbb{R}} |f(y)g(x-y)|\,dx\Big]dy = \int_{\mathbb{R}} |f(y)|\Big[\int_{\mathbb{R}} |g(x-y)|\,dx\Big]dy = \int_{\mathbb{R}} |f(y)|\,dy \int_{\mathbb{R}} |g(x)|\,dx = \|f\|_1\|g\|_1,$$

using translation invariance (Fact B.1). By Fubini, the function

$$x \mapsto \int_{\mathbb{R}} f(y)g(x-y)\,dy \tag{11}$$

is defined almost everywhere, and its L^1-norm satisfies

$$\int_{\mathbb{R}} \Big|\int_{\mathbb{R}} f(y)g(x-y)\,dy\Big|dx \le \int_{\mathbb{R}} \Big[\int_{\mathbb{R}} |f(y)g(x-y)|\,dx\Big]dy = \|f\|_1\|g\|_1.$$

Definition (convolution)

For $f, g \in L^1(\mathbb{R})$ the L^1-function (11) is denoted $f * g$ and called the **convolution** of f and g. We have just proved

$$\|f * g\|_1 \le \|f\|_1\|g\|_1, \qquad f, g \in L^1(\mathbb{R}). \tag{12}$$

Proposition 4.3.1

For $f, g, h \in L^1(\mathbb{R})$ and constants $\alpha, \beta \in \mathbb{C}$:

(a) $[(f * g) * h](x) = [f * (g * h)](x)$ for almost every x;

(b) $(f * g)(x) = (g * f)(x)$ for every x (for which either side is defined);

(c) $[(\alpha f + \beta g) * h](x) = [\alpha(f * h) + \beta(g * h)](x)$ for almost every x.

Proof. (a) Fix $x \in \mathbb{R}$. Then

$$[(f * g) * h](x) = \int_{\mathbb{R}} (f * g)(y)h(x-y)\,dy = \int_{\mathbb{R}} \Big[\int_{\mathbb{R}} f(z)g(y-z)\,dz\Big]h(x-y)\,dy,$$

and

$$\begin{aligned}[f * (g * h)](x) &= \int_{\mathbb{R}} f(z)(g * h)(x-z)\,dz = \int_{\mathbb{R}} f(z)\Big[\int_{\mathbb{R}} g(y)h(x-z-y)\,dy\Big]dz \\ &= \int_{\mathbb{R}} f(z)\Big[\int_{\mathbb{R}} g(y-z)h(x-y)\,dy\Big]dz\end{aligned}$$

(substituting $y \mapsto y - z$). Now

$$\int_{\mathbb{R}}\int_{\mathbb{R}} |f(z)g(y-z)h(x-y)|\,dy\,dz = [(|f| * |g|) * |h|](x),$$

and since $|f|*|g|$ and $|h|$ are L^1-functions, this value is finite for almost every x. Fix such an x and apply Fubini to $(y,z)\mapsto f(z)g(y-z)h(x-y)$; this proves (a).

(b) Directly,

$$(f*g)(x)=\int_{\mathbb{R}} f(y)g(x-y)\,dy=\int_{\mathbb{R}} f(-y)g(x+y)\,dy=\int_{\mathbb{R}} f(x-y)g(y)\,dy=(g*f)(x),$$

where the second equality uses $m(E)=m(-E)$ (reflection invariance) and the third uses translation invariance.

(c) For almost every x the functions $y\mapsto f(y)h(x-y)$ and $y\mapsto g(y)h(x-y)$ are L^1-functions, so Fact B.10 (linearity of the integral) gives (c). ∎

Definition (Banach algebra)

A Banach space X with a multiplication $(x,y)\mapsto xy$ that is associative, distributive and satisfies

$$\|xy\|\le\|x\|\|y\|,\qquad x,y\in X,$$

is called a **Banach algebra**. By (12) and Proposition 4.3.1(a),(c), $L^1(\mathbb{R})$ is a Banach algebra under convolution.

Proposition 4.3.2

If g is differentiable and $f,g,g'\in L^1(\mathbb{R})$, then $f*g$ is differentiable and

$$(f*g)'=f*g'.$$

Exercise 4.3.1

Prove Proposition 4.3.2.

Solution [Supplement]. *Step 1: g is bounded and $g(b)-g(a)=\int_a^b g'$.* A function differentiable everywhere with integrable derivative satisfies the fundamental theorem of calculus (see Ch. 5), so $g(x)=g(0)+\int_0^x g'$. Since $g'\in L^1$, the limits $g(\pm\infty)$ exist, and since $g\in L^1$ they are 0; hence g is bounded and $(f*g)(x)$ is defined for *every* x.

Step 2: an integral formula. For $x\in\mathbb{R}$, $h\neq 0$,

$$(f*g)(x+h)-(f*g)(x)=\int_{\mathbb{R}} f(y)\int_0^h g'(x-y+t)\,dt\,dy=\int_0^h (f*g')(x+t)\,dt,$$

by Fubini, which applies because $\int_{\mathbb{R}}\int_0^h|f(y)g'(x-y+t)|\,dt\,dy\le|h|\,\|f\|_1\|g'\|_1<\infty$.

Step 3. Hence $\dfrac{(f*g)(x+h)-(f*g)(x)}{h}$ is the average of $f*g'$ over the interval between x and $x+h$. Whenever $f*g'$ is continuous at x (for instance at every x if g' is bounded, since then $f*g'$ is continuous by Exercise 4.3.2), this tends to $(f*g')(x)$, which gives $(f*g)'=f*g'$. In general Step 2 shows $f*g$ is absolutely continuous with $(f*g)'=f*g'$ almost everywhere (Lebesgue differentiation, Ch. 5).

Alternative under bounded g'. If $|g'|\le M$, the mean value theorem gives $\left|f(y)\frac{g(x+h-y)-g(x-y)}{h}\right|\le M|f(y)|$, an integrable bound, and the difference quotient converges pointwise to $f(y)g'(x-y)$; the dominated convergence theorem gives $(f*g)'(x)=(f*g')(x)$ for every x. □

4.3.2 Convolution of L^p and L^q functions

Let $p, q > 1$ be conjugate exponents. For a function g and $s \in \mathbb{R}$ write $g_s(t) = g(t-s)$. If $f \in L^p(\mathbb{R})$ and $g \in L^q(\mathbb{R})$, Hölder's inequality gives

$$\int_{\mathbb{R}} |f(y)g(x-y)|\,dy = \int_{\mathbb{R}} |f(y)g_{-x}(-y)|\,dy \le \|f\|_p \|g_{-x}\|_q = \|f\|_p \|g\|_q.$$

Hence $(f * g)(x)$ is defined for every $x \in \mathbb{R}$ and $\|f * g\|_\infty \le \|f\|_p \|g\|_q$.

Exercise 4.3.2

If $f \in L^1(\mathbb{R})$ and $g \in L^\infty(\mathbb{R})$, show that $f * g$ is uniformly continuous.

Solution [Supplement]. Using $f * g = g * f$, for $x, y \in \mathbb{R}$,

$$\begin{aligned} |(f*g)(x) - (f*g)(y)| &= \Big| \int_{\mathbb{R}} g(z)(f(x-z) - f(y-z))dz \Big| \\ &\le \|g\|_\infty \int_{\mathbb{R}} |f(x-z) - f(y-z)|\,dz = \|g\|_\infty \|f_{x-y} - f\|_1, \end{aligned}$$

after the substitution $w = y - z$. By the continuity of translation in L^1 (Fact B.23), $\|f_s - f\|_1 \to 0$ as $s \to 0$. Hence for $\epsilon > 0$ there is $\delta > 0$ with $|(f*g)(x) - (f*g)(y)| < \epsilon$ whenever $|x - y| < \delta$. □

Proposition 4.3.3

Let $p, q \in (1, \infty)$ be conjugate exponents. Then $f * g \in C_0(\mathbb{R})$ for every $f \in L^p(\mathbb{R})$ and $g \in L^q(\mathbb{R})$.

Proof. By Hölder's inequality again, for $x, y \in \mathbb{R}$

$$|(f*g)(x) - (f*g)(y)| \le \int_{\mathbb{R}} |f(z)(g_{-x}(-z) - g_{-y}(-z))|\,dz \le \|f\|_p \|g_{-x} - g_{-y}\|_q,$$

so by Fact B.23 (continuity of translation in L^q, $q < \infty$), $f * g$ is continuous. Next, by Fact B.22 there are sequences $\langle f_n \rangle, \langle g_n \rangle$ in $C_c(\mathbb{R})$ with

$$\lim_{n\to\infty} \|f_n - f\|_p = 0, \qquad \lim_{n\to\infty} \|g_n - g\|_q = 0.$$

If the supports of f_n and g_n lie in $[-A, A]$ and $[-B, B]$, then the support of $f_n * g_n$ lies in $[-A - B, A + B]$, so $\langle f_n * g_n \rangle$ is a sequence in $C_c(\mathbb{R})$. Then

$$\begin{aligned} \|f_n * g_n - f * g\|_\infty &\le \|f_n * g_n - f_n * g\|_\infty + \|f_n * g - f * g\|_\infty \\ &\le \|f_n\|_p \|g_n - g\|_q + \|f_n - f\|_p \|g\|_q \to 0, \end{aligned}$$

so $\langle f_n * g_n \rangle$ converges uniformly to $f * g$. A uniform limit of functions in $C_c(\mathbb{R})$ belongs to $C_0(\mathbb{R})$. ■

4.3.3 Averaging kernels

It is usually hard to guess what $f * g$ looks like, but sometimes one can. Consider

$$k_r = \frac{1}{2r} \chi_{[-r,r]}, \qquad r > 0. \tag{13}$$

By definition

$$(k_r * f)(x) = \int_{-\infty}^{\infty} k_r(x-y) f(y)\,dy = \frac{1}{2r} \int_{x-r}^{x+r} f(y)\,dy,$$

the average of f over the neighborhood $[x-r, x+r]$ of x. So $k_r * f$ should be better behaved than f and approach f as $r \to 0$.

$k_r * f$ is uniformly continuous for $f \in L^1(\mathbb{R})$. By Fact B.15 (absolute continuity of the integral), for any $\epsilon > 0$ there is $\delta \in (0, 2r)$ with

$$E \in \mathcal{M},\ m(E) \le \delta \implies \int_E |f(t)|\,dt < 2r\epsilon.$$

If $|x-y| < \frac{\delta}{2} < r$, let $E = [x-r, x+r] \triangle [y-r, y+r]$, where $A \triangle B = (A \cup B) \setminus (A \cap B)$; then $m(E) = 2|x-y| \le \delta$. Therefore

$$|(k_r * f)(x) - (k_r * f)(y)| = \frac{1}{2r}\Big| \int_{x-r}^{x+r} f(t)\,dt - \int_{y-r}^{y+r} f(t)\,dt \Big| \le \frac{1}{2r}\int_E |f(t)|\,dt < \epsilon,$$

so $f * k_r$ is uniformly continuous on $\mathbb{R}$.

Uniform approximation. If f is uniformly continuous, for each $\epsilon > 0$ there is $\delta > 0$ with $|x-y| < \delta \Rightarrow |f(x) - f(y)| < \epsilon$. If $r < \delta$, then

$$|f(x) - (f * k_r)(x)| \le \frac{1}{2r}\int_{x-r}^{x+r} |f(x) - f(t)|\,dt < \epsilon,$$

so $f * k_r \to f$ uniformly as $r \to 0$.

Exercise 4.3.3

For $f = \chi_{[-1,1]}$ sketch $x \mapsto (f * k_r)(x)$ for $r = 1, \frac{1}{2}, \frac{1}{3}, \frac{1}{5}, \frac{1}{10}$. Do the same for $f(x) = \max\{1 - |x|, 0\}$.

Solution [Supplement]. *(i)* $f = \chi_{[-1,1]}$. $(f * k_r)(x) = \frac{1}{2r}\, m([x-r, x+r] \cap [-1, 1])$, an even function. For $r \le 1$:

$$(f * k_r)(x) = \begin{cases} 1, & |x| \le 1 - r, \\ \dfrac{1 + r - |x|}{2r}, & 1 - r \le |x| \le 1 + r, \\ 0, & |x| \ge 1 + r. \end{cases}$$

This is a trapezoid: flat top of height 1 on $[-(1-r), 1-r]$ and linear ramps of width $2r$ centered at ± 1, passing through the value $\frac{1}{2}$ at $x = \pm 1$. For $r = 1$ it is the triangle $(2 - |x|)/2$ on $[-2, 2]$; as $r = \frac{1}{2}, \frac{1}{3}, \frac{1}{5}, \frac{1}{10}$ the ramps steepen and the graph approaches the step $\chi_{[-1,1]}$. The convergence is pointwise (to $\frac{1}{2}$ at ± 1), not uniform: f is not continuous.

(ii) $f(x) = \max\{1 - |x|, 0\}$ *(tent).* For $r \le \frac{1}{2}$ and $x \ge 0$ (the function is even), a direct integration gives

$$(f * k_r)(x) = \begin{cases} 1 - \dfrac{x^2 + r^2}{2r}, & 0 \le x \le r, \\ 1 - x, & r \le x \le 1 - r, \\ \dfrac{(1 + r - x)^2}{4r}, & 1 - r \le x \le 1 + r, \\ 0, & x \ge 1 + r. \end{cases}$$

The tent's corners at 0 and ± 1 are replaced by parabolic arcs of width $2r$; the result is C^1, its maximum is $1 - \frac{r}{2}$, and $\|f * k_r - f\|_\infty = \frac{r}{2} \to 0$ (uniform convergence, since f is uniformly continuous). For $r = 1$: $(f * k_1)(x) = \frac{1}{2} - \frac{x^2}{4}$ for $|x| \le 1$ and $\frac{(2-|x|)^2}{4}$ for $1 \le |x| \le 2$, a flatter bell with peak $\frac{1}{2}$. $\square$

Exercise 4.3.4

Let $f_0 = \chi_{[-1,1]}$ and $f_{n+1} = f_n * f_0$ $(n = 0, 1, 2, \dots)$. Sketch how the graphs of this sequence change.

Solution [Supplement]. $f_1(x) = m([x-1, x+1] \cap [-1, 1]) = \max\{2 - |x|, 0\}$, a triangle of height 2 on $[-2, 2]$. Inductively, since $(f_n * f_0)(x) = \int_{x-1}^{x+1} f_n$:

- f_n is even, nonnegative, supported in $[-(n+1), n+1]$, increasing on $(-(n+1), 0]$;
- f_n is a piecewise polynomial of degree n with breakpoints at the integers of the same parity as $n+1$, and $f_n \in C^{n-1}$ (each convolution with f_0 gains one derivative: $f'_{n+1}(x) = f_n(x+1) - f_n(x-1)$);
- $\int f_n = \left(\int f_0\right)^{n+1} = 2^{n+1}$ by Fubini.

Thus the graphs go step $\to$ triangle $\to$ piecewise quadratic bell $\to \cdots$, becoming smoother and wider. After normalizing, $f_n/2^{n+1}$ is the density of a sum of $n+1$ independent uniform variables on $[-1, 1]$, which has variance $(n+1)/3$; by the central limit theorem the graphs look increasingly like the Gaussian curve $\frac{2^{n+1}}{\sqrt{2\pi(n+1)/3}} e^{-3x^2/(2(n+1))}$. □

Chapter 5. Fourier Transforms

Overview

Results from Chapters 1–3 are cited as *Fact B.k*; they are stated in the separate *Background* notes. Fourier's idea of representing an arbitrary periodic function by a trigonometric series not only contributed greatly to solving differential equations but stimulated all of analysis, including integration theory. This chapter studies Fourier series of periodic functions and Fourier transforms of integrable functions; Fubini's theorem and convolution are the key tools.

5.1 Fourier Series

Motivation. A periodic function should be the sum of its harmonics e^{inx}. Whether and in what sense this holds is the central question; approximate identities (Fejér's kernel) give a clean positive answer for Cesàro means and show that Fourier coefficients determine the function.

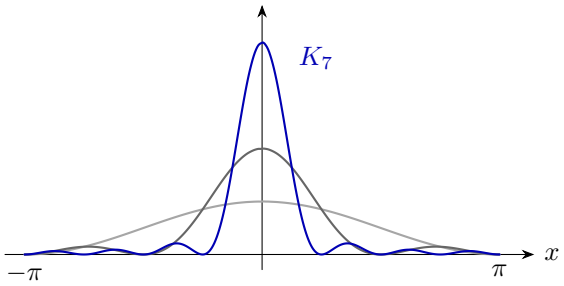


Figure. *Fejér kernels K_N ($N = 1, 3, 7$) are nonnegative, have integral 1 and concentrate at 0 as N grows. Such an approximate identity makes $K_N * f \to f$, which is why Cesàro means of Fourier series converge.*

5.1.1 The circle $\mathbb{T}$

Let $\mathbb{T}$ be the quotient of $\mathbb{R}$ by the subgroup $\{2\pi n : n = 0, \pm 1, \pm 2, \dots\}$, with the distance inherited from the real line. A 2π-periodic function can be regarded as a function on $\mathbb{T}$ and vice versa. In particular, a continuous function on $\mathbb{T}$ is a 2π-periodic continuous function, i.e. a continuous function on $[-\pi, \pi]$ with $f(-\pi) = f(\pi)$. Since $\mathbb{T} = (-\pi, \pi]$ we use Lebesgue measure, normalized to total mass 1:

$$\int_{\mathbb{T}} f = \frac{1}{2\pi} \int_{-\pi}^{\pi} f(x)\, dx.$$

The convolution of §4.3 is defined on $L^1(\mathbb{T})$ in the same way, making $L^1(\mathbb{T})$ a Banach algebra: for $f, g : \mathbb{T} \to \mathbb{C}$, $t \in \mathbb{T}$,

$$(f * g)(t) = \int_{\mathbb{T}} f(s) g(t - s)\, ds.$$

If $f, g \in L^1(\mathbb{T})$, then as in §4.3, $(f * g)(t)$ is defined for almost every t, $f * g \in L^1(\mathbb{T})$,

$$\|f * g\|_1 \le \|f\|_1 \|g\|_1, \qquad f, g \in L^1(\mathbb{T}),$$

and the identities of Proposition 4.3.1 hold.

5.1.2 Fourier coefficients

Definition (Fourier coefficients, Fourier series)

For $f \in L^1(\mathbb{T})$ and $n = 0, \pm 1, \pm 2, \dots$ define

$$\widehat{f}(n) = \int_{\mathbb{T}} f(t) e^{-int}\, dt,$$

so that $|\widehat{f}(n)| \le \|f\|_1$. With

$$u_n(t) = e^{int}, \qquad t \in \mathbb{T}, \tag{1}$$

one checks directly that

$$u_n * f = \widehat{f}(n) u_n, \qquad n \in \mathbb{Z},\ f \in L^1(\mathbb{T}); \tag{2}$$

in particular $(u_n * f)(0) = \widehat{f}(n)$. Also $\widehat{u_n}(m) = 1$ if $m = n$ and 0 if $m \neq n$. The series

$$\sum_{n=-\infty}^{\infty} \widehat{f}(n) u_n \tag{3}$$

is the **Fourier series** of $f \in L^1(\mathbb{T})$ and $\{\widehat{f}(n) : n \in \mathbb{Z}\}$ are its **Fourier coefficients**. Without suitable conditions on the coefficients, the series (3) need not make sense as a function.

Indeed (2): $(u_n * f)(t) = \int_{\mathbb{T}} f(s) e^{in(t-s)}\, ds = e^{int} \widehat{f}(n)$.

Exercise 5.1.1

If $\widehat{f} \in \ell^1(\mathbb{Z})$, show that the series (3) converges uniformly to a continuous function.

Solution [Supplement]. $|\widehat{f}(n) u_n(t)| = |\widehat{f}(n)|$ for all t, and $\sum_n |\widehat{f}(n)| < \infty$. By the Weierstrass M-test the partial sums $\sum_{|n| \le N} \widehat{f}(n) u_n$ (or any partial sums $\sum_{-M}^{N}$) form a uniformly Cauchy sequence of continuous functions on $\mathbb{T}$; their uniform limit is continuous. □

Proposition 5.1.1

For $f, g \in L^1(\mathbb{T})$, constants $a, b \in \mathbb{C}$ and $n \in \mathbb{Z}$,

$$\widehat{af + bg}(n) = a\widehat{f}(n) + b\widehat{g}(n), \qquad \widehat{f * g}(n) = \widehat{f}(n)\, \widehat{g}(n). \tag{4}$$

Moreover for each $t \in \mathbb{T}$

$$\widehat{f_t}(n) = \widehat{f}(n) u_n(-t), \qquad n \in \mathbb{Z}, \tag{5}$$

where $f_t(s) = f(s - t)$.

Proof. The first identity follows from Fact B.10 (linearity of the integral). For the second, by Fubini,

$$\begin{aligned} \widehat{f * g}(n) &= \int_{\mathbb{T}} (f * g)(t) e^{-int}\, dt = \int_{\mathbb{T}} e^{-int} \int_{\mathbb{T}} f(t - s) g(s)\, ds\, dt \\ &= \int_{\mathbb{T}} g(s) e^{-ins} \int_{\mathbb{T}} f(t - s) e^{-in(t-s)}\, dt\, ds = \int_{\mathbb{T}} g(s) e^{-ins}\, ds \int_{\mathbb{T}} f(t) e^{-int}\, dt = \widehat{g}(n) \widehat{f}(n). \end{aligned}$$

For (5),

$$\widehat{f_t}(n) = \int_{\mathbb{T}} f(s - t) e^{-ins}\, ds = \int_{\mathbb{T}} f(s) e^{-in(s+t)}\, ds = \widehat{f}(n) u_n(-t).$$

■

Proposition 5.1.2

If f is a 2π-periodic differentiable function with $f' \in L^1(\mathbb{T})$, then

$$\widehat{f'}(n) = in\widehat{f}(n), \qquad n \in \mathbb{Z}.$$

Proof. Integrating by parts (the boundary terms cancel by periodicity),

$$\widehat{f'}(n) = \int_{\mathbb{T}} f'(t)e^{-int}\,dt = in\int_{\mathbb{T}} f(t)e^{-int}\,dt = in\widehat{f}(n).$$

■

The next theorem is a variant of Fact B.21 and can be proved with part of the method of Fact B.22.

Theorem 5.1.3

For each $p \in [1,\infty)$, the space $C(\mathbb{T})$ of continuous functions is dense in $L^p(\mathbb{T})$.

For a function f on $\mathbb{T}$ and $t \in \mathbb{T}$ define the translate as in Fact B.19:

$$f_t(s) = f(s-t), \qquad s \in \mathbb{T}.$$

The next theorem is proved like Fact B.23, using Theorem 5.1.3 instead of Fact B.22; the computation is even simpler.

Theorem 5.1.4

For each $f \in L^p(\mathbb{T})$ $(1 \le p < \infty)$, the map

$$t \mapsto f_t : \mathbb{T} \to L^p(\mathbb{T})$$

is uniformly continuous.

Exercise 5.1.2

Prove Theorems 5.1.3 and 5.1.4.

Solution [Supplement]. *Theorem 5.1.3.* Simple functions $\sum_k c_k\chi_{E_k}$ with E_k measurable are dense in $L^p(\mathbb{T})$ (as in Fact B.21), so it suffices to approximate χ_E for measurable $E \subset (-\pi,\pi)$ (the point π is a null set). Given $\epsilon > 0$, by regularity of Lebesgue measure choose a compact K and an open U with $K \subset E \subset U \subset (-\pi,\pi)$ and $m(U \setminus K) < \epsilon$. Let

$$\varphi(x) = \frac{d(x,U^c)}{d(x,U^c) + d(x,K)},$$

which is continuous on $\mathbb{R}$, $0 \le \varphi \le 1$, $\varphi = 1$ on K and $\varphi = 0$ off U. Since $U \subset (-\pi,\pi)$ is open and $K \subset U$ compact, φ vanishes near $\pm\pi$ and extends to a continuous 2π-periodic function, i.e. $\varphi \in C(\mathbb{T})$. As $|\chi_E - \varphi| \le \chi_{U\setminus K}$,

$$\|\chi_E - \varphi\|_p^p \le \frac{m(U\setminus K)}{2\pi} < \epsilon.$$

Taking linear combinations proves density.

Theorem 5.1.4. Let $f \in L^p(\mathbb{T})$ and $\epsilon > 0$; choose $g \in C(\mathbb{T})$ with $\|f-g\|_p < \epsilon$. Since $\mathbb{T}$ is compact, g is uniformly continuous: there is $\delta > 0$ with $|g(u)-g(v)| < \epsilon$ whenever $d(u,v) < \delta$. Lebesgue measure on $\mathbb{T}$ is translation invariant, so $\|f_t - g_t\|_p = \|f-g\|_p$. If $d(t,s) < \delta$, then $d(u-t,u-s) < \delta$ for every u, hence $\|g_t - g_s\|_p \le \|g_t - g_s\|_\infty \le \epsilon$ (total mass 1), and

$$\|f_t - f_s\|_p \le \|f_t - g_t\|_p + \|g_t - g_s\|_p + \|g_s - f_s\|_p < 3\epsilon.$$

□

5.1.3 Approximate identities

We are now ready for the theorem that plays the central role in the study of Fourier series. Its equation (7) says that the sequence $\langle h_n \rangle$ acts like an identity in the Banach algebra $L^1(\mathbb{T})$.

Theorem 5.1.5

If a sequence $\langle h_n \rangle$ satisfies

$$h_n \geq 0, \qquad \int_{\mathbb{T}} h_n = 1, \qquad \lim_{n\to\infty} \int_{\mathbb{T}\setminus[-\delta,\delta]} h_n = 0 \quad (0 < \delta < \pi), \tag{6}$$

(the third condition meaning: for *every* $\delta \in (0,\pi)$ the limit is 0), then for every $f \in L^1(\mathbb{T})$

$$\lim_{n\to\infty} \|f - h_n * f\|_1 = 0. \tag{7}$$

Proof. Let $\epsilon > 0$. By Theorem 5.1.4 there is $\delta > 0$ with

$$|s| < \delta \implies \|f - f_s\|_1 < \epsilon,$$

and then N with

$$n \geq N \implies \int_{\mathbb{T}\setminus[-\delta,\delta]} h_n(s)\, ds < \frac{\epsilon}{2\|f\|_1}.$$

For every $t \in \mathbb{T}$,

$$f(t) - (h_n * f)(t) = f(t)\int_{\mathbb{T}} h_n(s)\, ds - \int_{\mathbb{T}} h_n(s) f(t-s)\, ds = \int_{\mathbb{T}} h_n(s)\big[f(t) - f_s(t)\big] ds.$$

Hence for $n \geq N$, using Tonelli,

$$\begin{aligned}
\|f - h_n * f\|_1 &\leq \int_{\mathbb{T}}\int_{\mathbb{T}} h_n(s)|f(t) - f_s(t)|\, ds\, dt = \int_{\mathbb{T}} h_n(s) \int_{\mathbb{T}} |f(t) - f_s(t)|\, dt\, ds \\
&= \int_{\mathbb{T}} h_n(s)\|f - f_s\|_1\, ds \\
&= \int_{[-\delta,\delta]} h_n(s)\|f - f_s\|_1\, ds + \int_{\mathbb{T}\setminus[-\delta,\delta]} h_n(s)\|f - f_s\|_1\, ds \\
&\leq \epsilon \int_{[-\delta,\delta]} h_n(s)\, ds + 2\|f\|_1 \int_{\mathbb{T}\setminus[-\delta,\delta]} h_n(s)\, ds \leq \epsilon + \epsilon = 2\epsilon.
\end{aligned}$$

■

The method of proof of Theorem 5.1.5 is very typical and applies in many situations. It applies verbatim to the next theorem, where the computation is even simpler and Tonelli–Fubini is not needed.

Theorem 5.1.6

If $\langle h_n \rangle$ satisfies (6), then $\lim_{n\to\infty}\|f - h_n * f\|_\infty = 0$ for every $f \in C(\mathbb{T})$.

Proof. Since f is uniformly continuous, choose $\delta > 0$ with

$$|s| < \delta \implies \|f - f_s\|_\infty < \epsilon,$$

and N with

$$n \geq N \implies \int_{\mathbb{T}\setminus[-\delta,\delta]} h_n(s)\, ds < \frac{\epsilon}{2\|f\|_\infty}.$$

Then for every $t \in \mathbb{T}$ and $n \geq N$,

$$|f(t) - (h_n * f)(t)| \leq \int_{\mathbb{T}} h_n(s)|f(t) - f_s(t)|\, ds = \Big(\int_{[-\delta,\delta]} + \int_{\mathbb{T}\setminus[-\delta,\delta]}\Big) h_n(s)|f(t) - f_s(t)|\, ds$$

$$\le \epsilon \int_{[-\delta,\delta]} h_n(s)\,ds + 2\|f\|_\infty \int_{\mathbb{T}\setminus[-\delta,\delta]} h_n(s)\,ds \le \epsilon + \epsilon = 2\epsilon.$$

■

Exercise 5.1.3

In the proof of Theorem 5.1.6, show that there is $\delta > 0$ with $|s| < \delta \Rightarrow \|f - f_s\|_\infty < \epsilon$.

Solution [Supplement]. A continuous function on the compact metric space $\mathbb{T}$ is uniformly continuous: there is $\delta > 0$ with $|f(u) - f(v)| < \epsilon/2$ whenever $d(u,v) < \delta$. If $|s| < \delta$, then $d(t, t-s) \le |s| < \delta$ for every t, so $|f(t) - f_s(t)| = |f(t) - f(t-s)| < \epsilon/2$ and $\|f - f_s\|_\infty \le \epsilon/2 < \epsilon$. □

Exercise 5.1.4

Use Theorem 5.1.6 to prove Theorem 5.1.5.

Solution [Supplement]. Let $f \in L^1(\mathbb{T})$, $\epsilon > 0$, and choose $g \in C(\mathbb{T})$ with $\|f - g\|_1 < \epsilon$ (Theorem 5.1.3). Since $\mathbb{T}$ has total mass 1, $\|\cdot\|_1 \le \|\cdot\|_\infty$, and $\|h_n\|_1 = 1$, so

$$\begin{aligned}\|f - h_n * f\|_1 &\le \|f - g\|_1 + \|g - h_n * g\|_1 + \|h_n * (g - f)\|_1 \\ &\le \epsilon + \|g - h_n * g\|_\infty + \|h_n\|_1 \|g - f\|_1 < 2\epsilon + \|g - h_n * g\|_\infty.\end{aligned}$$

By Theorem 5.1.6 the last term tends to 0, so $\limsup_n \|f - h_n * f\|_1 \le 2\epsilon$. □

5.1.4 The Fejér kernel

Sequences satisfying (6) are easy to find, e.g. $\langle n\pi\chi_{[-1/n,1/n]} : n = 1, 2, \dots \rangle$. A more important example: for $n = 1, 2, \dots$ put

$$K_n = \sum_{k=-n}^{n} \Big(1 - \frac{|k|}{n+1}\Big) u_k. \tag{8}$$

$\langle K_n \rangle$ is the **Fejér kernel**.

Exercise 5.1.5

Show that $\lim_{n\to\infty} \widehat{K_n}(k) = 1$ for every $k \in \mathbb{Z}$.

Solution [Supplement]. Since $\widehat{u_j}(k) = \delta_{jk}$, $\widehat{K_n}(k) = 1 - \frac{|k|}{n+1}$ for $|k| \le n$ and 0 for $|k| > n$. For fixed k and $n \ge |k|$ this is $1 - |k|/(n+1) \to 1$. □

Since $\int_{\mathbb{T}} u_n = 0$ for $n \in \mathbb{Z} \setminus \{0\}$, the second condition $\int_{\mathbb{T}} K_n = 1$ is immediate. Expanding the left side shows

$$\Big(-\frac{1}{4}u_{-1} + \frac{1}{2} - \frac{1}{4}u_1\Big) K_n(t) = \frac{1}{n+1}\Big(-\frac{1}{4}u_{-(n+1)} + \frac{1}{2} - \frac{1}{4}u_{n+1}\Big),$$

from which

$$K_n(t) = \frac{1}{n+1} \frac{\sin^2\!\big(\frac{n+1}{2}t\big)}{\sin^2\!\big(\frac{1}{2}t\big)}, \qquad n = 0, 1, 2, \dots \tag{9}$$

Exercise 5.1.6

Prove (9).

Solution [Supplement]. For $0 \le j, l \le n$, the number of pairs with $j - l = k$ is $n + 1 - |k|$ ($|k| \le n$). Hence

$$\Big|\sum_{j=0}^{n} e^{ijt}\Big|^2 = \sum_{j,l=0}^{n} e^{i(j-l)t} = \sum_{k=-n}^{n} (n + 1 - |k|) e^{ikt} = (n+1) K_n(t).$$

For $t \neq 0$,

$$\sum_{j=0}^{n} e^{ijt} = \frac{e^{i(n+1)t} - 1}{e^{it} - 1} = \frac{e^{i(n+1)t/2}}{e^{it/2}} \cdot \frac{\sin \frac{(n+1)t}{2}}{\sin \frac{t}{2}},$$

whose squared modulus is $\sin^2 \frac{(n+1)t}{2} / \sin^2 \frac{t}{2}$. This is (9). (Equivalently, $-\frac{1}{4}u_{-1} + \frac{1}{2} - \frac{1}{4}u_1 = \frac{1}{2}(1 - \cos t) = \sin^2 \frac{t}{2}$ and similarly on the right, which is the displayed identity.) □

Since $t \mapsto \sin^2(\frac{1}{2}t)$ is increasing on $[0, \pi]$ and K_n is even, for every $\delta > 0$

$$0 \leq K_n(t) \leq \frac{1}{n+1} \frac{1}{\sin^2(\frac{1}{2}\delta)}, \qquad t \in \mathbb{T} \setminus [-\delta, \delta],$$

so $\langle K_n \rangle$ satisfies all of (6).

Exercise 5.1.7

Sketch the graph of K_n for $n = 1, 2, 3, 5, 10$.

Solution [Supplement]. $K_1(t) = 1 + \cos t$, $K_2(t) = 1 + \frac{4}{3}\cos t + \frac{2}{3}\cos 2t$, and in general K_n is even, nonnegative, with maximum $K_n(0) = n + 1$ and zeros at $t = \frac{2\pi j}{n+1}$, $j = 1, \ldots, n$. Between the zeros there are small humps of height at most $\frac{1}{(n+1)\sin^2(t/2)}$. Each K_n has average 1 over $\mathbb{T}$. As $n = 1, 2, 3, 5, 10$ the central peak grows to height $2, 3, 4, 6, 11$ while its width $\frac{4\pi}{n+1}$ shrinks, and the graph away from 0 flattens toward 0: the mass concentrates at $t = 0$. □

5.1.5 Consequences

The key property of $\langle K_n \rangle$ is that each K_n is a finite linear combination of $\{u_n : n \in \mathbb{Z}\}$. Such periodic functions are called **trigonometric polynomials**; their vector space is $T(\mathbb{T})$. The Fourier coefficients of $P = \sum_{n=-N}^{N} a_n u_n$ are $\widehat{P}(n) = a_n$.

By (2), $K_n * f$ is a trigonometric polynomial for every $f \in L^1(\mathbb{T})$, and by Theorem 5.1.5, $K_n * f \to f$ in L^1. Hence:

Theorem 5.1.7

$T(\mathbb{T})$ is dense in $L^1(\mathbb{T})$.

If $\widehat{f}(n) = 0$ for all $n \in \mathbb{Z}$, then $K_n * f = 0$ by (2), so $\|f\|_1 = \lim \|f - K_n * f\|_1 = 0$:

Theorem 5.1.8 (uniqueness)

If $f \in L^1(\mathbb{T})$ and $\widehat{f} = 0$, then $f = 0$ almost everywhere.

If $a \in \ell^1(\mathbb{Z})$, then $\sum_{k\in\mathbb{Z}} |a_k u_k(t)| = \sum_k |a_k| < \infty$ for every t, so $g(t) = \sum_{k\in\mathbb{Z}} a_k u_k(t)$ makes sense, and by Fubini (for the counting measure, Exercise 4.2.3)

$$\widehat{g}(n) = \int_{\mathbb{T}} \Big(\sum_{k\in\mathbb{Z}} a_k u_{k-n} \Big) = \sum_{k\in\mathbb{Z}} a_k \int_{\mathbb{T}} u_{k-n} = a_n.$$

In particular, if $f \in L^1(\mathbb{T})$ and $\widehat{f} \in \ell^1(\mathbb{Z})$, then $g = \sum_k \widehat{f}(k) u_k$ and f have the same Fourier coefficients, so Theorem 5.1.8 gives:

Corollary 5.1.9

If $f \in L^1(\mathbb{T})$ and $\widehat{f} \in \ell^1(\mathbb{Z})$, then

$$f(t) = \sum_{n\in\mathbb{Z}} \widehat{f}(n) e^{int}$$

for almost every $t \in \mathbb{T}$.

On the other hand, given $\epsilon > 0$ take $P \in T(\mathbb{T})$ with $\|f - P\|_1 < \epsilon$ and let $N = \max\{|n| : \widehat{P}(n) \neq 0\}$. For $|n| > N$,

$$|\widehat{f}(n)| = |\widehat{f}(n) - \widehat{P}(n)| \leq \|f - P\|_1 < \epsilon.$$

This proves:

Theorem 5.1.10 (Riemann–Lebesgue)

$\lim_{n\to\pm\infty} \widehat{f}(n) = 0$ for every $f \in L^1(\mathbb{T})$.

Exercise 5.1.8

If a 2π-periodic function $f : \mathbb{R} \to \mathbb{C}$ is twice differentiable and $f'' \in L^1(\mathbb{T})$, show that $\widehat{f} \in \ell^1(\mathbb{Z})$.

Solution [Supplement]. f' is differentiable, hence continuous and so bounded on $\mathbb{T}$; thus $f' \in L^1(\mathbb{T})$. Applying Proposition 5.1.2 to f and to f',

$$\widehat{f''}(n) = in\,\widehat{f'}(n) = (in)^2 \widehat{f}(n) = -n^2 \widehat{f}(n).$$

Hence $|\widehat{f}(n)| = |\widehat{f''}(n)|/n^2 \leq \|f''\|_1/n^2$ for $n \neq 0$, and $\sum_n |\widehat{f}(n)| \leq |\widehat{f}(0)| + \|f''\|_1 \sum_{n\neq 0} n^{-2} < \infty$. □

Let $c_0(\mathbb{Z})$ be the vector space of functions $a : \mathbb{Z} \to \mathbb{C}$ with $\lim_{n\to\pm\infty} a(n) = 0$, with the norm of $\ell^\infty(\mathbb{Z})$.

Exercise 5.1.9

Show that the normed space $c_0(\mathbb{Z})$ is a Banach space.

Solution [Supplement]. $\ell^\infty(\mathbb{Z})$ is complete, so it suffices to show $c_0(\mathbb{Z})$ is closed in it. Let $a_k \in c_0(\mathbb{Z})$, $\|a_k - a\|_\infty \to 0$, and $\epsilon > 0$. Choose k with $\|a_k - a\|_\infty < \epsilon$, then N with $|a_k(n)| < \epsilon$ for $|n| > N$. For $|n| > N$, $|a(n)| \leq |a(n) - a_k(n)| + |a_k(n)| < 2\epsilon$. So $a \in c_0(\mathbb{Z})$. □

Summarizing:

Theorem 5.1.11

The map $f \mapsto \widehat{f}$ is a norm-decreasing, injective linear map from the Banach algebra $L^1(\mathbb{T})$ into $c_0(\mathbb{Z})$.

Proof. Linearity is (4); $\|\widehat{f}\|_\infty \leq \|f\|_1$ is the estimate $|\widehat{f}(n)| \leq \|f\|_1$; $\widehat{f} \in c_0(\mathbb{Z})$ is Theorem 5.1.10; injectivity is Theorem 5.1.8. ■

With the pointwise product $(ab)(n) = a(n)b(n)$, $c_0(\mathbb{Z})$ is a Banach algebra. The last identity of (4) says that $f \mapsto \widehat{f}$ preserves multiplication between $L^1(\mathbb{T})$ and $c_0(\mathbb{Z})$. A linear map between Banach algebras that preserves multiplication is a **homomorphism**.

Remark ($L^1(\mathbb{T})$ has no identity)

If the Banach algebra $L^1(\mathbb{T})$ had an identity e, then for every $n \in \mathbb{Z}$

$$\widehat{e}(n) = \widehat{e}(n)\widehat{u_n}(n) = \widehat{e * u_n}(n) = \widehat{u_n}(n) = 1,$$

contradicting $\widehat{e} \in c_0(\mathbb{Z})$. So $L^1(\mathbb{T})$ has no identity; Theorem 5.1.5 (an *approximate identity*) is the device that overcomes this.

Exercise 5.1.10

If $\langle h_n \rangle$ satisfies (6), show that $\|a\widehat{h_n} - a\|_\infty \to 0$ for every $a \in c_0(\mathbb{Z})$.

Solution [Supplement]. First, $|\widehat{h_n}(m)| \le \|h_n\|_1 = 1$. Next, for fixed m and $0 < \delta < \pi$,

$$|1 - \widehat{h_n}(m)| = \Big| \int_{\mathbb{T}} h_n(s)(1 - e^{-ims})\, ds \Big| \le \int_{[-\delta,\delta]} h_n(s)|ms|\, ds + 2\int_{\mathbb{T}\setminus[-\delta,\delta]} h_n \le |m|\delta + 2\int_{\mathbb{T}\setminus[-\delta,\delta]} h_n,$$

so $\limsup_n |1 - \widehat{h_n}(m)| \le |m|\delta$ for every δ, i.e. $\widehat{h_n}(m) \to 1$. Now let $\epsilon > 0$ and choose M with $|a(m)| < \epsilon$ for $|m| > M$. Then

$$\|a\widehat{h_n} - a\|_\infty \le \max_{|m|\le M} |a(m)||\widehat{h_n}(m) - 1| + \sup_{|m|>M} |a(m)| \cdot 2,$$

where the first term tends to 0 (finitely many m) and the second is at most 2ϵ. □

Finally, consider some consequences of Theorem 5.1.6. With the Fejér kernel, $K_n * f \in T(\mathbb{T})$ and $K_n * f \to f$ uniformly for $f \in C(\mathbb{T})$:

Corollary 5.1.12

$T(\mathbb{T})$ is dense in $C(\mathbb{T})$ (in the uniform norm).

If $f \in L^p(\mathbb{T})$ $(1 \le p < \infty)$, Theorem 5.1.3 gives $g \in C(\mathbb{T})$ with $\|f - g\|_p < \epsilon$, and Corollary 5.1.12 gives $P \in T(\mathbb{T})$ with $\|g - P\|_\infty < \epsilon$. Then

$$\|f - P\|_p \le \|f - g\|_p + \|g - P\|_p < \epsilon + \|g - P\|_\infty < 2\epsilon.$$

Corollary 5.1.13

$T(\mathbb{T})$ is dense in $L^p(\mathbb{T})$ for $1 \le p < \infty$.

Corollary 5.1.12 closely resembles the Weierstrass theorem that a continuous function on a bounded closed interval can be uniformly approximated by polynomials, and in fact implies it.

Theorem 5.1.14 (Weierstrass)

The space of polynomial functions on $[0, 1]$ is dense in $C[0, 1]$.

Proof. Extend a continuous $f : [0, 1] \to \mathbb{C}$ to a continuous periodic $g : \mathbb{T} \to \mathbb{C}$. By Corollary 5.1.12 choose $\{a_k \in \mathbb{C} : k = 0, \pm 1, \ldots, \pm n\}$ with

$$\Big| f(t) - \sum_{k=-n}^{n} a_k u_k(t) \Big| < \epsilon, \qquad t \in [0, 1].$$

Since the Taylor series of the exponential converges uniformly on $[0,1]$, for each k there is $M(k) \in \mathbb{N}$ with

$$\Big| \sum_{\ell=0}^{M(k)} \frac{(ikt)^\ell}{\ell!} - u_k(t) \Big| \le \frac{\epsilon}{(2n+1)|a_k|+1}, \qquad t \in [0,1].$$

Put

$$P(t) = \sum_{k=-n}^{n} a_k \sum_{\ell=0}^{M(k)} \frac{(ikt)^\ell}{\ell!}, \qquad t \in [0,1].$$

Then $|f(t) - P(t)| < 2\epsilon$ (Exercise 5.1.11). ■

Exercise 5.1.11

Complete the proof of Theorem 5.1.14.

Solution [Supplement]. *Extension.* Since $[0,1] \subset (-\pi,\pi)$, define g on $[-\pi,\pi]$ by $g = f$ on $[0,1]$, g linear from $f(1)$ at $t=1$ to 0 at $t=2$, $g=0$ on $[2,\pi]\cup[-\pi,-1]$, and g linear from 0 at $t=-1$ to $f(0)$ at $t=0$. Then g is continuous with $g(-\pi)=g(\pi)$, so $g \in C(\mathbb{T})$.

Estimate. P is a polynomial in t. For $t \in [0,1]$,

$$\begin{aligned} |f(t)-P(t)| &\le \Big|f(t) - \sum_{k=-n}^{n} a_k u_k(t)\Big| + \sum_{k=-n}^{n} |a_k| \Big|u_k(t) - \sum_{\ell=0}^{M(k)} \frac{(ikt)^\ell}{\ell!}\Big| \\ &< \epsilon + \sum_{k=-n}^{n} \frac{|a_k|\epsilon}{(2n+1)|a_k|+1} < \epsilon + (2n+1)\cdot\frac{\epsilon}{2n+1} = 2\epsilon. \end{aligned}$$

(If f is real-valued, $\operatorname{Re} P$ is a real polynomial with $|f - \operatorname{Re} P| \le |f-P|$.) □

5.2 The Fourier Integral

Motivation. For nonperiodic functions on $\mathbb{R}$ the sum over frequencies becomes an integral. We transfer the series theory to the transform: continuity and decay of $\hat f$, inversion via Gauss and Fejér kernels, and the rules turning differentiation into multiplication.

5.2.1 Definition and continuity

Definition (Fourier transform)

For $f \in L^1(\mathbb{R})$ and $\alpha \in \mathbb{R}$ define

$$\widehat{f}(\alpha) = \int_{\mathbb{R}} f(x) e^{-i\alpha x}\, dx.$$

$\widehat f$ is the **Fourier transform** of f.

Clearly $|\widehat f(\alpha)| \le \|f\|_1$ for all $\alpha \in \mathbb{R}$. If $\alpha_n \to \alpha$, then

$$|\widehat f(\alpha_n) - \widehat f(\alpha)| \le \int_{\mathbb{R}} |f(x)||e^{-i\alpha_n x} - e^{-i\alpha x}|\, dx,$$

and since $|e^{-i\alpha_n x} - e^{-i\alpha x}| \le 2$ and $e^{-i\alpha_n x} \to e^{-i\alpha x}$, the Lebesgue convergence theorem gives $\widehat f(\alpha_n) \to \widehat f(\alpha)$. So $\widehat f$ *is always a bounded continuous function.* Writing

$$u_\alpha(x) = e^{i\alpha x}, \qquad x \in \mathbb{R},$$

one checks immediately that

$$u_\alpha * f = \widehat{f}(\alpha) u_\alpha, \qquad \alpha \in \mathbb{R}.$$

Example 1

For $f(x) = e^{-|x|/a}$ $(a > 0)$,

$$\widehat{f}(\alpha) = \int_{\mathbb{R}} e^{-|x|/a} e^{-i\alpha x}\, dx = \int_{-\infty}^{0} e^{(1/a - i\alpha)x}\, dx + \int_{0}^{\infty} e^{(-1/a - i\alpha)x}\, dx = \frac{2a}{1 + a^2\alpha^2}.$$

For $g = \chi_{[a,b]}$,

$$\widehat{g}(\alpha) = \int_{-\infty}^{\infty} \chi_{[a,b]} e^{-i\alpha x}\, dx = \frac{e^{-i\alpha a} - e^{-i\alpha b}}{i\alpha},$$

and in particular

$$\widehat{\chi_{[-a,a]}}(\alpha) = \frac{2\sin \alpha a}{\alpha}, \qquad a > 0.$$

Exercise 5.2.1

In Example 1, describe how the graphs of f and $\widehat{f}$ change as $a \to \infty$. Do the same for $\widehat{\chi_{[-a,a]}}$.

Solution [Supplement]. As $a \to \infty$, $f(x) = e^{-|x|/a}$ spreads out and increases to the constant 1. Its transform $\widehat{f}(\alpha) = \frac{2a}{1+a^2\alpha^2}$ is a Cauchy-type bump with peak $2a$ at $\alpha = 0$ and half-width $1/a$; it becomes taller and narrower while keeping total integral $\int \widehat{f} = 2\pi f(0) = 2\pi$, i.e. it concentrates at $\alpha = 0$ (like $2\pi\delta_0$). Similarly $\widehat{\chi_{[-a,a]}}(\alpha) = \frac{2\sin a\alpha}{\alpha}$ has central peak $2a$, zeros at $\alpha = k\pi/a$ $(k \neq 0)$, and oscillations decaying like $2/|\alpha|$. As $a \to \infty$ the peak grows, the zeros crowd together and the oscillation becomes faster: the transform of a wide function is a narrow one. (Note $\frac{2\sin a\alpha}{\alpha} \notin L^1(\mathbb{R})$.) □

The basic properties of Proposition 5.1.1 hold for Fourier integrals as well.

Proposition 5.2.1

For $f, g \in L^1(\mathbb{R})$ and constants $a, b \in \mathbb{C}$,

$$\widehat{af + bg}(\alpha) = a\widehat{f}(\alpha) + b\widehat{g}(\alpha), \qquad \widehat{f * g}(\alpha) = \widehat{f}(\alpha)\,\widehat{g}(\alpha), \qquad \alpha \in \mathbb{R}.$$

Moreover for each $\alpha \in \mathbb{R}$

$$\widehat{fu_\alpha} = (\widehat{f})_\alpha, \qquad \widehat{f_x}(\alpha) = \widehat{f}(\alpha) u_\alpha(-x), \quad x \in \mathbb{R}.$$

Proof. For example, for every $\beta \in \mathbb{R}$

$$\widehat{fu_\alpha}(\beta) = \int_{\mathbb{R}} f(x) u_\alpha(x) e^{-i\beta x}\, dx = \int_{\mathbb{R}} f(x) e^{-i(\beta-\alpha)x}\, dx = \widehat{f}(\beta - \alpha),$$

so $\widehat{fu_\alpha} = (\widehat{f})_\alpha$. The rest is Exercise 5.2.2. ■

Exercise 5.2.2

Complete the proof of Proposition 5.2.1.

Solution [Supplement]. Linearity follows from linearity of the integral. For the convolution, $(x, y) \mapsto f(x - y)g(y)e^{-i\alpha x}$ is integrable on $\mathbb{R}^2$ (its absolute value integrates to $\|f\|_1\|g\|_1$), so by

Fubini

$$\widehat{f * g}(\alpha) = \int_{\mathbb{R}} g(y) e^{-i\alpha y} \int_{\mathbb{R}} f(x-y) e^{-i\alpha(x-y)}\, dx\, dy = \widehat{g}(\alpha)\widehat{f}(\alpha).$$

For translation, $\widehat{f_x}(\alpha) = \int_{\mathbb{R}} f(y-x)e^{-i\alpha y}\, dy = \int_{\mathbb{R}} f(y)e^{-i\alpha(y+x)}\, dy = \widehat{f}(\alpha)e^{-i\alpha x} = \widehat{f}(\alpha)u_\alpha(-x)$. □

5.2.2 Differentiation

Proposition 5.2.2

Let $f \in L^1(\mathbb{R})$.

(a) Let $g(x) = -ixf(x)$. If $g \in L^1(\mathbb{R})$, then $\widehat{f}$ is differentiable and

$$\widehat{f}\,'(\alpha) = \widehat{g}(\alpha), \qquad \alpha \in \mathbb{R}.$$

(b) If f is differentiable and $f' \in L^1(\mathbb{R})$, then

$$\widehat{f'}(\alpha) = i\alpha\widehat{f}(\alpha), \qquad \alpha \in \mathbb{R}.$$

Proof. (a) The difference quotient is

$$\frac{\widehat{f}(\alpha+h) - \widehat{f}(\alpha)}{h} = \int_{\mathbb{R}} f(x)e^{-i\alpha x}\frac{e^{-ihx}-1}{h}\, dx,$$

and

$$\left| f(x)e^{-i\alpha x}\frac{e^{-ihx}-1}{h} \right| \le |xf(x)|.$$

Since $x \mapsto |xf(x)|$ is integrable and $\lim_{h\to 0}\frac{e^{-ihx}-1}{h} = -ix$, the Lebesgue convergence theorem gives, for every real sequence $h_n \to 0$,

$$\lim_{n\to\infty} \frac{\widehat{f}(\alpha+h_n) - \widehat{f}(\alpha)}{h_n} = \widehat{g}(\alpha).$$

(b) As in Proposition 5.1.2, integrate by parts: $\widehat{f'}(\alpha) = \int_{\mathbb{R}} f'(x)e^{-i\alpha x}\, dx = i\alpha \int_{\mathbb{R}} f(x)e^{-i\alpha x}\, dx$; the boundary terms vanish because $f(x) \to 0$ as $x \to \pm\infty$ (f has limits at $\pm\infty$ since $f' \in L^1$, and they must be 0 since $f \in L^1$). ■

5.2.3 Riemann–Lebesgue lemma

As for Fourier series, $\lim_{\alpha\to\pm\infty}\widehat{f}(\alpha) = 0$ for $f \in L^1(\mathbb{R})$. Using $e^{\pi i} = -1$,

$$\widehat{f}(\alpha) = -\int_{\mathbb{R}} f(x)e^{-i\alpha(x+\pi/\alpha)}\, dx = -\int_{\mathbb{R}} f\Big(x - \frac{\pi}{\alpha}\Big)e^{-i\alpha x}\, dx,$$

and hence

$$2\widehat{f}(\alpha) = \int_{\mathbb{R}} [f(x) - f_{\pi/\alpha}(x)]e^{-i\alpha x}\, dx.$$

By Fact B.23,

$$0 \le \lim_{\alpha\to\pm\infty}|\widehat{f}(\alpha)| \le \frac{1}{2}\lim_{\alpha\to\pm\infty}\|f - f_{\pi/\alpha}\|_1 = 0.$$

Since $\widehat{f}$ is already known to be continuous:

Theorem 5.2.3 (Riemann–Lebesgue)

$\widehat{f} \in C_0(\mathbb{R})$ for every $f \in L^1(\mathbb{R})$.

5.2.4 Approximate identities on $\mathbb{R}$

Using Fact B.23 instead of Theorem 5.1.4 in the proof of Theorem 5.1.5 yields:

Theorem 5.2.4

If a family $\{h_\lambda : \lambda \in (0,\infty)\}$ satisfies

$$h_\lambda \geq 0, \qquad \int_{\mathbb{R}} h_\lambda = 1, \qquad \lim_{\lambda\to\infty} \int_{\mathbb{R}\setminus[-\delta,\delta]} h_\lambda = 0 \quad (\delta > 0), \tag{10}$$

then:

(a) $\lim_{\lambda\to\infty} \|f - h_\lambda * f\|_1 = 0$ for every $f \in L^1(\mathbb{R})$;

(b) $\lim_{\lambda\to\infty} \|f - h_\lambda * f\|_\infty = 0$ for every bounded uniformly continuous $f : \mathbb{R} \to \mathbb{C}$.

Exercise 5.2.3

Prove Theorem 5.2.4.

Solution [Supplement]. (a) Let $\epsilon > 0$. By Fact B.23 choose $\delta > 0$ with $|s| < \delta \Rightarrow \|f - f_s\|_1 < \epsilon$, and Λ with $\int_{\mathbb{R}\setminus[-\delta,\delta]} h_\lambda < \epsilon/(2\|f\|_1)$ for $\lambda \geq \Lambda$. Since $f(t) - (h_\lambda * f)(t) = \int_{\mathbb{R}} h_\lambda(s)[f(t) - f_s(t)]\,ds$, Tonelli gives for $\lambda \geq \Lambda$

$$\|f - h_\lambda * f\|_1 \leq \int_{\mathbb{R}} h_\lambda(s)\|f - f_s\|_1\,ds \leq \epsilon \int_{[-\delta,\delta]} h_\lambda + 2\|f\|_1 \int_{\mathbb{R}\setminus[-\delta,\delta]} h_\lambda < 2\epsilon.$$

(b) Choose δ with $|s| < \delta \Rightarrow \|f - f_s\|_\infty < \epsilon$ (uniform continuity) and Λ with $\int_{\mathbb{R}\setminus[-\delta,\delta]} h_\lambda < \epsilon/(2\|f\|_\infty)$ for $\lambda \geq \Lambda$. Then for every t,

$$|f(t) - (h_\lambda * f)(t)| \leq \int_{\mathbb{R}} h_\lambda(s)|f(t) - f_s(t)|\,ds \leq \epsilon + 2\|f\|_\infty \cdot \frac{\epsilon}{2\|f\|_\infty} = 2\epsilon.$$

□

If $h : \mathbb{R} \to \mathbb{C}$ satisfies

$$h \geq 0, \qquad \int_{\mathbb{R}} h = 1, \tag{11}$$

put

$$h_\lambda(x) = \lambda h(\lambda x), \qquad x \in \mathbb{R},\ \lambda > 0. \tag{12}$$

Then $\{h_\lambda : \lambda > 0\}$ satisfies (10): $h_\lambda \geq 0$ and $\int h_\lambda = 1$ are clear, and for fixed $\delta > 0$, as $\lambda \to \infty$,

$$\int_{\mathbb{R}\setminus[-\delta,\delta]} h_\lambda(x)\,dx = \int_{\mathbb{R}\setminus[-\lambda\delta,\lambda\delta]} h(x)\,dx \to 0.$$

Exercise 5.2.4

Show that $\widehat{h_\lambda}(\alpha) = \widehat{h}\left(\frac{\alpha}{\lambda}\right)$ for every $\alpha \in \mathbb{R}$.

Solution [Supplement]. Substituting $y = \lambda x$: $\widehat{h_\lambda}(\alpha) = \int_{\mathbb{R}} \lambda h(\lambda x) e^{-i\alpha x}\,dx = \int_{\mathbb{R}} h(y) e^{-i(\alpha/\lambda)y}\,dy = \widehat{h}(\alpha/\lambda)$. □

The case where h satisfying (11) has the form

$$h(x) = \frac{1}{2\pi}\int_{\mathbb{R}} p(\alpha)u_\alpha(x)\,d\alpha \tag{13}$$

is very useful. Here $p : \mathbb{R} \to \mathbb{C}$ is assumed to satisfy

$$0 \le p \le 1, \qquad p(0) = \lim_{\alpha\to 0} p(\alpha) = 1, \qquad p \in L^1(\mathbb{R}). \tag{14}$$

A simple example satisfying all requirements:

$$H(x) = \frac{1}{2\pi}\int_{\mathbb{R}} e^{-|\alpha|}e^{i\alpha x}\,d\alpha = \frac{1}{\pi}\,\frac{1}{1+x^2}, \tag{15}$$

and then

$$H_\lambda(x) = \frac{1}{2\pi}\int_{\mathbb{R}} e^{-|\alpha/\lambda|}e^{i\alpha x}\,d\alpha = \frac{1}{2\pi}\,\frac{2\lambda}{1+\lambda^2x^2}.$$

$\{H_\lambda : \lambda > 0\}$ is the **Poisson kernel**.

5.2.5 Smooth functions are dense

Let $C_c^n(\mathbb{R})$ be the functions in $C_c(\mathbb{R})$ that are n times differentiable with continuous $f^{(n)}$, and $C_c^\infty(\mathbb{R}) = \bigcap_{n=1}^\infty C_c^n(\mathbb{R})$. If $h \in C_c^\infty(\mathbb{R})$ satisfies (11), then $h_\lambda \in C_c^\infty(\mathbb{R})$, and by repeated use of Proposition 4.3.2, $h_\lambda * f \in C_c^\infty(\mathbb{R})$ for every $f \in C_c(\mathbb{R})$. Moreover by Theorem 5.2.4

$$\lim_{\lambda\to\infty}\|f - h_\lambda * f\|_1 = \lim_{\lambda\to\infty}\|f - h_\lambda * f\|_\infty = 0.$$

Since $C_c(\mathbb{R})$ is dense in $L^1(\mathbb{R})$ and in $C_0(\mathbb{R})$:

Proposition 5.2.5

$C_c^\infty(\mathbb{R})$ is dense in the normed spaces $L^1(\mathbb{R})$ and $C_0(\mathbb{R})$.

Exercise 5.2.5

Give an example of $h \in C_c^\infty(\mathbb{R})$ satisfying (11).

Solution [Supplement]. Let $\varphi(x) = \exp(-\frac{1}{1-x^2})$ for $|x| < 1$ and $\varphi(x) = 0$ for $|x| \ge 1$. Since $\psi(t) = e^{-1/t}$ $(t > 0)$, $\psi(t) = 0$ $(t \le 0)$ is C^∞ (every derivative on $t > 0$ is $P(1/t)e^{-1/t}$ for a polynomial P, which tends to 0 as $t \downarrow 0$), and $\varphi(x) = \psi(1-x^2)$, $\varphi \in C_c^\infty(\mathbb{R})$, $\varphi \ge 0$, $\int\varphi > 0$. Take $h = \varphi/\int_{\mathbb{R}}\varphi$. □

5.2.6 The inversion formula

For h given by (13) and $\{h_\lambda\}$ by (12), compute $h_\lambda * f$ for $f \in L^1(\mathbb{R})$ using Fubini:

$$\begin{aligned}(h_\lambda * f)(x) &= \int_{\mathbb{R}} f(x-y)\frac{\lambda}{2\pi}\int_{\mathbb{R}} p(\alpha)e^{i\alpha\lambda y}\,d\alpha\,dy = \frac{1}{2\pi}\int_{\mathbb{R}} f(x-y)\int_{\mathbb{R}} p(\tfrac{\alpha}{\lambda})e^{i\alpha y}\,d\alpha\,dy\\ &= \frac{1}{2\pi}\int_{\mathbb{R}} p(\tfrac{\alpha}{\lambda})\int_{\mathbb{R}} f(x-y)e^{i\alpha y}\,dy\,d\alpha = \frac{1}{2\pi}\int_{\mathbb{R}} p(\tfrac{\alpha}{\lambda})\int_{\mathbb{R}} f(y)e^{i\alpha(x-y)}\,dy\,d\alpha\\ &= \frac{1}{2\pi}\int_{\mathbb{R}} p(\tfrac{\alpha}{\lambda})\widehat{f}(\alpha)e^{i\alpha x}\,d\alpha.\end{aligned}$$

(Fubini applies since $\iint|f(x-y)||p(\alpha/\lambda)|\,d\alpha\,dy = \lambda\|f\|_1\|p\|_1 < \infty$.)

From now on assume p, h satisfy (14), (11) and (13). Then

$$(h_\lambda * f)(x) = \frac{1}{2\pi}\int_{\mathbb{R}} p\Big(\frac{\alpha}{\lambda}\Big)\widehat{f}(\alpha)e^{i\alpha x}\,d\alpha.$$

Since

$$\Big|p\Big(\frac{\alpha}{\lambda}\Big)\widehat{f}(\alpha)e^{i\alpha x}\Big| \le |\widehat{f}(\alpha)|, \qquad \lim_{\lambda\to\infty} p\Big(\frac{\alpha}{\lambda}\Big) = 1,$$

if $\widehat{f} \in L^1(\mathbb{R})$ the Lebesgue convergence theorem gives

$$\lim_{\lambda\to\infty}(h_\lambda * f)(x) = \frac{1}{2\pi}\int_{\mathbb{R}} \widehat{f}(\alpha)e^{i\alpha x}\,d\alpha, \qquad x \in \mathbb{R}.$$

On the other hand $\lim_{\lambda\to\infty}\|h_\lambda * f - f\|_1 = 0$ by Theorem 5.2.4, so by Fact B.18 there is a sequence $\lambda_n \to \infty$ with

$$\lim_{n\to\infty}(h_{\lambda_n} * f)(x) = f(x)$$

for almost every x. This proves the analogue of Corollary 5.1.9 (with, e.g., $p(\alpha) = e^{-|\alpha|}$ and $h = H$ from (15)):

Theorem 5.2.6 (inversion formula)

If $f \in L^1(\mathbb{R})$ and $\widehat{f} \in L^1(\mathbb{R})$, then

$$f(x) = \frac{1}{2\pi}\int_{\mathbb{R}} \widehat{f}(\alpha)e^{i\alpha x}\,d\alpha \tag{16}$$

for almost every x.

The right side of (16) is a continuous function of x (as in the continuity of $\widehat{f}$). Hence if f is continuous, (16) holds for every $x \in \mathbb{R}$.

Example 2

Applying the inversion formula to Example 1 (with $a = 1$),

$$e^{-|x|} = \int_{-\infty}^{\infty}\frac{1}{\pi(1+\alpha^2)}e^{i\alpha x}\,d\alpha = \int_{-\infty}^{\infty}\frac{\cos\alpha x}{\pi(1+\alpha^2)}\,d\alpha + i\int_{-\infty}^{\infty}\frac{\sin\alpha x}{\pi(1+\alpha^2)}\,d\alpha.$$

Since the left side is real (and the cosine integrand is even),

$$\int_0^{\infty}\frac{\cos\alpha x}{1+\alpha^2}\,d\alpha = \frac{\pi}{2}e^{-|x|}.$$

If $\widehat{f} = 0$, then $\widehat{f} \in L^1$ and (16) gives $f = 0$ almost everywhere, the analogue of Theorem 5.1.8:

Theorem 5.2.7 (uniqueness)

If $f \in L^1(\mathbb{R})$ and $\widehat{f} = 0$, then $f = 0$ almost everywhere.

5.2.7 Fejér and Gauss kernels

Two more examples of p satisfying (14), (11), (13). Recalling the definition (8) of the Fejér kernel, for $\lambda > 0$ put

$$K_\lambda(x) = \frac{1}{2\pi}\int_{\mathbb{R}}\max\Big\{1-\frac{|\alpha|}{\lambda},0\Big\}u_\alpha(x)\,d\alpha = \frac{1}{2\pi}\int_{-\lambda}^{\lambda}\Big(1-\frac{|\alpha|}{\lambda}\Big)e^{i\alpha x}\,d\alpha.$$

Then

$$\lambda K_1(\lambda x) = \frac{1}{2\pi}\int_{-1}^{1}(1-|\alpha|)e^{i\alpha\lambda x}\lambda\,d\alpha = K_\lambda(x),$$

so with $\lambda = 1$ in mind define

$$K(x) = \frac{1}{2\pi}\int_{-1}^{1}(1-|\alpha|)e^{i\alpha x}\,d\alpha.$$

A direct computation (integration by parts) gives

$$K_\lambda(x) = \frac{1}{\pi}\int_0^\lambda \Big(1-\frac{\alpha}{\lambda}\Big)\cos\alpha x\,d\alpha = \frac{1}{\pi\lambda x^2}(1-\cos\lambda x) = \frac{1}{2\pi\lambda}\,\frac{\sin^2(\frac{\lambda x}{2})}{(\frac{x}{2})^2},$$

so $K \geq 0$. Replacing α by $-\alpha$ in the defining integral of K_λ does not change it, so with $f_\lambda(\alpha) = \max\{1-\frac{|\alpha|}{\lambda}, 0\}$ we have $K_\lambda = \frac{1}{2\pi}\widehat{f_\lambda}$. Since f_λ is integrable and $\widehat{f_\lambda} = 2\pi K_\lambda = O(x^{-2})$ is integrable, the inversion formula gives

$$\int_{-\infty}^{\infty} K_\lambda(x)e^{i\alpha x}\,dx = \frac{1}{2\pi}\int_{-\infty}^{\infty}\widehat{f_\lambda}(x)e^{i\alpha x}\,dx = f_\lambda(\alpha) = \max\Big\{1-\frac{|\alpha|}{\lambda},0\Big\}.$$

Since f_λ is continuous, this holds for every $\alpha\in\mathbb{R}$; in particular for $\lambda = 1$, $\alpha = 0$,

$$\int_{-\infty}^{\infty} K(x)\,dx = 1.$$

The family $\{K_\lambda : \lambda > 0\}$ obtained this way is also called the **Fejér kernel**.

Finally let

$$G(x) = \frac{1}{2\pi}\int_{\mathbb{R}} e^{-\alpha^2/2}u_\alpha(x)\,d\alpha.$$

By Fact B.11 (differentiation under the integral) and integration by parts,

$$G'(x) = \frac{1}{2\pi}\int_{\mathbb{R}} i\alpha e^{-\alpha^2/2}e^{i\alpha x}\,d\alpha = -xG(x)$$

(indeed $\int i\alpha e^{-\alpha^2/2}e^{i\alpha x}\,d\alpha = -i\int(e^{-\alpha^2/2})'e^{i\alpha x}\,d\alpha = i\cdot ix\int e^{-\alpha^2/2}e^{i\alpha x}\,d\alpha$). Since $G(0) = \frac{1}{2\pi}\sqrt{2\pi} = \frac{1}{\sqrt{2\pi}}$, solving the ODE gives

$$G(x) = \frac{1}{\sqrt{2\pi}}e^{-x^2/2},$$

so $G\geq 0$ and $\int_{\mathbb{R}} G = 1$. The family $\{G_\lambda : \lambda > 0\}$ is the **Gauss kernel**.

5.2.8 Transforms of the kernels

The inversion formula says that if g and $\widehat{g}$ are continuous L^1-functions, then

$$g(x) = \frac{1}{2\pi}\widehat{\widehat{g}}(-x), \qquad x\in\mathbb{R}. \tag{17}$$

Suppose $f, g\in L^1(\mathbb{R})$, g continuous, satisfy

$$f(x) = \frac{1}{2\pi}\int_{\mathbb{R}} g(\alpha)e^{i\alpha x}\,d\alpha, \qquad x\in\mathbb{R}.$$

Then $f(x) = \frac{1}{2\pi}\widehat{g}(-x)$, so $\widehat{g}\in L^1(\mathbb{R})$. By the inversion formula for g (valid everywhere, g being continuous),

$$g(y) = \frac{1}{2\pi}\int_{\mathbb{R}}\widehat{g}(\alpha)e^{i\alpha y}\,d\alpha = \int_{\mathbb{R}} f(-\alpha)e^{i\alpha y}\,d\alpha = \int_{\mathbb{R}} f(\beta)e^{-i\beta y}\,d\beta = \widehat{f}(y).$$

So $\widehat{f} = g$. Applying this to the three kernels:

$$\widehat{K_\lambda}(\alpha) = \max\Big\{1-\frac{|\alpha|}{\lambda},0\Big\}, \qquad \widehat{P_\lambda}(\alpha) = e^{-|\alpha/\lambda|}, \qquad \widehat{G_\lambda}(\alpha) = e^{-\alpha^2/2\lambda^2},$$

where $P_\lambda = H_\lambda$ is the Poisson kernel.

Exercise 5.2.6

For $f \in L^1(\mathbb{R})$ express $\widehat{K_\lambda * f}$ in terms of $\widehat{f}$. Use this to show that $\{f \in L^1(\mathbb{R}) : \widehat{f} \in C_c(\mathbb{R})\}$ is dense in $L^1(\mathbb{R})$, and compare with Theorem 5.1.7 for Fourier series.

Solution [Supplement]**.** By Proposition 5.2.1,

$$\widehat{K_\lambda * f}(\alpha) = \widehat{K_\lambda}(\alpha)\widehat{f}(\alpha) = \max\left\{1 - \frac{|\alpha|}{\lambda}, 0\right\}\widehat{f}(\alpha),$$

which is continuous and vanishes outside $[-\lambda, \lambda]$; so $K_\lambda * f$ belongs to the set in question. Since $K_\lambda(x) = \lambda K(\lambda x)$ with $K \geq 0$, $\int K = 1$, the family satisfies (10), and Theorem 5.2.4 gives $\|K_\lambda * f - f\|_1 \to 0$. Hence the set is dense. Comparison: on $\mathbb{T}$, $\widehat{K_n * f}(k) = (1 - \frac{|k|}{n+1})\widehat{f}(k)$ vanishes for $|k| > n$, i.e. $K_n * f$ is a trigonometric polynomial. "Compactly supported Fourier transform" on $\mathbb{R}$ is the analogue of "finitely many nonzero Fourier coefficients" (trigonometric polynomials) on $\mathbb{T}$, and both density statements are proved by the same Fejér-kernel argument. □

5.2.9 The Fourier transform as a homomorphism

By now we know that $f \mapsto \widehat{f}$ is an injective linear map from $L^1(\mathbb{R})$ into $C_0(\mathbb{R})$. If $f \in C_c^2(\mathbb{R})$, then $f'' \in L^1(\mathbb{R})$ and by Proposition 5.2.2(b) $\widehat{f''}(\alpha) = -\alpha^2\widehat{f}(\alpha)$; since $\widehat{f''}$ is bounded and $\widehat{f}$ is bounded and continuous, $|\widehat{f}(\alpha)| \leq \min\{\|f\|_1, \|f''\|_1/\alpha^2\}$, so $\widehat{f} \in L^1(\mathbb{R})$. By (17), f is the Fourier transform of another L^1-function ($f(x) = \widehat{k}(x)$ with $k(\alpha) = \frac{1}{2\pi}\widehat{f}(-\alpha)$). Thus $C_c^2(\mathbb{R})$ lies in the range of $f \mapsto \widehat{f} : L^1(\mathbb{R}) \to C_0(\mathbb{R})$, and $C_c^2(\mathbb{R}) \supset C_c^\infty(\mathbb{R})$ is dense in $C_0(\mathbb{R})$ by Proposition 5.2.5:

Theorem 5.2.8

The Fourier transform $f \mapsto \widehat{f}$ is an injective, norm-decreasing homomorphism from the Banach algebra $L^1(\mathbb{R})$ into $C_0(\mathbb{R})$, and its range is dense.

Here the multiplication on $C_0(\mathbb{R})$ is the pointwise product $(fg)(x) = f(x)g(x)$. The map $f \mapsto \widehat{f} : L^1(\mathbb{T}) \to c_0(\mathbb{Z})$ of Theorem 5.1.11 is also called a Fourier transform in this broader sense.

Exercise 5.2.7

Show that the range of $f \mapsto \widehat{f} : L^1(\mathbb{T}) \to c_0(\mathbb{Z})$ is dense.

Solution [Supplement]**.** Every finitely supported $a : \mathbb{Z} \to \mathbb{C}$ is the transform of the trigonometric polynomial $\sum_n a(n)u_n \in L^1(\mathbb{T})$. For $a \in c_0(\mathbb{Z})$ let $a_N = a\chi_{[-N,N]}$; then $\|a - a_N\|_\infty = \sup_{|n|>N}|a(n)| \to 0$. So finitely supported sequences, all in the range, are dense in $c_0(\mathbb{Z})$. □

5.2.10 Poisson summation

How are Fourier series and Fourier integrals related? Given $f \in L^1(\mathbb{R})$, as in Fact B.12 the series

$$F(t) = \sum_{n=-\infty}^{\infty} f(t - 2n\pi) \tag{18}$$

converges absolutely for almost every $t \in \mathbb{T}$, and $F \in L^1(\mathbb{T})$ (indeed $\int_{\mathbb{T}} \sum_n |f(t-2n\pi)|\,dt = \frac{1}{2\pi}\|f\|_1$ by term-by-term integration). By Levi's theorem (term-by-term integration),

$$\widehat{F}(n) = \int_{\mathbb{T}} F(t)e^{-int}\,dt = \sum_{k=-\infty}^{\infty} \frac{1}{2\pi}\int_{-\pi}^{\pi} f(t + 2k\pi)e^{-int}\,dt = \frac{1}{2\pi}\int_{\mathbb{R}} f(t)e^{-int}\,dt = \frac{1}{2\pi}\widehat{f}(n). \tag{19}$$

Hence if the Fourier series of F converges at $t \in \mathbb{T}$ (to $F(t)$), we get the **Poisson summation formula**

$$\sum_{n=-\infty}^{\infty} f(t-2n\pi) = \frac{1}{2\pi} \sum_{n=-\infty}^{\infty} \widehat{f}(n) e^{int}.$$

Exercise 5.2.8

Use $f(x) = e^{-|x|}$ to prove

$$\sum_{n=-\infty}^{\infty} \frac{1}{n^2+1} = \pi \, \frac{e^{2\pi}+1}{e^{2\pi}-1}.$$

Solution [Supplement]. Here $\widehat{f}(\alpha) = \frac{2}{1+\alpha^2}$ (Example 1), so $\widehat{F}(n) = \frac{1}{\pi}\frac{1}{1+n^2}$ and $\widehat{F} \in \ell^1(\mathbb{Z})$. The series $F(t) = \sum_n e^{-|t-2n\pi|}$ converges uniformly on $[-\pi,\pi]$ (for $|n| \geq 1$ its terms are at most $e^{-(2|n|-1)\pi}$), so F is continuous. By Corollary 5.1.9 and Exercise 5.1.1, $F(t) = \sum_n \widehat{F}(n) e^{int}$ a.e., both sides continuous, hence for every t. At $t = 0$:

$$\sum_n e^{-2\pi|n|} = \frac{1}{\pi} \sum_n \frac{1}{1+n^2}.$$

The left side is $1 + \frac{2e^{-2\pi}}{1-e^{-2\pi}} = \frac{1+e^{-2\pi}}{1-e^{-2\pi}} = \frac{e^{2\pi}+1}{e^{2\pi}-1}$, which gives the formula. □

Exercise 5.2.9

For $f \in L^1(\mathbb{R})$ and $\lambda > 0$ define, as in (18),

$$F_\lambda(t) = \sum_{n=-\infty}^{\infty} f_\lambda(t-2n\pi), \qquad f_\lambda(x) = \lambda f(\lambda x).$$

Show that

$$\widehat{F_\lambda}(n) = \frac{1}{2\pi} \widehat{f}\Big(\frac{n}{\lambda}\Big), \qquad \lim_{\lambda\to\infty} \|F_\lambda\|_{L^1(\mathbb{T})} = \frac{1}{2\pi}\|f\|_{L^1(\mathbb{R})}.$$

Use this to prove Theorem 5.2.7.

Solution [Supplement]. By (19) and Exercise 5.2.4, $\widehat{F_\lambda}(n) = \frac{1}{2\pi}\widehat{f_\lambda}(n) = \frac{1}{2\pi}\widehat{f}(n/\lambda)$. Upper bound: $\|F_\lambda\|_{L^1(\mathbb{T})} \leq \frac{1}{2\pi}\sum_n \int_{-\pi}^{\pi} |f_\lambda(t-2n\pi)|\,dt = \frac{1}{2\pi}\|f_\lambda\|_1 = \frac{1}{2\pi}\|f\|_1$. Lower bound: separating the $n=0$ term,

$$\|F_\lambda\|_{L^1(\mathbb{T})} \geq \frac{1}{2\pi}\Big[\int_{-\pi}^{\pi} |f_\lambda(t)|\,dt - \sum_{n\neq 0}\int_{-\pi}^{\pi} |f_\lambda(t-2n\pi)|\,dt\Big] = \frac{1}{2\pi}\Big[\int_{|x|\leq\pi} |f_\lambda| - \int_{|x|>\pi} |f_\lambda|\Big]$$
$$= \frac{1}{2\pi}\Big[\int_{|y|\leq\lambda\pi} |f| - \int_{|y|>\lambda\pi} |f|\Big] \longrightarrow \frac{1}{2\pi}\|f\|_1 \quad (\lambda\to\infty).$$

Proof of Theorem 5.2.7: if $\widehat{f} = 0$, then $\widehat{F_\lambda}(n) = 0$ for all n and λ, so $F_\lambda = 0$ a.e. by Theorem 5.1.8, i.e. $\|F_\lambda\|_{L^1(\mathbb{T})} = 0$. Letting $\lambda \to \infty$ gives $\|f\|_1 = 0$. □

Chapter 6. Normed Spaces

Overview

Results from Chapters 1–3 are cited as *Fact B.k*; they are stated in the separate *Background* notes. Several normed spaces have appeared so far; this chapter studies the basic properties of general normed spaces. The dual space — the space of bounded linear maps into the scalars — is essential for understanding a normed space, and we identify the duals of the ℓ^p-spaces and of Hilbert spaces. We also study elementary properties of bounded linear maps between Banach spaces (Baire category, open mapping, closed graph, uniform boundedness) and apply them to Fourier transforms.

6.1 Dual Spaces

Motivation. A normed space is often best understood through the continuous linear functionals on it: measures act on functions, sequences in ℓ^q act on ℓ^p. Identifying these dual spaces, and knowing there are enough functionals to separate points (Hahn–Banach), is the basic tool of functional analysis.

6.1.1 Bounded linear maps

Definition (operator norm)

For a linear map $T : X \to Y$ between normed spaces put

$$\|T\| = \sup\{\|T(x)\| : x \in X,\ \|x\| \le 1\}.$$

If $\|T\| < \infty$, T is a **bounded linear map** and $\|T\|$ is its **norm**.

The norm of T measures how much T stretches the unit ball $\{x \in X : \|x\| \le 1\}$ of X inside Y. For example, $z \mapsto \alpha z : \mathbb{C} \to \mathbb{C}$ has norm $|\alpha|$.

Exercise 6.1.1

Show that a linear map $T : X \to Y$ is bounded if and only if there is $M > 0$ with

$$x \in X \implies \|T(x)\| \le M\|x\|,$$

and that in this case $\|T\| \le M$.

Solution [Supplement]. If T is bounded, take $M = \|T\| + 1$ (or $\|T\|$ if positive): for $x \neq 0$, $x/\|x\|$ has norm 1, so $\|T(x)\| = \|x\|\,\|T(x/\|x\|)\| \le \|T\|\|x\|$; for $x = 0$ both sides vanish. Conversely, if $\|T(x)\| \le M\|x\|$ for all x, then $\|T(x)\| \le M$ whenever $\|x\| \le 1$, so $\|T\| \le M < \infty$. □

Exercise 6.1.2

Prove that $\|T\| = \sup\left\{\dfrac{\|T(x)\|}{\|x\|} : x \in X,\ x \neq 0\right\}$.

Solution [Supplement]. Let S be the right side. For $x \neq 0$, $\|T(x)\|/\|x\| = \|T(x/\|x\|)\| \le \|T\|$, so $S \le \|T\|$. Conversely, if $0 < \|x\| \le 1$ then $\|T(x)\| \le \|T(x)\|/\|x\| \le S$, and $T(0) = 0$; so $\|T\| \le S$. □

Exercise 6.1.3

Show that the set of all bounded linear maps between two normed spaces is a vector space, and that $T \mapsto \|T\|$ is a norm on it.

Solution [Supplement]. If S, T are bounded and $a \in \mathbb{C}$, then for $\|x\| \le 1$

$$\|(S+T)(x)\| \le \|S(x)\| + \|T(x)\| \le \|S\| + \|T\|, \qquad \|(aT)(x)\| = |a|\|T(x)\| \le |a|\|T\|,$$

so $S+T$, aT are bounded with $\|S+T\| \le \|S\| + \|T\|$ and $\|aT\| \le |a|\|T\|$; applying the latter to a^{-1} gives $\|aT\| = |a|\|T\|$ for $a \neq 0$. Clearly $\|T\| \ge 0$, and $\|T\| = 0$ forces $T(x) = 0$ for $\|x\| \le 1$, hence for all x by scaling, i.e. $T = 0$. □

Proposition 6.1.1

For a linear map $T : X \to Y$ between normed spaces, the following are equivalent:

(a) T is bounded;

(b) T is continuous;

(c) T is continuous at some point $x_0 \in X$.

Proof. The inequality $\|T(x) - T(y)\| \le \|T\|\|x - y\|$ gives (a)⇒(b), and (b)⇒(c) is trivial. If T is continuous at x_0, there is $\delta > 0$ with $\|x - x_0\| < \delta \Rightarrow \|T(x) - T(x_0)\| < \epsilon$ (take $\epsilon = 1$, say). By linearity this is the same as $\|y\| < \delta \Rightarrow \|T(y)\| < \epsilon$, from which $\|T\| \le \epsilon/\delta$. ■

Proposition 6.1.2

For normed spaces X, Y, the set $B(X,Y)$ of bounded linear maps from X to Y is a normed space. If Y is a Banach space, so is $B(X,Y)$.

Proof. That $B(X,Y)$ is a normed space is Exercise 6.1.3. Assume Y is a Banach space and let $\langle T_n \rangle$ be Cauchy in $B(X,Y)$. For each $x \in X$, $\|T_n(x) - T_m(x)\| \le \|T_n - T_m\|\|x\|$, so $\langle T_n(x) \rangle$ is Cauchy in Y and has a limit, which we call $T(x)$. This gives a map $x \mapsto T(x)$, and $T \in B(X,Y)$ with $\|T_n - T\| \to 0$ (Exercise 6.1.4). ■

Exercise 6.1.4

Complete the proof of Proposition 6.1.2.

Solution [Supplement]. T is linear as a pointwise limit of linear maps. Given $\epsilon > 0$ choose N with $\|T_n - T_m\| < \epsilon$ for $m, n \ge N$. For $\|x\| \le 1$ and $n, m \ge N$, $\|T_n(x) - T_m(x)\| < \epsilon$; letting $m \to \infty$ gives $\|T_n(x) - T(x)\| \le \epsilon$. Hence $T_N - T$ is bounded, so $T = T_N - (T_N - T) \in B(X,Y)$, and $\|T_n - T\| \le \epsilon$ for $n \ge N$. □

Definition (dual space)

A linear map with values in the scalars $\mathbb{C}$ is a **linear functional**. The normed space of all bounded linear functionals on X is denoted $X^* = B(X, \mathbb{C})$. By Proposition 6.1.2, X^* is always a Banach space; it is the **dual space** of X.

Example 1

$f \mapsto \int_0^1 f$ is a bounded linear functional on $(C[0,1], \|\cdot\|_{\sup})$. However

$$f \mapsto \int_{\mathbb{R}} f(x)\,dx \tag{1}$$

is not even a linear functional on $(C_0(\mathbb{R}), \|\cdot\|_{\sup})$, since the integral of $f \in C_0(\mathbb{R})$ need not be finite. Restricting the domain to $C_c(\mathbb{R})$ gives a linear functional which is not bounded: choosing $f_n \in C_c(\mathbb{R})$ with $0 \le f_n \le 1$ and $f_n = 1$ on $[-n, n]$ gives $\|f_n\|_{\sup} = 1$ but $\int_{\mathbb{R}} f_n \to \infty$. On the other hand, (1) is a bounded linear functional on $(L^1(\mathbb{R}), \|\cdot\|_1)$ by inequality Fact B.9 ($|\int f| \le \int |f|$).

Example 2

Let (X, μ) be a measure space, p, q conjugate exponents, and $g \in L^q(\mu)$ $(1 \le q \le \infty)$. Consider

$$\varphi_g : f \mapsto \int_X fg\,d\mu, \qquad f \in L^p(\mu).$$

By Hölder, $|\varphi_g(f)| \le \|g\|_q \|f\|_p$, so φ_g is a bounded linear functional on $L^p(\mu)$ with

$$\|\varphi_g\| \le \|g\|_q, \qquad g \in L^q(\mu).$$

In fact equality holds.
Case $p = \infty$, $q = 1$. For fixed $g \in L^1(\mu)$ we want $f \in L^\infty(\mu)$ with $\|f\|_\infty \|g\|_1 = |\int_X gf|$. Take

$$f(x) = \begin{cases} \dfrac{|g(x)|}{g(x)}, & g(x) \neq 0, \\ 0, & g(x) = 0; \end{cases}$$

this function is written $\operatorname{sgn} g$. Then $gf = |g|$, so $\int_X gf = \|g\|_1 = \|f\|_\infty \|g\|_1$ (if $g \neq 0$).
Case $p = 1$, $q = \infty$. Fix $g \in L^\infty(\mu)$ with $\|g\|_\infty \neq 0$. For small $\epsilon > 0$ the set $\{x \in X : |g(x)| > \|g\|_\infty - \epsilon\}$ has positive measure; choose a subset E of it with $0 < \mu(E) < \infty$ (possible if (X, μ) is σ-finite) and put $f = \chi_E \cdot \operatorname{sgn} g \in L^1(\mu)$. Then

$$\int_X gf = \int_E g \cdot \operatorname{sgn} g = \int_E |g| \ge \mu(E)(\|g\|_\infty - \epsilon) = \|f\|_1(\|g\|_\infty - \epsilon),$$

so $\|\varphi_g\| \ge \|g\|_\infty - \epsilon$, and hence $\|\varphi_g\| = \|g\|_\infty$.
Case $1 < q < \infty$. For fixed $g \in L^q(\mu)$ put $f = |g|^{q/p} \operatorname{sgn} g$. Then $|f|^p = |g|^q$, so $\|g\|_q^{q-1} = \|f\|_p$ (as $q/p = q - 1$), and $fg = |g|^q$. Hence

$$\int_X fg = \int |g|^q = \|g\|_q^q = \|g\|_q \|f\|_p,$$

and $\|\varphi_g\| = \|g\|_q$.

So for any measure space (X, μ) and $q \in [1, \infty]$,

$$g \mapsto \varphi_g : L^q(\mu) \to L^p(\mu)^* \tag{2}$$

is a linear map that preserves norms (for $p = 1$, $q = \infty$ assume (X, μ) is σ-finite). A norm-preserving linear map is an **isometry**; an isometry is automatically injective.

Exercise 6.1.5

For the bounded function $g(x) = x$ on $[0,1]$, show that there is no $f \in L^1[0,1]$, $f \neq 0$, with $\|f\|_1\|g\|_\infty = \int_0^1 fg$.

Solution [Supplement]. Here $\|g\|_\infty = 1$. For any $f \in L^1[0,1]$,

$$\Big|\int_0^1 f(x)x\,dx\Big| \le \int_0^1 |f(x)|x\,dx \le \int_0^1 |f(x)|\,dx = \|f\|_1\|g\|_\infty.$$

Equality would force $\int_0^1 |f(x)|(1-x)\,dx = 0$, so $|f(x)|(1-x) = 0$ a.e.; since $1-x > 0$ on $[0,1)$, $f = 0$ a.e. So the supremum $\|\varphi_g\| = 1$ is not attained. □

6.1.2 The duals of ℓ^p and c_0

In many cases (2) is surjective; we examine ℓ^p. For $n = 1, 2, \ldots$ define the sequence

$$e_n(m) = \begin{cases} 1, & m = n, \\ 0, & m \neq n. \end{cases} \tag{3}$$

If $\varphi \in (\ell^p)^*$ and $\varphi = \varphi_g$, then

$$\varphi(e_n) = \varphi_g(e_n) = \sum_{i=1}^{\infty} g(i)e_n(i) = g(n), \qquad n \in \mathbb{N}.$$

So for arbitrary $\varphi \in (\ell^p)^*$ define $g(n) = \varphi(e_n)$. Assume $1 \le p < \infty$; it suffices to show $g \in \ell^q$ (and $\varphi = \varphi_g$) to prove (2) is surjective.

$p = 1$, $q = \infty$. If $\varphi \in (\ell^1)^*$, then since $e_n \in \ell^1$, $|\varphi(e_n)| \le \|\varphi\|\|e_n\|_1 = \|\varphi\|$ for each n, so $g \in \ell^\infty$ and $\|g\|_\infty \le \|\varphi\|$.

$1 < p < \infty$. Fix $N \in \mathbb{N}$ and, as in Example 2, put

$$f(n) = \begin{cases} |g(n)|^{q-1} \cdot \operatorname{sgn} g(n), & n \le N, \\ 0, & n > N. \end{cases}$$

Then $\|f\|_p^p = \sum_{n=1}^N |f(n)|^p = \sum_{n=1}^N |g(n)|^q$, so

$$\sum_{n=1}^{N} |g(n)|^q = \sum_{n=1}^{\infty} f(n)g(n) = \varphi(f) \le \|\varphi\|\|f\|_p = \|\varphi\|\Big(\sum_{n=1}^{N} |g(n)|^q\Big)^{1/p}.$$

Thus $\big(\sum_{n=1}^N |g(n)|^q\big)^{1/q} \le \|\varphi\|$ for every N, so $\|g\|_q \le \|\varphi\|$ and $g \in \ell^q$.

Finally, to see $\varphi = \varphi_g$, let A be the vector space of finite linear combinations of the e_n. Since $\varphi(e_n) = g(n) = \varphi_g(e_n)$, φ and φ_g agree on A; A is dense in ℓ^p ($p < \infty$) and both are continuous on ℓ^p, so $\varphi_g = \varphi$.

Theorem 6.1.3

If $1 \le p < \infty$ and p, q are conjugate exponents, then $g \mapsto \varphi_g : \ell^q \to (\ell^p)^*$ is a surjective isometry.

Using the Hahn–Banach theorem proved below, one sees that $g \mapsto \varphi_g$ is not surjective for $p = \infty$, i.e. $(\ell^\infty)^* \supsetneq \ell^1$. Since $c_0 = c_0(\mathbb{N})$ is a subspace of ℓ^∞, restricting bounded linear functionals of ℓ^∞ to c_0 gives bounded linear functionals of c_0, so we may consider the linear map

$$g \mapsto \varphi_g : \ell^1 \to (c_0)^*. \tag{4}$$

Exercise 6.1.6

Show that (4) is a surjective isometry.

Solution [Supplement]. *Isometry.* $|\varphi_g(x)| = |\sum g(n)x(n)| \le \|g\|_1\|x\|_\infty$, so $\|\varphi_g\| \le \|g\|_1$. For $N \in \mathbb{N}$ let $x_N(n) = \operatorname{sgn} g(n)$ for $n \le N$ and 0 for $n > N$. Then $x_N \in c_0$, $\|x_N\|_\infty \le 1$, and $\varphi_g(x_N) = \sum_{n\le N}|g(n)|$. Letting $N \to \infty$ gives $\|\varphi_g\| \ge \|g\|_1$. *Surjectivity.* Given $\varphi \in (c_0)^*$, put $g(n) = \varphi(e_n)$. With x_N defined as above for this g, $\sum_{n\le N}|g(n)| = \varphi(x_N) \le \|\varphi\|$, so $g \in \ell^1$ with $\|g\|_1 \le \|\varphi\|$. Now φ and φ_g agree on finite linear combinations of the e_n, which are dense in c_0 (for $x \in c_0$, $\|x - x\chi_{[1,N]}\|_\infty = \sup_{n>N}|x(n)| \to 0$). Both are continuous, so $\varphi = \varphi_g$. □

Summarizing:

$$(c_0)^* = \ell^1, \qquad (\ell^p)^* = \ell^q \quad (1 \le p < \infty),$$

and therefore in many cases $X = X^{**}$.

6.1.3 The canonical embedding $X \to X^{**}$

For each $x \in X$ define

$$T_x(\varphi) = \varphi(x), \qquad \varphi \in X^*. \tag{5}$$

Clearly T_x is a linear functional on X^* and $\|T_x\| \le \|x\|$, so $T_x \in X^{**}$. To show that $x \mapsto T_x$ is injective we must show

$$x \neq 0 \implies T_x \neq 0,$$

i.e. that for $x \neq 0$ there is $\varphi \in X^*$ with $\varphi(x) \neq 0$. Put differently: is there for $x \neq y$ a bounded linear functional $\varphi \in X^*$ with $\varphi(x) \neq \varphi(y)$ — does X have enough bounded linear functionals? The next theorem answers this affirmatively and is fundamental for the relationship between X and X^*.

Theorem 6.1.4 (Hahn–Banach)

Let X be a normed space and Y a subspace. For every $\psi \in Y^*$ there is $\varphi \in X^*$ with

$$\varphi|_Y = \psi, \qquad \|\varphi\| = \|\psi\|.$$

Proof. Although everything so far was over complex vector spaces, it is convenient to prove this first for real vector spaces. So assume X, Y are real and ψ is a real linear functional. If $\psi = 0$ put $\varphi = 0$; so assume $\psi \neq 0$, and after multiplying by a scalar, $\|\psi\| = 1$. The proof has three independent steps. Step 1 is the case where X has dimension one more than Y; it is the heart of the proof. Step 2 uses Step 1 repeatedly, via the axiom of choice. Step 3 extends the real result to complex spaces.

Step 1 (one more dimension). For $x_0 \in X \setminus Y$ consider

$$Y_1 = \{ax_0 + y : a \in \mathbb{R},\ y \in Y\},$$

and ask how to extend ψ to $\psi_1 : Y_1 \to \mathbb{R}$. Since $\psi_1(y) = \psi(y)$ is already fixed, we only need to choose $\psi_1(x_0)$. Any choice gives a linear functional; the point is to get $\|\psi_1\| = 1$, i.e.

$$|a\psi_1(x_0) + \psi(y)| \le \|ax_0 + y\|, \qquad a \in \mathbb{R},\ y \in Y. \tag{6}$$

For $y, y' \in Y$,

$$\psi(y) - \psi(y') = \psi(y - y') \le \|y - y'\| \le \|y - x_0\| + \|y' - x_0\|,$$

so

$$\psi(y) - \|y - x_0\| \le \psi(y') + \|y' - x_0\|, \qquad y, y' \in Y.$$

Hence with

$$A = \sup\{\psi(y) - \|y - x_0\| : y \in Y\}, \qquad B = \inf\{\psi(y') + \|y' - x_0\| : y' \in Y\},$$

we have $A \le B$. Choose any number in $[A, B]$ as $\psi_1(x_0)$. Then for all $y, y' \in Y$

$$\psi(y) - \|y - x_0\| \le \psi_1(x_0) \le \psi(y') + \|y' - x_0\|,$$

which is equivalent to $|\psi_1(x_0) - \psi(y)| \le \|x_0 - y\|$ for all $y \in Y$. This inequality also holds with $-\frac{y}{a} \in Y$ in place of y ($a \neq 0$); multiplying by $|a|$ gives (6) (and (6) is trivial for $a = 0$).

Step 2 (Zorn). Define

$$\mathcal{P} = \{(Y_\iota, \psi_\iota) : \ Y_\iota \text{ a subspace of } X \text{ containing } Y,\ \psi_\iota \in Y_\iota^*,\ \psi_\iota|_Y = \psi,\ \|\psi_\iota\| = 1\}.$$

For $(Y_1, \psi_1), (Y_2, \psi_2) \in \mathcal{P}$ define $(Y_1, \psi_1) \le (Y_2, \psi_2)$ if $Y_1 \subset Y_2$ and $\psi_1 = \psi_2|_{Y_1}$. This makes $\mathcal{P}$ a partially ordered set, so by the axiom of choice (equivalently, the Hausdorff maximality principle) there is a maximal totally ordered subset $\Omega \subset \mathcal{P}$ (maximal meaning: if $\mathcal{N}$ is totally ordered in $\mathcal{P}$ and $\mathcal{N} \supset \Omega$, then $\mathcal{N} = \Omega$). Then

$$Z = \bigcup \{Y_\iota : (Y_\iota, \psi_\iota) \in \Omega\}$$

is a subspace of X, and defining $\rho(x) = \psi_\iota(x)$ when $x \in Y_\iota$ gives a linear functional ρ on Z with $\|\rho\| = 1$ (Exercise 6.1.7). It remains to show $Z = X$. If $Z \subsetneq X$, Step 1 gives $(Z_1, \rho_1) \in \mathcal{P}$ with $Z_1 \supsetneq Z$; then $(Z_1, \rho_1) \notin \Omega$ but $\Omega \cup \{(Z_1, \rho_1)\}$ is totally ordered, contradicting maximality. So $Z = X$, which proves the real case.

Step 3 (complex case). Let X, Y be complex and ψ a complex linear functional. A complex vector space is closed under multiplication by reals, so it may be regarded as a real vector space. Writing $\psi = u + iv$, u is a real linear functional on the real space Y, with $\|u\| \le \|\psi\|$. By the real case there is a real linear functional U on X with $U|_Y = u$ and $\|U\| = \|u\|$. For $x \in X$ define

$$\varphi(x) = U(x) - iU(ix).$$

Clearly $\varphi(x + y) = \varphi(x) + \varphi(y)$, and

$$\varphi(ix) = U(ix) - iU(-x) = i\big[U(x) - iU(ix)\big] = i\varphi(x),$$

so φ is complex linear. For $y \in Y$,

$$\varphi(y) = U(y) - iU(iy) = u(y) - iu(iy),$$

so to show $\varphi|_Y = \psi$ it suffices that $\psi(y) = u(y) - iu(iy)$. For any $\alpha \in \mathbb{C}$ one checks immediately $\alpha = \operatorname{Re}\alpha - i\operatorname{Re}(i\alpha)$; since ψ is complex linear,

$$\psi(y) = \operatorname{Re}\psi(y) - i\operatorname{Re}(i\psi(y)) = u(y) - i\operatorname{Re}(\psi(iy)) = u(y) - iu(iy),$$

so $\varphi|_Y = \psi$.

Finally we show $\|\varphi\| = \|\psi\|$. Since $\|U\| = \|u\| \le \|\psi\|$, it suffices to show $\|\varphi\| \le \|U\|$ (the reverse $\|\varphi\| \ge \|\psi\|$ is clear because φ extends ψ). For $x \in X$ choose $\alpha \in \mathbb{C}$ with $|\alpha| = 1$ and $\alpha\varphi(x) = |\varphi(x)|$. Then $\varphi(\alpha x)$ is real, so

$$|\varphi(x)| = \varphi(\alpha x) = U(\alpha x) \le \|U\|\|\alpha x\| = \|U\|\|x\|.$$

Thus $\|\varphi\| \le \|U\|$, completing the proof. ∎

Exercise 6.1.7

In the proof of the Hahn–Banach theorem, show that Z is a subspace and that ρ is a linear functional on Z with $\|\rho\| = 1$.

Solution [Supplement]. *Z is a subspace.* Let $x, x' \in Z$, say $x \in Y_\iota$, $x' \in Y_\kappa$ with $(Y_\iota, \psi_\iota), (Y_\kappa, \psi_\kappa) \in \Omega$. Since Ω is totally ordered, one contains the other, say $Y_\iota \subset Y_\kappa$; then $ax + bx' \in Y_\kappa \subset Z$. *$\rho$ is well defined.* If $x \in Y_\iota \cap Y_\kappa$ and $Y_\iota \subset Y_\kappa$, then $\psi_\iota = \psi_\kappa|_{Y_\iota}$, so $\psi_\iota(x) = \psi_\kappa(x)$. *Linearity.* With $x, x' \in Y_\kappa$ as above, $\rho(ax + bx') = \psi_\kappa(ax + bx') = a\psi_\kappa(x) + b\psi_\kappa(x') = a\rho(x) + b\rho(x')$. *Norm.* For $x \in Y_\iota$, $|\rho(x)| = |\psi_\iota(x)| \le \|x\|$, so $\|\rho\| \le 1$; and $\rho|_Y = \psi$ with $\|\psi\| = 1$ gives $\|\rho\| \ge 1$. □

For $x \in X$, $x \neq 0$, apply Hahn–Banach to the one-dimensional subspace $Y = \{\alpha x : \alpha \in \mathbb{C}\}$ and the functional $\psi(\alpha x) = \alpha\|x\|$ (which has $\|\psi\| = 1$):

Corollary 6.1.5

If $x \neq 0$ is a point of a normed space X, there is $\varphi \in X^*$ with $\varphi(x) = \|x\|$ and $\|\varphi\| = 1$.

In particular, by Corollary 6.1.5 the norm of T_x defined in (5) equals $\|x\|$, so $x \mapsto T_x$ is an isometry from X into X^{**}. Hence a normed space X may always be regarded as a subspace of X^{**}. If $x \mapsto T_x : X \to X^{**}$ is surjective, X is called **reflexive**. The Banach spaces ℓ^p $(1 < p < \infty)$ are typical reflexive spaces. However $(c_0)^{**} = \ell^\infty$, so c_0 is not reflexive.

The next corollary is useful for deciding whether a given subspace is closed. The closure of a subset Y is written $\overline{Y}$.

Corollary 6.1.6

For a subspace Y of a normed space X and a point $x \in X$, the following are equivalent:

(a) $x \in \overline{Y}$;

(b) if $\varphi \in X^*$ and $\varphi|_Y = 0$, then $\varphi(x) = 0$.

Proof. Every $\varphi \in X^*$ is continuous, so (a)⇒(b) is clear. If $x \notin \overline{Y}$, there is $\delta > 0$ with

$$y \in Y \implies \|x - y\| > \delta.$$

Consider the subspace $Z = \{\alpha x + y : \alpha \in \mathbb{C},\ y \in Y\}$ spanned by x and Y, and define $\psi(\alpha x + y) = \alpha$ (well defined since $x \notin Y$); ψ is a linear functional on Z. For $\alpha x + y \in Z$ with $\alpha \neq 0$,

$$|\psi(\alpha x + y)|\delta = |\alpha|\delta \le |\alpha|\left\|x + \frac{1}{\alpha}y\right\| = \|\alpha x + y\|,$$

so $\psi \in Z^*$ with $\|\psi\| \le \frac{1}{\delta}$. Also $\psi(x) = 1 \neq 0$. Extending ψ to X by Hahn–Banach gives $\varphi \in X^*$ with $\varphi|_Y = 0$ but $\varphi(x) \neq 0$. ■

Exercise 6.1.8

Show that there is $\varphi \in (\ell^\infty)^*$ with $\varphi|_{c_0} = 0$ but $\varphi \neq 0$. Show that $(\ell^\infty)^* \supsetneq \ell^1$, and that the Banach space ℓ^1 is not reflexive.

Solution [Supplement]. c_0 is a closed subspace of ℓ^∞ (Exercise 5.1.9) and the constant sequence $\mathbf{1} \notin c_0$, so $\mathbf{1} \notin \overline{c_0}$. By Corollary 6.1.6 there is $\varphi \in (\ell^\infty)^*$ with $\varphi|_{c_0} = 0$ and $\varphi(\mathbf{1}) \neq 0$. Such φ is not of the form φ_g, $g \in \ell^1$: otherwise $g(n) = \varphi_g(e_n) = \varphi(e_n) = 0$ for all n (as $e_n \in c_0$), so $g = 0$ and $\varphi = 0$. Hence the isometry $g \mapsto \varphi_g : \ell^1 \to (\ell^\infty)^*$ is not surjective: $(\ell^\infty)^* \supsetneq \ell^1$. ℓ^1 not reflexive: $(\ell^1)^* = \ell^\infty$ (Theorem 6.1.3), so $(\ell^1)^{**} = (\ell^\infty)^*$. If $x \mapsto T_x$ were onto, then $\varphi = T_g$ for some $g \in \ell^1$, i.e. $\varphi(h) = T_g(\varphi_h) = \sum_n g(n)h(n)$ for $h \in \ell^\infty$ (identifying h with $\varphi_h \in (\ell^1)^*$); so $\varphi = \varphi_g$, which we just ruled out. □

Exercise 6.1.9

Show that a Banach space X is reflexive if and only if X^* is reflexive. Show that the Banach space ℓ^∞ is not reflexive.

Solution [Supplement]**.** Write $J_X : X \to X^{**}$ and $J_{X^*} : X^* \to X^{***}$ for the canonical isometries.

($\Rightarrow$) Let X be reflexive and $\Phi \in X^{***}$. Put $\varphi = \Phi \circ J_X \in X^*$. For $\xi \in X^{**}$ write $\xi = J_X(x)$; then $\Phi(\xi) = \Phi(J_X x) = \varphi(x) = J_X(x)(\varphi) = \xi(\varphi) = J_{X^*}(\varphi)(\xi)$. So $\Phi = J_{X^*}(\varphi)$ and X^* is reflexive.

($\Leftarrow$) Let X^* be reflexive and suppose $J_X(X) \subsetneq X^{**}$. Since X is complete, $J_X(X)$ is a closed subspace, so by Corollary 6.1.6 there is $\Phi \in X^{***}$ with $\Phi \neq 0$ and $\Phi|_{J_X(X)} = 0$. By reflexivity of X^*, $\Phi = J_{X^*}(\varphi)$ for some $\varphi \in X^*$. Then $\varphi(x) = J_X(x)(\varphi) = \Phi(J_X x) = 0$ for every x, so $\varphi = 0$ and $\Phi = 0$, a contradiction. Finally $\ell^\infty = (\ell^1)^*$ and ℓ^1 is not reflexive (Exercise 6.1.8), so ℓ^∞ is not reflexive. □

6.2 Linear Maps between Banach Spaces

Motivation. Completeness has strong consequences for linear maps, all coming from the Baire category theorem: bijective bounded maps have bounded inverses, closed graphs force boundedness, and pointwise bounded families are uniformly bounded. A striking application is a continuous function whose Fourier series diverges at a point.

So far completeness has played little role. This section shows how completeness matters when the domain or range of a bounded linear map is a Banach space. The following is the key property of complete metric spaces. For a point x of a metric space (X, d) and $r > 0$, write

$$N_X(x, r) = \{y \in X : d(x, y) < r\}$$

for the neighborhood of x, or just $N(x, r)$. When X is a normed space, $N_X(0, r)$ is written X_r.

Theorem 6.2.1 (Baire)

If a sequence $\langle C_n \rangle$ of closed subsets of a complete metric space X has union X, then some C_n contains a nonempty open set.

Proof. Suppose not. Then there is a neighborhood $N(x_1, r_1)$ with $C_1 \cap \overline{N(x_1, r_1)} = \emptyset$ (as $C_1 \neq X$ is closed). Here $\overline{A}$ is the closure of A. By assumption $N(x_1, r_1) \not\subset C_2$, so there is a neighborhood $N(x_2, r_2)$ with

$$\overline{N(x_2, r_2)} \subset N(x_1, r_1) \setminus C_2$$

($N(x_1, r_1) \setminus C_2$ is a nonempty open set). Repeating, we get neighborhoods $\langle N(x_n, r_n) \rangle$ with

$$N(x_{n+1}, r_{n+1}) \subset N(x_n, r_n), \qquad \overline{N(x_n, r_n)} \cap \Big[\bigcup_{k=1}^{n} C_k\Big] = \emptyset$$

for each n. We may also arrange $r_n \to 0$; then $\langle x_n \rangle$ is a Cauchy sequence (for $m > n$, $x_m \in N(x_n, r_n)$). Its limit x lies in every $\overline{N(x_n, r_n)}$, so $x \notin \bigcup_{n=1}^{\infty} C_n = X$, a contradiction. ■

Corollary 6.2.2

In a complete metric space, the intersection of a sequence of dense open sets is nonempty.

Exercise 6.2.1

Prove Corollary 6.2.2.

Solution [Supplement]. Let U_n be dense open sets and $C_n = U_n^c$, closed. Since U_n is dense, C_n contains no nonempty open set. If $\bigcap_n U_n = \emptyset$, then $\bigcup_n C_n = X$, and Baire's theorem gives some C_n containing a nonempty open set — a contradiction. (The same argument applied to the complete space $\overline{N(x,r)}$ shows $\bigcap U_n$ is in fact dense.) □

The core of this section is the open mapping theorem and the uniform boundedness principle; together with Hahn–Banach these are the three basic tools for studying Banach spaces.

6.2.1 The open mapping theorem

Theorem 6.2.3 (open mapping theorem)

Let $T : X \to Y$ be a surjective bounded linear map between Banach spaces. Then there is $\delta > 0$ with $T(X_1) \supset Y_\delta$.

Proof. First, $X = \bigcup_n X_n$ and T is surjective, so $Y = \bigcup_n \overline{T(X_n)}$. By Baire's theorem there are $n \in \mathbb{N}$ and a neighborhood $N_Y(y_0, r)$ with $N_Y(y_0, r) \subset \overline{T(X_n)}$. If $\|y\| < r$, then $y + y_0 \in N_Y(y_0, r)$, so $y + y_0 \in \overline{T(X_n)}$. Choosing x_0 with $T(x_0) = y_0$,

$$y \in \overline{T(X_n)} - T(x_0) \subset \overline{T(X_{n+\|x_0\|})}.$$

Putting $\delta = \dfrac{r}{n + \|x_0\|}$ we get $Y_\delta \subset \overline{T(X_1)}$, and by homogeneity

$$Y_s \subset \overline{T(X_{s/\delta})}, \qquad s \in (0, \infty). \tag{7}$$

We now use (7) to show $Y_\delta \subset T(X_1)$ up to a factor. Fix $\epsilon \in (0,1)$. If $y \in Y_1$, by (7) there is $x_1 \in X_{1/\delta}$ with $\|y - T(x_1)\| < \epsilon$. Since $y - T(x_1) \in Y_\epsilon$, (7) with $s = \epsilon$ gives $x_2 \in X_{\epsilon/\delta}$ with $\|y - T(x_1) - T(x_2)\| < \epsilon^2$. Continuing, we obtain $\langle x_n \rangle$ with

$$\|x_n\| < \frac{\epsilon^{n-1}}{\delta}, \qquad \|y - T(x_1) - T(x_2) - \cdots - T(x_n)\| < \epsilon^n.$$

Hence $\sum_{n=1}^{\infty} x_n$ converges (absolutely, in the Banach space X) to some $x \in X$ with $\|x\| \le \frac{1}{\delta(1-\epsilon)}$, and

$$y = \lim_{n\to\infty} T(x_1 + x_2 + \cdots + x_n) = T(x).$$

So $y \in T(X_{1/\delta(1-\epsilon)})$; thus $Y_1 \subset T(X_{1/\delta(1-\epsilon)})$, i.e. $Y_{\delta(1-\epsilon)} \subset T(X_1)$. ■

A map $f : X \to Y$ between metric spaces is an **open map** if $f(U)$ is open in Y for every open $U \subset X$. The next corollary explains the name.

Corollary 6.2.4

A surjective bounded linear map $T : X \to Y$ between Banach spaces is always an open map.

Exercise 6.2.2

Prove Corollary 6.2.4.

Solution [Supplement]. Let $U \subset X$ be open and $y = T(x) \in T(U)$ with $x \in U$. Choose $r > 0$ with $N_X(x, r) = x + X_r \subset U$. By Theorem 6.2.3, $T(X_r) = rT(X_1) \supset rY_\delta = Y_{r\delta}$. Hence

$$N_Y(y, r\delta) = y + Y_{r\delta} \subset T(x) + T(X_r) = T(x + X_r) \subset T(U),$$

so $T(U)$ is open. □

Corollary 6.2.5

If a bounded linear map $T : X \to Y$ between Banach spaces is bijective, then its inverse $T^{-1} : Y \to X$ is also bounded.

Proof. Choose $\delta > 0$ with $Y_\delta \subset T(X_1)$. Then $T^{-1}(Y_\delta) \subset X_1$, so $\|T^{-1}\| \le \frac{1}{\delta}$. ■

6.2.2 The Fourier transform $L^1(\mathbb{T}) \to c_0(\mathbb{Z})$ is not onto

For $n = 1, 2, \dots$ let

$$D_n(t) = \sum_{k=-n}^{n} u_k(t) = \frac{\sin(n + \frac{1}{2})t}{\sin \frac{1}{2}t}, \qquad t \in \mathbb{T}. \tag{8}$$

$\langle D_n \rangle$ is the **Dirichlet kernel**.

Exercise 6.2.3

Prove the second equality in (8).

Solution [Supplement]**.** For $t \neq 0 \pmod{2\pi}$, the geometric sum gives

$$\sum_{k=-n}^{n} e^{ikt} = e^{-int}\frac{e^{i(2n+1)t} - 1}{e^{it} - 1} = \frac{e^{i(n+1)t} - e^{-int}}{e^{it} - 1} = \frac{e^{i(n+\frac{1}{2})t} - e^{-i(n+\frac{1}{2})t}}{e^{it/2} - e^{-it/2}} = \frac{\sin(n + \frac{1}{2})t}{\sin \frac{1}{2}t},$$

after multiplying numerator and denominator by $e^{-it/2}$. At $t = 0$ both sides equal $2n + 1$ (as a limit). □

Then $\widehat{D_n} = \sum_{k=-n}^{n} \chi_{\{k\}}$, so $\|\widehat{D_n}\|_\infty = 1$. On the other hand, since $|\sin t| \le |t|$,

$$\begin{aligned}\|D_n\|_1 &\ge \frac{1}{2\pi}\int_{-\pi}^{\pi} \Big|\frac{\sin(n + \frac{1}{2})t}{\frac{t}{2}}\Big| dt = \frac{2}{\pi}\int_0^{\pi} \frac{1}{t}\Big|\sin\Big(n + \frac{1}{2}\Big)t\Big| dt = \frac{2}{\pi}\int_0^{(n+\frac{1}{2})\pi} \frac{1}{t}|\sin t|\, dt \\ &\ge \frac{2}{\pi}\sum_{k=1}^{n} \frac{1}{k\pi}\int_{(k-1)\pi}^{k\pi} |\sin t|\, dt = \frac{4}{\pi^2}\sum_{k=1}^{n}\frac{1}{k},\end{aligned}$$

so $\lim_{n\to\infty}\|D_n\|_1 = \infty$. With $E_n = D_n/\|D_n\|_1$ we have $\|E_n\|_1 = 1$ but $\|\widehat{E_n}\|_\infty \to 0$. So the Fourier transform $f \mapsto \widehat{f}$ cannot satisfy the conclusion of the open mapping theorem (no $\delta > 0$ with $\|\widehat{f}\|_\infty \ge \delta\|f\|_1$, which Corollary 6.2.5 would give since $f \mapsto \widehat{f}$ is injective). Hence *the Fourier transform $L^1(\mathbb{T}) \to c_0(\mathbb{Z})$ is not surjective.*

6.2.3 The closed graph theorem

Another application of the open mapping theorem is the closed graph theorem, which is essential for proving that a given linear map is bounded. The hypothesis says that the graph of T is closed.

Theorem 6.2.6 (closed graph theorem)

A linear map $T : X \to Y$ between Banach spaces satisfying

$$x_n \in X,\ \lim_{n\to\infty} x_n = x,\ \lim_{n\to\infty} T(x_n) = y \implies T(x) = y$$

is bounded.

Proof. For $x \in X$ put $\|x\|' = \|x\| + \|T(x)\|$; then $x \mapsto \|x\|'$ is a norm on X. If $\langle x_n \rangle$ is Cauchy for $(X, \|\cdot\|')$, then $\langle x_n \rangle$ is Cauchy in the Banach space $(X, \|\cdot\|)$ and $\langle T(x_n) \rangle$ is Cauchy in the Banach space Y. So there are $x \in X$, $y \in Y$ with

$$\lim_{n\to\infty} \|x_n - x\| = 0, \qquad \lim_{n\to\infty} \|T(x_n) - y\| = 0.$$

By hypothesis $T(x) = y$, so

$$\lim_{n\to\infty} \|x_n - x\|' = \lim_{n\to\infty} \|x_n - x\| + \lim_{n\to\infty} \|T(x_n) - T(x)\| = 0,$$

and $(X, \|\cdot\|')$ is a Banach space. Since $\|x\| \le \|x\|'$, the identity $\iota : (X, \|\cdot\|') \to (X, \|\cdot\|)$ is a bounded bijection, so by Corollary 6.2.5 its inverse $\iota^{-1} : (X, \|\cdot\|) \to (X, \|\cdot\|')$ is bounded. Hence there is $M > 0$ with $\|x\|' \le M\|x\|$, and

$$\|Tx\| + \|x\| = \|x\|' \le M\|x\|, \qquad x \in X,$$

so $\|Tx\| \le (M-1)\|x\|$. ■

Exercise 6.2.4

In the proof of the closed graph theorem, show that $x \mapsto \|x\|'$ is a norm.

Solution [Supplement]. $\|x\|' \ge 0$, and $\|x\|' = 0$ forces $\|x\| = 0$, i.e. $x = 0$. By linearity of T, $\|ax\|' = |a|\|x\| + |a|\|Tx\| = |a|\|x\|'$ and

$$\|x + z\|' = \|x + z\| + \|Tx + Tz\| \le \|x\| + \|z\| + \|Tx\| + \|Tz\| = \|x\|' + \|z\|'.$$

□

6.2.4 The uniform boundedness principle

Another application of Baire's theorem is the uniform boundedness principle, also called the Banach–Steinhaus theorem.

Theorem 6.2.7 (uniform boundedness principle)

Let $\{T_\alpha : \alpha \in A\}$ be a family of bounded linear maps from a Banach space X to a normed space Y. If for every $x \in X$

$$M(x) = \sup\{\|T_\alpha(x)\| : \alpha \in A\} < \infty,$$

then $\sup\{\|T_\alpha\| : \alpha \in A\} < \infty$.

Proof. For $n = 1, 2, \dots$ put

$$C_n = \bigcap_{\alpha \in A} \{x \in X : \|T_\alpha(x)\| \le n\}.$$

Each $x \mapsto \|T_\alpha(x)\|$ is continuous, so C_n is closed, and by hypothesis $X = \bigcup_n C_n$. By Baire's theorem some C_n contains a neighborhood $N(x_0, r)$, i.e.

$$\alpha \in A,\ \|x - x_0\| < r \implies \|T_\alpha(x)\| \le n.$$

If $\|x\| < r$, then $\|(x + x_0) - x_0\| < r$, so for every $\alpha \in A$

$$\|T_\alpha(x)\| - \|T_\alpha(x_0)\| \le \|T_\alpha(x + x_0)\| \le n,$$

hence $\|T_\alpha(x)\| \le n + \|T_\alpha(x_0)\| \le n + M(x_0)$. Therefore

$$\sup_{\alpha\in A}\|T_\alpha\| \le \frac{1}{r}(n + M(x_0)).$$

■

The next corollary is very useful for defining new bounded linear maps from a given sequence.

Corollary 6.2.8

Let $\langle T_n\rangle$ be a sequence of bounded linear maps from a Banach space X to a normed space Y such that $\langle T_n(x)\rangle$ converges in Y for every $x \in X$. Then $T(x) = \lim_{n\to\infty} T_n(x)$ defines a bounded linear map $T : X \to Y$.

Proof. T is clearly linear, and $\{T_n\}$ satisfies the hypothesis of the uniform boundedness principle (convergent sequences are bounded). So with $M = \sup_n\|T_n\| < \infty$, for each $x \in X$

$$\|T(x)\| = \lim_{n\to\infty}\|T_n(x)\| \le \sup_n\|T_n\|\|x\| = M\|x\|,$$

so $\|T\| \le M$. ■

6.2.5 A continuous function whose Fourier series diverges

Using the uniform boundedness principle we show that the Fourier series of a continuous periodic function need not converge pointwise. By (2) of §5.1 the partial sums of the Fourier series are

$$\sum_{k=-n}^{n}\widehat{f}(k)u_k = \sum_{k=-n}^{n} u_k * f = D_n * f.$$

Fix $t_0 \in \mathbb{T}$ and n, and consider the linear functional on the Banach space $C(\mathbb{T})$

$$T_n : f \mapsto (D_n * f)(t_0), \qquad f \in C(\mathbb{T}).$$

Then

$$|T_n(f)| \le \int_{\mathbb{T}}|f(t)D_n(t_0 - t)|\,dt \le \|f\|_\infty\|D_n\|_1, \qquad f \in C(\mathbb{T}),$$

so $\|T_n\| \le \|D_n\|_1$; in fact $\|T_n\| = \|D_n\|_1$. The function $t \mapsto \operatorname{sgn} D_n(t_0 - t)$ has only finitely many discontinuities, so we can find continuous functions $\langle h_i\rangle$ with $\|h_i\|_\infty \le 1$ converging pointwise to it (except at those finitely many points). By the Lebesgue convergence theorem

$$T_n(h_i) = \int_{\mathbb{T}} h_i(t)D_n(t_0 - t)\,dt \to \int_{\mathbb{T}}|D_n(t_0 - t)|\,dt = \|D_n\|_1,$$

so $\|T_n\| = \|D_n\|_1$. If the Fourier series of every continuous function converged at t_0, then $\langle T_n(f)\rangle$ would converge to $f(t_0)$ for each $f \in C(\mathbb{T})$, and the uniform boundedness principle would make $\langle\|T_n\|\rangle$ bounded — but we know $\|D_n\|_1 \to \infty$. Hence *there is a continuous function whose Fourier series does not converge pointwise.*

6.3 Hilbert Spaces

Motivation. Inner products bring Euclidean geometry (angles, orthogonal projections, Pythagoras) to infinite dimensions. In a Hilbert space every functional is an inner product, every vector expands in an orthonormal basis, and for $L^2(\mathbb{T})$ this is exactly the theory of Fourier series.

6.3.1 Inner products

Definition (inner product)

For a complex vector space X, a function $(x,y)\mapsto\langle x,y\rangle : X\times X\to\mathbb{C}$ satisfying

(I1) $x\mapsto\langle x,y\rangle : X\to\mathbb{C}$ is a linear functional for each $y\in X$;

(I2) $\langle x,y\rangle=\overline{\langle y,x\rangle}$ for all $x,y\in X$;

(I3) $\langle x,x\rangle\geq 0$ for all $x\in X$;

(I4) $x=0 \iff \langle x,x\rangle=0$

is an **inner product**, and a vector space with an inner product is an **inner product space**.

Example 1

On $L^2(X,\mu)$,

$$\langle f,g\rangle=\int_X f\overline{g}\,d\mu,\qquad f,g\in L^2(X,\mu),$$

is an inner product.

For x,y in an inner product space X, choose $\alpha\in\mathbb{C}$ with $|\alpha|=1$ and $\alpha\langle y,x\rangle=|\langle x,y\rangle|$. Then for every real t,

$$0\leq\langle x-t\alpha y,x-t\alpha y\rangle=\langle x,x\rangle-t\alpha\langle y,x\rangle-t\overline{\alpha}\langle x,y\rangle+t^2\langle y,y\rangle=\langle x,x\rangle-2t|\langle x,y\rangle|+t^2\langle y,y\rangle.$$

A nonnegative real quadratic in t has nonpositive discriminant, so

$$|\langle x,y\rangle|^2\leq\langle x,x\rangle\langle y,y\rangle,\qquad x,y\in X.$$

This is the **Schwarz inequality**; for $L^2(X,\mu)$ it is a special case of Hölder's inequality. For $x\in X$ put

$$\|x\|=\langle x,x\rangle^{1/2};\tag{9}$$

$x\mapsto\|x\|$ is easily seen to be a norm. For instance the triangle inequality $\|x+y\|\leq\|x\|+\|y\|$ follows from Schwarz: $\|x+y\|^2=\|x\|^2+2\,\mathrm{Re}\langle x,y\rangle+\|y\|^2\leq(\|x\|+\|y\|)^2$. Hence every inner product space is a normed space. A complete inner product space is a **Hilbert space**. Conversely the norm determines the inner product via the **polarization identity**

$$\langle x,y\rangle=\frac{1}{4}\sum_{k=0}^{3}i^k\|x+i^ky\|^2,\tag{10}$$

and the norm of an inner product space satisfies the **parallelogram law**

$$\|x+y\|^2+\|x-y\|^2=2\|x\|^2+2\|y\|^2.\tag{11}$$

Exercise 6.3.1

Prove (10) and (11) for vectors x,y of an inner product space X.

Solution [Supplement]. For $c\in\mathbb{C}$ with $|c|=1$,

$$\|x+cy\|^2=\|x\|^2+\overline{c}\langle x,y\rangle+c\langle y,x\rangle+\|y\|^2.$$

(11): add the cases $c=1$ and $c=-1$; the cross terms cancel. (10): multiply the case $c=i^k$ by i^k and sum over $k=0,1,2,3$:

$$\sum_k i^k\|x+i^ky\|^2=\Big(\sum_k i^k\Big)(\|x\|^2+\|y\|^2)+\Big(\sum_k i^k\overline{i^k}\Big)\langle x,y\rangle+\Big(\sum_k i^{2k}\Big)\langle y,x\rangle=0+4\langle x,y\rangle+0,$$

since $\sum_{k=0}^{3}i^k=0$, $i^k\overline{i^k}=1$, and $\sum_k(-1)^k=0$. □

6.3.2 The dual of a Hilbert space

We now determine the dual of a Hilbert space H. By the Schwarz inequality, for each $y \in H$

$$\phi_y : x \mapsto \langle x, y\rangle, \qquad x \in H,$$

is a bounded linear functional with $\|\phi_y\| = \|y\|$ (take $x = y$ for equality). Hence

$$y \mapsto \phi_y : H \to H^* \tag{12}$$

is an isometric injection. Note that this map is not linear: it preserves sums, but $\phi_{\alpha y} = \overline{\alpha}\phi_y$ for $\alpha \in \mathbb{C}$. Such a map is called **conjugate linear**.

We want to show (12) is surjective: for every $\phi \in H^*$ find $y \in H$ with $\phi(x) = \langle x, y\rangle$. Intuitively, it suffices to find a vector perpendicular to the subspace $\ker\phi$; the next theorem guarantees such vectors exist.

Theorem 6.3.1 (closest point)

Let C be a closed convex subset of a Hilbert space H and $x_0 \in H \setminus C$. Put

$$d = \inf\{\|x_0 - y\| : y \in C\}.$$

Then there is a unique $y_0 \in C$ with $\|x_0 - y_0\| = d$.

Proof. We may assume $x_0 = 0$ (translate C by $-x_0$). Replacing x, y by $\frac{1}{2}x, \frac{1}{2}y$ in the parallelogram law gives

$$\frac{1}{4}\|x - y\|^2 = \frac{1}{2}\|x\|^2 + \frac{1}{2}\|y\|^2 - \left\|\frac{x+y}{2}\right\|^2.$$

If $x, y \in C$ then $\frac{x+y}{2} \in C$ by convexity, so $\|\frac{x+y}{2}\| \geq d$ and

$$\|x - y\|^2 \leq 2\|x\|^2 + 2\|y\|^2 - 4d^2, \qquad x, y \in C. \tag{13}$$

Uniqueness. If $x, y \in C$ and $\|x\| = \|y\| = d$, (13) gives $x = y$. *Existence.* Take a sequence $\langle y_n\rangle$ in C with $\|y_n\| \to d$. By (13),

$$\|y_n - y_m\|^2 \leq 2\|y_n\|^2 + 2\|y_m\|^2 - 4d^2 \to 0,$$

so $\langle y_n\rangle$ is Cauchy. As H is complete and C closed, $y_0 = \lim_n y_n \in C$, and $\|y_0\| = \lim_n \|y_n\| = d$. ■

Now let E be a closed subspace of H and $x_0 \in H \setminus E$. By Theorem 6.3.1 there is $y_0 \in E$ with

$$y \in E \implies \|x_0 - y_0\| \leq \|x_0 - y\|.$$

Put $z_0 = x_0 - y_0$. For every $y \in E$ and $\alpha \in \mathbb{C}$,

$$\langle z_0, z_0\rangle = \|x_0 - y_0\|^2 \leq \|x_0 - y_0 - \alpha y\|^2 = \langle z_0 - \alpha y, z_0 - \alpha y\rangle,$$

so

$$0 \leq -\alpha\langle y, z_0\rangle - \overline{\alpha}\langle z_0, y\rangle + |\alpha|^2\langle y, y\rangle.$$

Substituting $\alpha = t\langle z_0, y\rangle$ $(t \in \mathbb{R})$,

$$0 \leq -2t|\langle y, z_0\rangle|^2 + t^2|\langle y, z_0\rangle|^2\langle y, y\rangle$$

for every $t \in \mathbb{R}$; taking small $t > 0$ forces $\langle y, z_0\rangle = 0$ for every $y \in E$. Two vectors with $\langle x, y\rangle = 0$ are **orthogonal**, written $x \perp y$; for a subset E, $E^\perp$ denotes the set of vectors orthogonal to every vector of E. Since $z_0 \neq 0$ (as $x_0 \notin E$) and $z_0 \in E^\perp$:

Corollary 6.3.2

If E is a closed proper subspace of a Hilbert space H, then $E^\perp \supsetneq \{0\}$.

Return to $\phi \in H^*$. If $\phi \neq 0$, apply Corollary 6.3.2 to the closed proper subspace $\ker\phi$ to get $z \in (\ker\phi)^\perp$ with $\|z\| = 1$. Since $\phi(x)z - \phi(z)x \in \ker\phi$, for every $x \in H$

$$0 = \langle \phi(x)z - \phi(z)x, z\rangle = \phi(x) - \phi(z)\langle x, z\rangle = \phi(x) - \langle x, \overline{\phi(z)}z\rangle.$$

So with $y = \overline{\phi(z)}z$,

$$\phi(x) = \langle x, y\rangle, \qquad x \in H.$$

This proves:

Theorem 6.3.3 (Riesz representation)

For a Hilbert space H, the map $y \mapsto \phi_y$ defined by (12) is a surjective, isometric, conjugate linear map $H \to H^*$.

Exercise 6.3.2

Show that every Hilbert space is reflexive.

Solution [Supplement]**.** By Theorem 6.3.3, H^* is itself a Hilbert space with inner product $\langle \phi_y, \phi_z\rangle_{H^*} = \langle z, y\rangle$ (this is an inner product inducing the dual norm, since $y \mapsto \phi_y$ is a conjugate-linear isometric bijection). Let $\Lambda \in H^{**}$. By Theorem 6.3.3 applied to H^*, there is $\phi_z \in H^*$ with $\Lambda(\phi) = \langle \phi, \phi_z\rangle_{H^*}$ for all $\phi \in H^*$. Then for every y,

$$\Lambda(\phi_y) = \langle \phi_y, \phi_z\rangle_{H^*} = \langle z, y\rangle = \phi_y(z) = T_z(\phi_y).$$

Since every element of H^* is some ϕ_y, $\Lambda = T_z$. So $x \mapsto T_x$ is onto. □

Exercise 6.3.3

Show that $E^\perp$ is a closed subspace for any subset E of a Hilbert space. Show that $\overline{E} = E^{\perp\perp}$ for any subspace E.

Solution [Supplement]**.** $E^\perp = \bigcap_{y\in E} \ker\phi_y$ is an intersection of kernels of bounded linear functionals, hence a closed subspace. For a subspace E: every $x \in E$ is orthogonal to $E^\perp$, so $E \subset E^{\perp\perp}$, and since $E^{\perp\perp}$ is closed, $\overline{E} \subset E^{\perp\perp}$. Conversely let $x \in E^{\perp\perp}$. $\overline{E}$ is a closed subspace; if $x \notin \overline{E}$, the closest-point argument gives $y_0 \in \overline{E}$ with $z_0 = x - y_0 \neq 0$ and $z_0 \perp \overline{E}$, so $z_0 \in E^\perp$. Since $x \in E^{\perp\perp}$ and $y_0 \in \overline{E} \subset E^{\perp\perp}$, also $z_0 \in E^{\perp\perp}$. So $z_0 \perp z_0$, i.e. $z_0 = 0$ — a contradiction. Hence $E^{\perp\perp} = \overline{E}$. □

6.3.3 Orthonormal sets

A subset $\{u_\alpha : \alpha \in A\}$ of an inner product space H satisfying

$$\langle u_\alpha, u_\beta\rangle = \begin{cases} 1, & \alpha = \beta, \\ 0, & \alpha \neq \beta \end{cases}$$

is an **orthonormal set**; the subspace it spans (finite linear combinations) is written P_A.

Proposition 6.3.4

Let $\{u_\alpha : \alpha \in F\}$ be a finite subset of an orthonormal set $\{u_\alpha : \alpha \in A\}$ in an inner product space H. For fixed $x \in H$,

$$\Big\|x - \sum_{\alpha\in F} \langle x, u_\alpha\rangle u_\alpha\Big\| \le \|x - y\|, \qquad y \in P_F, \tag{14}$$

with equality only for $y = \sum_{\alpha\in F}\langle x, u_\alpha\rangle u_\alpha$. Moreover

$$\sum_{\alpha\in F} |\langle x, u_\alpha\rangle|^2 \le \|x\|^2, \qquad x \in H.$$

Proof. Write $z = \sum_{\alpha\in F}\langle x, u_\alpha\rangle u_\alpha$. For each $\alpha \in F$, $\langle x - z, u_\alpha\rangle = \langle x, u_\alpha\rangle - \langle x, u_\alpha\rangle = 0$. If $y \in P_F$ then $y - z \in P_F$, so $\langle x - z, y - z\rangle = 0$, and

$$\|x - y\|^2 = \|x - z\|^2 + \|z - y\|^2, \qquad y \in P_F,$$

which proves the first part (with equality iff $y = z$). Also $\|z\|^2 = \sum_{\alpha\in F}|\langle x, u_\alpha\rangle|^2$, so putting $y = 0$ gives $\|x\|^2 = \|x - z\|^2 + \|z\|^2 \ge \|z\|^2$, the second inequality. ∎

Inequality (14) says that the coefficients c_α making $\sum_\alpha c_\alpha u_\alpha$ closest to x are $c_\alpha = \langle x, u_\alpha\rangle$. Since the second inequality holds for every finite $F \subset A$, Fact B.28 gives:

Corollary 6.3.5 (Bessel's inequality)

For any orthonormal set $\{u_\alpha : \alpha \in A\}$ in an inner product space H,

$$\sum_{\alpha\in A} |\langle x, u_\alpha\rangle|^2 \le \|x\|^2, \qquad x \in H.$$

We want to know for which orthonormal sets Bessel's inequality is an equality. In finite dimensions the orthonormal set must be a basis; we explain the analogue for general Hilbert spaces.

Theorem 6.3.6

For an orthonormal set $\mathcal{U} = \{u_\alpha : \alpha \in A\}$ in a Hilbert space H the following are equivalent:

(a) for every $x \in H$,
$$\sum_{\alpha\in A} |\langle x, u_\alpha\rangle|^2 = \|x\|^2; \tag{15}$$

(b) for all $x, y \in H$,
$$\sum_{\alpha\in A} \langle x, u_\alpha\rangle\overline{\langle y, u_\alpha\rangle} = \langle x, y\rangle; \tag{16}$$

(c) if $\mathcal{W} \supset \mathcal{U}$ is an orthonormal set, then $\mathcal{W} = \mathcal{U}$ (maximality);

(d) P_A is dense in H.

Proof. For each $x \in H$ define $\widehat{x} : A \to \mathbb{C}$ by

$$\widehat{x}(\alpha) = \langle x, u_\alpha\rangle, \qquad \alpha \in A.$$

By Bessel's inequality $\widehat{x} \in \ell^2(A)$, and we get a bounded linear map

$$\Phi : x \mapsto \widehat{x} : H \to \ell^2(A). \tag{17}$$

(a) says Φ preserves norms; such a map preserves inner products by the polarization identity (10). The left side of (16) is exactly $\langle \widehat{x}, \widehat{y}\rangle$, so (a)⇒(b).

(b)⇒(c): if $\mathcal{W} \supsetneq \mathcal{U}$ is orthonormal, pick $u \in \mathcal{W} \setminus \mathcal{U}$; then (16) with $x = y = u$ reads $0 = \sum_\alpha |\langle u, u_\alpha\rangle|^2 = \|u\|^2 = 1$, a contradiction.

(c)⇒(d): if P_A is not dense in H, Corollary 6.3.2 gives a unit vector $u \in (P_A)^\perp = (\overline{P_A})^\perp$, and $\mathcal{U} \cup \{u\}$ is a larger orthonormal set, so (c) fails.

(d)⇒(a): For $x \in P_A$ (a finite linear combination of the u_α) (15) holds, so Φ is an isometry on the dense subspace P_A. For arbitrary $x \in H$ choose $x_n \in P_A$ with $x_n \to x$. Since Φ is bounded,

$$\sum_{\alpha\in A} |\langle x, u_\alpha\rangle|^2 = \|\widehat{x}\|_2^2 = \lim_{n\to\infty} \|\widehat{x_n}\|_2^2 = \lim_{n\to\infty} \|x_n\|^2 = \|x\|^2.$$

■

Identity (16) is **Parseval's identity**. An orthonormal set $\{u_\alpha : \alpha \in A\}$ satisfying the conditions of Theorem 6.3.6 is an **orthonormal basis** of H. Given an orthonormal basis, the linear map Φ of (17) preserves inner products. A linear map between Hilbert spaces preserving inner products is a **Hilbert space isomorphism**. Let us show Φ is surjective. Given $y \in \ell^2(A)$, choose finitely supported $y_n \in \ell^2(A)$ with $\|y_n - y\|_2 \to 0$. Then $\Phi(x_n) = y_n$ for some $x_n \in P_A$, and $\|x_n - x_m\| = \|y_n - y_m\|_2$, so $\langle x_n\rangle$ is Cauchy in H. If $x \in H$ is its limit, $\Phi(x) = \lim_n \Phi(x_n) = \lim_n y_n = y$. Hence:

Theorem 6.3.7

If $\{u_\alpha : \alpha \in A\}$ is an orthonormal basis of a Hilbert space H, then $x \mapsto \widehat{x}$ is a bijective Hilbert space isomorphism $H \to \ell^2(A)$.

Thus the structure of a Hilbert space is completely determined by the cardinality of an orthonormal basis. Of course, which orthonormal set is convenient depends on the situation.

6.3.4 Fourier series in $L^2(\mathbb{T})$

The set $\{u_n : n \in \mathbb{Z}\}$ of §5.1 (1) is clearly an orthonormal set in $L^2(\mathbb{T})$, and $\widehat{f}(n) = \langle f, u_n\rangle$ for $f \in L^2(\mathbb{T})$. Corollary 5.1.13 says that Theorem 6.3.6(d) holds in this case, so $\{u_n : n \in \mathbb{Z}\}$ is an orthonormal basis of $L^2(\mathbb{T})$. Writing out the identities:

$$\sum_{n=-\infty}^{\infty} \widehat{f}(n)\overline{\widehat{g}(n)} = \frac{1}{2\pi}\int_{-\pi}^{\pi} f(t)\overline{g(t)}\,dt, \qquad f, g \in L^2(\mathbb{T}),$$

also called **Parseval's identity** (the series converges absolutely for $f, g \in L^2(\mathbb{T})$, so the limit is unambiguous). Moreover

$$\lim_{n\to\infty} \Big\| f - \sum_{k=-n}^{n} \widehat{f}(k)u_k \Big\|_2 = 0, \qquad f \in L^2(\mathbb{T}). \tag{18}$$

(The partial sums of the Fourier series converge in $\|\cdot\|_2$, so by Fact B.18 some subsequence converges almost everywhere. In fact, in the 1960s it was proved that the Fourier series of an L^2 periodic function converges almost everywhere (Carleson). On the other hand, there is an L^1 periodic function whose Fourier series diverges everywhere (Kolmogorov).)

In Theorem 6.3.6 completeness of the Hilbert space played no big role, as the proof shows. But applying Theorem 6.3.7 to Fourier series says that for every $a \in \ell^2(\mathbb{Z})$ there is an L^2-periodic function with Fourier coefficients a; here the completeness of $L^2(\mathbb{T})$ is decisive. Together with Fact B.17 this is called the **Riesz–Fischer theorem**. If $\int_{\mathbb{T}} |f|^2 < \infty$ were understood as an improper Riemann integral, this conclusion would be unobtainable — one of the big reasons the Lebesgue integral is more useful than the Riemann integral.

6.3.5 The Plancherel transform

By Fact B.29, $L^1(\mathbb{T}) \supset L^2(\mathbb{T})$, so restricting the Fourier transform to $L^2(\mathbb{T})$ gives the Hilbert space isomorphism $f \mapsto \widehat{f}$ and Parseval's identity (18). However $L^1(\mathbb{R})$ no longer contains $L^2(\mathbb{R})$, so to get the corresponding theorem we start from $C_c(\mathbb{R}) \subset L^1(\mathbb{R}) \cap L^2(\mathbb{R})$.

If $f \in C_c(\mathbb{R})$ and $\operatorname{supp} f \subset (-\pi, \pi)$, then Parseval for periodic functions and (19) of §5.2 (with $F = f$ on $(-\pi, \pi)$, $\widehat{F}(n) = \frac{1}{2\pi}\widehat{f}(n)$) give

$$\frac{1}{2\pi}\int_{-\pi}^{\pi} |f(t)|^2\, dt = \sum_{n=-\infty}^{\infty} \Big|\frac{1}{2\pi}\widehat{f}(n)\Big|^2.$$

This also holds for $g(x) = f(x)e^{-i\alpha x}$ (with $\widehat{g}(n) = \widehat{f}(n+\alpha)$), so

$$\int_{\mathbb{R}} |f(x)|^2\, dx = \frac{1}{2\pi}\sum_{n=-\infty}^{\infty} |\widehat{f}(n+\alpha)|^2.$$

Integrating both sides over $\alpha \in [0, 1)$,

$$\int_{\mathbb{R}} |f(x)|^2\, dx = \frac{1}{2\pi}\int_0^1 \sum_{n=-\infty}^{\infty} |\widehat{f}(n+\alpha)|^2\, d\alpha = \frac{1}{2\pi}\int_{\mathbb{R}} |\widehat{f}(\alpha)|^2\, d\alpha.$$

In general, if $\operatorname{supp} f \subset (-\pi\lambda, \pi\lambda)$, put $g(x) = \sqrt{\lambda} f(\lambda x)$; then $\operatorname{supp} g \subset (-\pi, \pi)$ and $\widehat{g}(\alpha) = \frac{1}{\sqrt{\lambda}}\widehat{f}(\alpha/\lambda)$, so for every $f \in C_c(\mathbb{R})$

$$\int_{\mathbb{R}} |f(x)|^2\, dx = \int_{\mathbb{R}} |g(x)|^2 = \frac{1}{2\pi}\int_{\mathbb{R}} |\widehat{g}(\alpha)|^2\, d\alpha = \frac{1}{2\pi}\int_{\mathbb{R}} |\widehat{f}(\alpha)|^2\, d\alpha.$$

Let $\widehat{\mathbb{R}}$ denote the real line with $\frac{1}{2\pi}$ times Lebesgue measure. Then

$$\int_{\mathbb{R}} |f(x)|^2\, dx = \int_{\widehat{\mathbb{R}}} |\widehat{f}(\alpha)|^2\, d\alpha, \qquad f \in C_c(\mathbb{R}), \tag{19}$$

and the inversion formula reads

$$f(x) = \int_{\widehat{\mathbb{R}}} \widehat{f}(\alpha) e^{i\alpha x}\, d\alpha.$$

Regarding $C_c(\mathbb{R})$ as a subspace of $L^2(\mathbb{R})$, (19) gives an isometric linear map

$$f \mapsto \widehat{f} : C_c(\mathbb{R}) \to L^2(\widehat{\mathbb{R}}).$$

Since $C_c(\mathbb{R})$ is dense in $L^2(\mathbb{R})$, this map extends to an isometric linear map

$$\mathcal{F} : L^2(\mathbb{R}) \to L^2(\widehat{\mathbb{R}}).$$

Consistency on $L^1 \cap L^2$. If $f \in L^1(\mathbb{R}) \cap L^2(\mathbb{R})$, both $\widehat{f}$ and $\mathcal{F}(f)$ are defined; is $\widehat{f} = \mathcal{F}(f)$? Review the proofs of Fact B.20, Fact B.21 and Fact B.22. If $p \neq q \in [1, \infty)$ and $f \in L^p[a, b] \cap L^q[a, b]$, there is a simple function $s : [a, b] \to \mathbb{C}$ with simultaneously

$$\|f - s\|_p < \epsilon, \qquad \|f - s\|_q < \epsilon.$$

Moreover, choosing in the proof of Fact B.21 a continuous g with

$$\mu(\{x : g(x) \neq s(x)\}) < \min\Big\{\Big(\frac{\epsilon}{2\|s\|_\infty}\Big)^p, \Big(\frac{\epsilon}{2\|s\|_\infty}\Big)^q\Big\}, \qquad |g| \leq \|s\|_\infty,$$

we get simultaneously

$$\|f - g\|_p < 2\epsilon, \qquad \|f - g\|_q < 2\epsilon. \tag{20}$$

Examining the proof of Fact B.22 in the same way, for every $f \in L^p(\mathbb{R}) \cap L^q(\mathbb{R})$ there is $g \in C_c(\mathbb{R})$ with (20). Hence if $f \in L^1(\mathbb{R}) \cap L^2(\mathbb{R})$ there is a sequence $\langle g_n \rangle$ in $C_c(\mathbb{R})$ with

$$\|f - g_n\|_1 \to 0, \qquad \|f - g_n\|_2 \to 0,$$

and then

$$\lim_{n\to\infty} \|\widehat{f} - \widehat{g_n}\|_\infty = \lim_{n\to\infty} \|\mathcal{F}(f) - \mathcal{F}(g_n)\|_2 = 0.$$

Since $\widehat{g_n} = \mathcal{F}(g_n)$, we conclude $\widehat{f} = \mathcal{F}(f)$ (Exercise 6.3.4).

Exercise 6.3.4

If a sequence $\langle h_n \rangle$ in $C_0(\mathbb{R})$ satisfies $\lim_n \|f - h_n\|_\infty = \lim_n \|g - h_n\|_2 = 0$, show that $f = g$ almost everywhere.

Solution [Supplement]. Since $\|g - h_n\|_2 \to 0$, by Fact B.18 a subsequence $h_{n_k} \to g$ almost everywhere. But $h_{n_k} \to f$ uniformly, in particular pointwise everywhere. Hence $f = g$ almost everywhere. □

Surjectivity. By the proof of Theorem 5.2.8, every $h \in C_c^2(\mathbb{R})$ is the Fourier transform of a bounded L^1-function k ($k = \frac{1}{2\pi}\widehat{h}(-\cdot)$, which is bounded and integrable). A bounded L^1-function is an L^2-function, so $h = \widehat{k} = \mathcal{F}(k)$ lies in the range of $\mathcal{F}$. Hence the range of $\mathcal{F}$ is dense in $L^2(\widehat{\mathbb{R}})$; being the isometric image of a complete space it is closed, so $\mathcal{F} : L^2(\mathbb{R}) \to L^2(\widehat{\mathbb{R}})$ is a surjective isometry, the **Plancherel transform**. An isometry between Hilbert spaces is of course a Hilbert space isomorphism (polarization). Summarizing:

Theorem 6.3.8 (Plancherel)

There is a unique Hilbert space isomorphism $\mathcal{F} : L^2(\mathbb{R}) \to L^2(\widehat{\mathbb{R}})$ with $\mathcal{F}(f) = \widehat{f}$ for every $f \in L^1(\mathbb{R}) \cap L^2(\mathbb{R})$.

(Uniqueness: $\mathcal{F}$ is determined on the dense subspace $C_c(\mathbb{R}) \subset L^1 \cap L^2$ and is continuous.)

If $f \in L^2(\mathbb{R})$, then $f_A = \chi_{[-A,A]} f \in L^1(\mathbb{R}) \cap L^2(\mathbb{R})$ and

$$\widehat{f_A}(\alpha) = \int_{-A}^{A} f(x) e^{-i\alpha x}\, dx.$$

Since $\lim_{A\to\infty} \|f - f_A\|_2 = 0$,

$$\lim_{A\to\infty} \|\mathcal{F}(f) - \widehat{f_A}\|_2 = 0,$$

the analogue of (18) for Fourier series. Finally, the statement that the Plancherel transform is a Hilbert space isomorphism reads

$$\int_{\mathbb{R}} f(x)\overline{g(x)}\, dx = \frac{1}{2\pi} \int_{\mathbb{R}} \mathcal{F}f(\alpha)\overline{\mathcal{F}g(\alpha)}\, d\alpha,$$

also called **Parseval's identity**.

Example 2

Applying Parseval to $f(x) = g(x) = e^{-|x|}$ (with $\widehat{f}(\alpha) = \frac{2}{1+\alpha^2}$),

$$\int_{-\infty}^{\infty} (e^{-|x|})^2\, dx = \frac{1}{2\pi} \int_{-\infty}^{\infty} \left[\frac{2}{1+\alpha^2}\right]^2 d\alpha,$$

so

$$\int_{-\infty}^{\infty} \frac{1}{(1+\alpha^2)^2}\, d\alpha = \pi \int_0^{\infty} e^{-2x}\, dx = \frac{\pi}{2}.$$

Exercise 6.3.5

Compute $\displaystyle\int_{-\infty}^{\infty} \frac{\sin a\alpha \, \sin b\alpha}{\alpha^2}\, d\alpha$ for $a, b > 0$.

Solution [Supplement]**.** By Example 1 of §5.2, $\widehat{\chi_{[-a,a]}}(\alpha) = \frac{2\sin a\alpha}{\alpha}$, real-valued. Parseval with $f = \chi_{[-a,a]}$, $g = \chi_{[-b,b]}$ (both in $L^1 \cap L^2$):

$$\int_{\mathbb{R}} \chi_{[-a,a]}\chi_{[-b,b]}\, dx = \frac{1}{2\pi} \int_{\mathbb{R}} \frac{2\sin a\alpha}{\alpha} \cdot \frac{2\sin b\alpha}{\alpha}\, d\alpha.$$

The left side is $2\min\{a, b\}$. Therefore

$$\int_{-\infty}^{\infty} \frac{\sin a\alpha \, \sin b\alpha}{\alpha^2}\, d\alpha = \pi \min\{a, b\}.$$

(For $a = b$ this gives the classical $\int_{\mathbb{R}} \frac{\sin^2 a\alpha}{\alpha^2}\, d\alpha = \pi a$.) □

Chapter 7. Complex Measures

Overview

Results from Chapters 1–3 are cited as *Fact B.k*; they are stated in the separate *Background* notes. So far we have considered positive measures and integrals with respect to them. This chapter allows measures to take negative or complex values. If a positive measure is given on a σ-algebra, every other measure is the sum of a part expressible as an integral with respect to that measure and a part that is not (singular) — this is the Lebesgue–Radon–Nikodym theorem, the core of measure theory. Applying it to Lebesgue measure restricted to the Borel sets, we find all Borel measures on the real line.

7.1 Signed Measures and Complex Measures

Motivation. Charge, mass differences and functionals on $C(X)$ lead to measures that can be negative or complex. The Hahn and Jordan decompositions show that such a measure is just a difference of positive measures, so the positive theory still applies.

7.1.1 Definitions

A function $\lambda : \mathcal{S} \to [0, \infty]$ on a σ-algebra $\mathcal{S}$ of a set X satisfying

$$\{A_n \in \mathcal{S} : n = 1, 2, \dots\} \text{ pairwise disjoint } \implies \lambda\Big(\bigcup_{n=1}^{\infty} A_n\Big) = \sum_{n=1}^{\infty} \lambda(A_n) \tag{1}$$

with $\lambda(\emptyset) = 0$ was called a positive measure.

Definition (signed measure, complex measure)

A function $\lambda : \mathcal{S} \to \mathbb{R}^*$ satisfying (1) and $\lambda(\emptyset) = 0$ is a **signed measure**. A function $\lambda : \mathcal{S} \to \mathbb{C}$ satisfying (1) is a **complex measure**.

In both definitions the convergence of $\sum \lambda(A_n)$ is part of the assumption; for signed measures, convergence to $+\infty$ or $-\infty$ is allowed. If $E, F \in \mathcal{S}$ with $\lambda(E) = \infty$ and $\lambda(F) = -\infty$, then in

$$\lambda(E \cup F) = \lambda(E \setminus F) + \lambda(F \setminus E) + \lambda(E \cap F)$$

both ∞ and $-\infty$ would appear on the right. Hence *a signed measure cannot take both values* $\pm\infty$. Sums and scalar multiples of complex measures are complex measures.

Example 1

For a measurable function h on $\mathbb{R}$ define

$$\lambda(E) = \int_E h(x)\,dx, \qquad E \in \mathcal{M}. \tag{2}$$

If $h \geq 0$, λ is a positive measure by Fact B.7. If $h : \mathbb{R} \to \mathbb{R}^*$ is measurable and at least one of h_+, h_- is integrable, λ defined by (2) is a signed measure. If $h : \mathbb{R} \to \mathbb{C}$ is integrable, λ is a complex measure.

Given a complex measure $\lambda : \mathcal{S} \to \mathbb{C}$, let $\lambda_1(E), \lambda_2(E)$ be the real and imaginary parts of $\lambda(E)$. Then λ_1, λ_2 are signed measures and

$$\lambda = \lambda_1 + i\lambda_2. \tag{3}$$

So we first study signed measures, showing that a signed measure is a difference of positive measures — just as a real function is $f = f_+ - f_-$.

7.1.2 The Hahn decomposition

Definition (positive and negative sets)

Let λ be a signed measure on a σ-algebra $\mathcal{S}$ of X. A set $P \in \mathcal{S}$ is **positive** if

$$E \in \mathcal{S},\ E \subset P \implies \lambda(E) \geq 0,$$

and **negative** if

$$E \in \mathcal{S},\ E \subset P \implies \lambda(E) \leq 0.$$

For example, for the signed measure of Example 1, $\{x : h(x) \geq 0\}$ and $\{x : h(x) < 0\}$ are positive and negative sets. A measurable subset of a positive set is positive. If $\{P_n : n = 1, 2, \dots\}$ are positive sets, their union P is positive: for $E \subset P$, $E \in \mathcal{S}$, put

$$E_n = (E \cap P_n) \setminus (P_1 \cup \cdots \cup P_{n-1}), \qquad n = 1, 2, \dots \quad (P_0 = \emptyset);$$

then $E_n \subset P_n$, so $\lambda(E_n) \geq 0$, and $\lambda(E) = \sum_n \lambda(E_n) \geq 0$.

Theorem 7.1.1 (Hahn decomposition)

If λ is a signed measure on a σ-algebra $(X, \mathcal{S})$, then X splits into a positive set and a negative set. If (P_1, Q_1) and (P_2, Q_2) are two such splittings, then for every $E \in \mathcal{S}$

$$\lambda(E \cap P_1) = \lambda(E \cap P_2), \qquad \lambda(E \cap Q_1) = \lambda(E \cap Q_2).$$

Proof. First assume λ does not take the value $+\infty$. Define

$$\alpha = \sup\{\lambda(P) : P \text{ is a positive set}\};$$

$\emptyset$ is positive, so $\alpha \geq 0$. Choose positive sets $\langle P_n \rangle$ with $\alpha = \lim_n \lambda(P_n)$; their union P is positive. For each n, $P \setminus P_n \subset P$ is positive, so $\lambda(P \setminus P_n) \geq 0$ and

$$\lambda(P_n) \leq \lambda(P_n) + \lambda(P \setminus P_n) = \lambda(P) \leq \alpha.$$

Hence $\lambda(P) = \alpha < \infty$. Now put $Q = X \setminus P$. If a measurable $E \subset Q$ is positive, then $E \cup P$ is positive, so

$$\alpha \geq \lambda(E \cup P) = \lambda(E) + \lambda(P) = \lambda(E) + \alpha \geq \alpha,$$

and therefore $\lambda(E) = 0$. That is,

$$E \in \mathcal{S},\ E \subset Q,\ E \text{ positive} \implies \lambda(E) = 0. \tag{4}$$

It remains to show Q is negative. Take any $E \in \mathcal{S}$ with $E \subset Q$. If E is positive, then $\lambda(E) = 0$ by (4) and we are done. If E is not positive, it has subsets of negative measure; let n_1 be the smallest natural number for which there is $F_1 \subset E$ with $\lambda(F_1) < -\frac{1}{n_1}$, and choose such F_1. If $E \setminus F_1$ is positive, then $\lambda(E \setminus F_1) = 0$ by (4), so $\lambda(E) = \lambda(F_1) < 0$ and we are done. If $E \setminus F_1$ is not positive,

repeat: if $E \setminus (F_1 \cup \cdots \cup F_{k-1})$ is positive, its measure is 0 and $\lambda(E) = \sum_{i=1}^{k-1} \lambda(F_i) < 0$; otherwise choose the smallest natural number n_k and $F_k \in \mathcal{S}$ with

$$F_k \subset E \setminus (F_1 \cup \cdots \cup F_{k-1}), \qquad \lambda(F_k) < -\frac{1}{n_k}.$$

If the process never stops, put

$$F = E \setminus \Big[\bigsqcup_{k=1}^{\infty} F_k \Big].$$

Clearly $\langle n_k \rangle$ is nondecreasing.

If $\langle n_k \rangle$ is bounded, then $\lambda(\bigsqcup_k F_k) = -\infty$, and since $\lambda(F) < \infty$, $\lambda(E) = \lambda(\bigsqcup_k F_k) + \lambda(F) = -\infty$, and we are done.

If $\langle n_k \rangle$ is unbounded, we show F is positive. Suppose F contains $G \in \mathcal{S}$ with $\lambda(G) < 0$. Since $n_k \to \infty$, we can choose k with $\frac{1}{n_k - 1} < -\lambda(G)$. Then

$$G \subset F \subset E \setminus (F_1 \cup \cdots \cup F_{k-1}), \qquad \lambda(G) < -\frac{1}{n_k - 1},$$

contradicting the minimality in the choice of n_k. So F is positive, $\lambda(F) = 0$ by (4), and $\lambda(E) = \sum_{k=1}^{\infty} \lambda(F_k) < 0$. Hence Q is negative.

The case where λ does not take $-\infty$ is proved in the same way (apply the above to $-\lambda$).

For the second part, see Exercise 7.1.1. ■

Exercise 7.1.1

Prove the second part of the Hahn decomposition theorem.

Solution [Supplement]**.** Let $E \in \mathcal{S}$. The set $E \cap (P_1 \setminus P_2) = E \cap P_1 \cap Q_2$ is a subset of the positive set P_1 and of the negative set Q_2, so its measure is both ≥ 0 and ≤ 0, i.e. 0. Likewise $\lambda(E \cap (P_2 \setminus P_1)) = 0$. Hence

$$\lambda(E \cap P_1) = \lambda(E \cap P_1 \cap P_2) + \lambda(E \cap P_1 \cap Q_2) = \lambda(E \cap P_1 \cap P_2) = \lambda(E \cap P_2).$$

The same argument with Q_1, Q_2 gives $\lambda(E \cap Q_1) = \lambda(E \cap Q_2)$. □

7.1.3 Jordan decomposition and total variation

Given a Hahn decomposition $X = P \sqcup Q$ for λ, define for $E \in \mathcal{S}$

$$\lambda_+(E) = \lambda(E \cap P), \qquad \lambda_-(E) = -\lambda(E \cap Q).$$

By Theorem 7.1.1 these do not depend on the decomposition. λ_+ and λ_- are positive measures and

$$\lambda(E) = \lambda_+(E) - \lambda_-(E), \qquad E \in \mathcal{S}.$$

Also define

$$|\lambda|(E) = \lambda_+(E) + \lambda_-(E);$$

$|\lambda|$ is a positive measure as well.

Exercise 7.1.2

If λ is defined by (2), show that for every $E \in \mathcal{M}$

$$\lambda_+(E) = \int_E h_+, \qquad \lambda_-(E) = \int_E h_-, \qquad |\lambda|(E) = \int_E |h|.$$

Solution [Supplement]. $P = \{h \geq 0\}$, $Q = \{h < 0\}$ is a Hahn decomposition: for $E \subset P$, $\lambda(E) = \int_E h \geq 0$, and for $E \subset Q$, $\lambda(E) \leq 0$. Since $h\chi_P = h_+$ and $h\chi_Q = -h_-$,

$$\lambda_+(E) = \int_{E\cap P} h = \int_E h_+, \qquad \lambda_-(E) = -\int_{E\cap Q} h = \int_E h_-,$$

and $|\lambda|(E) = \int_E (h_+ + h_-) = \int_E |h|$. □

Proposition 7.1.2

If λ is a signed measure on a σ-algebra $\mathcal{S}$ of X, then for every $E \in \mathcal{S}$

$$|\lambda|(E) = \sup\Big\{ \sum_{n=1}^{\infty} |\lambda(E_n)| : \ E_n \in \mathcal{S}, \ E = \bigsqcup_{n=1}^{\infty} E_n \Big\}. \tag{5}$$

Proof. Let α be the right side of (5). If (P, Q) is a Hahn decomposition, then

$$|\lambda|(E) = \lambda_+(E) + \lambda_-(E) = |\lambda(E \cap P)| + |\lambda(E \cap Q)| \leq \alpha.$$

Conversely, if $E_n \in \mathcal{S}$ and $E = \bigsqcup_{n=1}^{\infty} E_n$, then

$$\sum_{n=1}^{\infty} |\lambda(E_n)| = \sum_{n=1}^{\infty} |\lambda_+(E_n) - \lambda_-(E_n)| \leq \sum_{n=1}^{\infty} (\lambda_+(E_n) + \lambda_-(E_n)) = \sum_{n=1}^{\infty} |\lambda|(E_n) = |\lambda|(E),$$

so $\alpha \leq |\lambda|(E)$. ■

7.1.4 Total variation of a complex measure

From now on let λ be a complex measure on a σ-algebra $(X, \mathcal{S})$ and define $|\lambda|(E)$ for $E \in \mathcal{S}$ by (5). Clearly $|\lambda(E)| \leq |\lambda|(E)$ for every $E \in \mathcal{S}$.

Proposition 7.1.3

If λ is a complex measure on a σ-algebra $\mathcal{S}$ of X, then $E \mapsto |\lambda|(E)$ is a positive measure.

Proof. For a disjoint sequence $\langle E_n \rangle$ in $\mathcal{S}$ with union E we must show

$$|\lambda|(E) = \sum_{n=1}^{\infty} |\lambda|(E_n).$$

If $E = \bigsqcup_{m=1}^{\infty} F_m$, then for all m, n

$$F_m = \bigsqcup_{n=1}^{\infty} (E_n \cap F_m), \qquad E_n = \bigsqcup_{m=1}^{\infty} (E_n \cap F_m).$$

Hence

$$\sum_{m=1}^{\infty} |\lambda(F_m)| = \sum_{m=1}^{\infty} \Big| \sum_{n=1}^{\infty} \lambda(E_n \cap F_m) \Big| \leq \sum_{m=1}^{\infty} \sum_{n=1}^{\infty} |\lambda(E_n \cap F_m)|$$

$$= \sum_{n=1}^{\infty}\sum_{m=1}^{\infty}|\lambda(E_n \cap F_m)| \le \sum_{n=1}^{\infty}|\lambda|(E_n),$$

and taking the supremum over partitions $\{F_m\}$ gives $|\lambda|(E) \le \sum_n |\lambda|(E_n)$. For the reverse, let $\epsilon > 0$ and choose for each E_n a partition $\{E_{n,m} : m \in \mathbb{N}\}$ with

$$|\lambda|(E_n) < \sum_{m=1}^{\infty}|\lambda(E_{n,m})| + \frac{\epsilon}{2^n}, \qquad n = 1, 2, \dots$$

Then $\{E_{n,m} : m, n = 1, 2, \dots\}$ is a partition of E, so

$$\sum_{n=1}^{\infty}|\lambda|(E_n) \le \sum_{m,n=1}^{\infty}|\lambda(E_{n,m})| + \epsilon \le |\lambda|(E) + \epsilon.$$

■

By (3), a complex measure λ is a linear combination of positive measures:

$$\lambda = (\lambda_{1+} - \lambda_{1-}) + i(\lambda_{2+} - \lambda_{2-}). \tag{6}$$

If $\{E_n\}$ is a partition of $E \in \mathcal{S}$, then

$$\sum_{n=1}^{\infty}|\lambda(E_n)| \le \sum_{n=1}^{\infty}\left(\lambda_{1+}(E_n)+\lambda_{1-}(E_n)+\lambda_{2+}(E_n)+\lambda_{2-}(E_n)\right) = \lambda_{1+}(E)+\lambda_{1-}(E)+\lambda_{2+}(E)+\lambda_{2-}(E),$$

so for every $E \in \mathcal{S}$

$$|\lambda|(E) \le \lambda_{1+}(E) + \lambda_{1-}(E) + \lambda_{2+}(E) + \lambda_{2-}(E).$$

Since λ is a complex measure, the values on the right are always finite. Hence $|\lambda|(E) < \infty$ for every $E \in \mathcal{S}$; in particular $|\lambda|(X) < \infty$, and we define

$$\|\lambda\| = |\lambda|(X).$$

Note that positive measures are in general not complex measures. A positive measure with $\mu(X) < \infty$ is a **finite positive measure**; finite positive measures are of course complex measures.

Exercise 7.1.3

For a σ-algebra $\mathcal{S}$ on X, show that the set of all complex measures on $\mathcal{S}$ is a vector space and that $\lambda \mapsto \|\lambda\|$ is a norm on it.

Solution [Supplement]**.** If λ, ρ are complex measures and $a \in \mathbb{C}$, then $(\lambda + \rho)(\emptyset) = 0$ and for disjoint A_n,

$$(\lambda+\rho)\Big(\bigsqcup A_n\Big) = \sum \lambda(A_n) + \sum \rho(A_n) = \sum (\lambda+\rho)(A_n)$$

(sum of convergent series); similarly for $a\lambda$. So they form a vector space. Norm: $\|\lambda\| \ge 0$. If $\|\lambda\| = 0$ then $|\lambda(E)| \le |\lambda|(E) \le |\lambda|(X) = 0$ for all E, so $\lambda = 0$. For any partition $\{E_n\}$ of X, $\sum|a\lambda(E_n)| = |a|\sum|\lambda(E_n)|$, so $|a\lambda| = |a||\lambda|$ and $\|a\lambda\| = |a|\|\lambda\|$. Finally $\sum|(\lambda+\rho)(E_n)| \le \sum|\lambda(E_n)| + \sum|\rho(E_n)| \le \|\lambda\| + \|\rho\|$; taking the supremum gives $\|\lambda + \rho\| \le \|\lambda\| + \|\rho\|$. □

7.2 The Radon–Nikodym Theorem and the Lebesgue Decomposition

Motivation. When can one measure be written as $\nu(E) = \int_E f\,d\mu$, i.e. when does ν have a density? Exactly when ν ignores the null sets of μ. This is the theorem behind probability densities and conditional expectation, and it identifies the dual of L^p.

7.2.1 Absolute continuity and singularity

Definition

Let μ be a positive measure and λ a signed or complex measure on a σ-algebra $\mathcal{S}$ of X.

- λ is **absolutely continuous** with respect to μ, written $\lambda \ll \mu$, if
$$E \in \mathcal{S},\ \mu(E) = 0 \implies \lambda(E) = 0.$$
- λ is **singular** with respect to μ, written $\lambda \perp \mu$, if there are disjoint $A, B \in \mathcal{S}$ with
$$\lambda(E) = \lambda(A \cap E), \qquad \mu(E) = \mu(B \cap E), \qquad E \in \mathcal{S}. \tag{7}$$

The measure defined by (2) on the real line is absolutely continuous with respect to Lebesgue measure. The Dirac measure, on the other hand, is singular with respect to Lebesgue measure.

A linear combination of two complex measures absolutely continuous with respect to μ is again absolutely continuous. Let λ_1, λ_2 be singular with respect to μ, with disjoint $A_i, B_i \in \mathcal{S}$ such that for $i = 1, 2$
$$\lambda_i(E) = \lambda_i(A_i \cap E), \qquad \mu(E) = \mu(B_i \cap E), \qquad E \in \mathcal{S}.$$
Put $A = A_1 \cup A_2$, $B = B_1 \cap B_2$. Then A, B are disjoint and
$$(\lambda_1 + \lambda_2)(A \cap E) = (\lambda_1 + \lambda_2)(E), \qquad \mu(B \cap E) = \mu(E)$$
(indeed $\lambda_1(A \cap E) = \lambda_1(A_1 \cap A \cap E) = \lambda_1(A_1 \cap E)$, and $\mu(E) = \mu(B_1 \cap E) = \mu(B_2 \cap B_1 \cap E)$). So linear combinations of singular measures are singular.

If a complex measure λ is both absolutely continuous and singular with respect to μ, take disjoint A, B satisfying (7). For each $E \in \mathcal{S}$, $\mu(E \cap A) = \mu(E \cap A \cap B) = 0$, so $\lambda(E \cap A) = 0$ by $\lambda \ll \mu$. Hence $\lambda(E) = \lambda(E \cap A) = 0$, i.e. $\lambda = 0$.

The first part (8) of the next theorem says that, for a fixed positive measure μ on a σ-algebra, every complex measure is the sum of an absolutely continuous measure and a singular measure; this is the **Lebesgue decomposition**. The second part (9) says that the absolutely continuous part is in fact given by a function, the **Radon–Nikodym derivative** of λ with respect to μ. Together this is the Radon–Nikodym theorem, one of the most important theorems of this whole course; it makes precise the slogan "measures are generalized functions".

Theorem 7.2.1 (Lebesgue–Radon–Nikodym)

Let μ be a σ-finite positive measure and λ a complex measure on a σ-algebra $\mathcal{S}$ of X. Then there are unique measures λ_a, λ_s with
$$\lambda = \lambda_a + \lambda_s, \qquad \lambda_a \ll \mu, \qquad \lambda_s \perp \mu. \tag{8}$$
Moreover there is a unique $h \in L^1(X, \mu)$ with
$$\lambda_a(E) = \int_E h \, d\mu, \qquad E \in \mathcal{S}. \tag{9}$$
If μ and λ are σ-finite positive measures, there are unique measures λ_a, λ_s satisfying (8) and a unique measurable $h : X \to [0, \infty]$ satisfying (9).

Proof. First let λ be a complex measure; we start with uniqueness. If (λ_a, λ_s) and (λ_a', λ_s') both satisfy (8), then $\lambda_a - \lambda_a' = \lambda_s' - \lambda_s$ is both absolutely continuous and singular with respect to μ, hence 0. Uniqueness of $h \in L^1(X, \mu)$ follows from Fact B.8(b) (if $\int_E h = 0$ for all E, then $h = 0$ a.e.).

Step 1: μ and λ finite positive measures. Then $\nu = \mu + \lambda$ is also a finite positive measure. For every $f \in L^2(\nu)$, by the Cauchy–Schwarz inequality,

$$\Big|\int_X f\,d\lambda\Big| \le \int_X |f|\,d\lambda \le \int_X |f|\,d\nu \le \Big(\int_X |f|^2\,d\nu\Big)^{1/2}\nu(X)^{1/2},$$

and since $\nu(X) < \infty$, $f \mapsto \int_X f\,d\lambda$ is a bounded linear functional on $L^2(\nu)$. By Theorem 6.3.3 (Riesz) there is $g \in L^2(\nu)$ with

$$\int_X f\,d\lambda = \int_X fg\,d\nu, \qquad f \in L^2(\nu). \tag{10}$$

As $\nu(X) < \infty$, Fact B.29 gives $g \in L^1(\nu)$. Moreover, putting $f = \chi_E$,

$$\nu(E) > 0 \implies \frac{1}{\nu(E)}\int_E g\,d\nu = \frac{\lambda(E)}{\nu(E)} \in [0,1],$$

so by Fact B.27, $0 \le g \le 1$ ν-almost everywhere. Hence we may assume $0 \le g \le 1$ while keeping (10). Put

$$A = \{x \in X : 0 \le g(x) < 1\}, \qquad B = \{x \in X : g(x) = 1\},$$

and for $E \in \mathcal{S}$

$$\lambda_a(E) = \lambda(E \cap A), \qquad \lambda_s(E) = \lambda(E \cap B).$$

Rewriting (10) (using $\nu = \mu + \lambda$),

$$\int_X (1-g) f\,d\lambda = \int_X fg\,d\mu, \qquad f \in L^2(\nu). \tag{11}$$

Putting $f = \chi_B$,

$$0 = \int_B (1-g)\,d\lambda = \int_B g\,d\mu = \mu(B),$$

so $\lambda_s \perp \mu$ (with the disjoint sets B and $A = X \setminus B$: $\lambda_s(E) = \lambda_s(B \cap E)$ and $\mu(E) = \mu(A \cap E)$). Next, putting $f = (1 + g + \cdots + g^n)\chi_E$ in (11),

$$\int_E (1 - g^{n+1})\,d\lambda = \int_E g(1 + g + \cdots + g^n)\,d\mu.$$

Let $n \to \infty$. Since $(1-g^{n+1})|_B = 0$ and $1 - g^{n+1} \nearrow 1$ on A, the left side tends to $\lambda(E \cap A) = \lambda_a(E)$ (monotone convergence). If $h = \lim_n g(1 + g + \cdots + g^n)$, the right side tends to $\int_E h\,d\mu$ by the monotone convergence theorem. This gives (8) and (9), and in particular $\lambda_a \ll \mu$. Since $\lambda_a(X) < \infty$, $h \in L^1(\mu)$.

Step 2: μ σ-finite, λ finite positive. Write $X = \bigsqcup_n X_n$ with $\mu(X_n) < \infty$ and choose positive numbers $\langle \alpha_n \rangle$ with $\sum_n \alpha_n \mu(X_n) < \infty$. For $E \in \mathcal{S}$ put

$$\mu_0(E) = \sum_{n=1}^{\infty} \alpha_n \mu(E \cap X_n); \tag{12}$$

μ_0 is clearly a finite positive measure. By Step 1 there are λ_a, λ_s with

$$\lambda = \lambda_a + \lambda_s, \qquad \lambda_a \ll \mu_0, \qquad \lambda_s \perp \mu_0.$$

If $\mu(E) = 0$, then $\mu(E \cap X_n) = 0$ for every n, so $\mu_0(E) = 0$; hence $\lambda_a \ll \mu_0$ gives $\lambda_a \ll \mu$. If $\mu_0(E) = \mu_0(B \cap E)$, then $\mu_0(E \setminus B) = 0$, so $\mu((E \setminus B) \cap X_n) = 0$ for every n (as $\alpha_n > 0$). Hence $\mu(E \setminus B) = 0$ and $\mu(E) = \mu(E \cap B)$; so $\lambda_s \perp \mu_0$ gives $\lambda_s \perp \mu$. This proves (8). By Step 1 there is $h_0 \in L^1(\mu_0)$ with

$$\lambda_a(E) = \int_E h_0\,d\mu_0, \qquad E \in \mathcal{S}.$$

Put

$$k = \sum_{n=1}^{\infty} \alpha_n \chi_{X_n}. \tag{13}$$

Then $\int_E d\mu_0 = \int_E k\, d\mu$, so

$$\int_X f\, d\mu_0 = \int_X fk\, d\mu$$

for every simple f, hence (by the definition of the integral) for every measurable $f : X \to [0,\infty]$. Applying this to $f = h_0\chi_E$ and putting $h = h_0 k$ gives (9), and $h \in L^1(\mu)$ since $\int h\, d\mu = \int h_0\, d\mu_0 < \infty$ (for $h_0 \geq 0$).

So the theorem holds when μ is a σ-finite positive measure and λ a finite positive measure. If λ is a complex measure, it is a linear combination of finite positive measures by (6), and the same conclusion follows (apply the result to each of the four and combine; uniqueness was shown above). ■

Exercise 7.2.1

Prove the Lebesgue–Radon–Nikodym theorem when λ is a σ-finite positive measure. Also prove it when λ is a signed measure that is a difference of σ-finite positive measures. Note that in the uniqueness proof one cannot subtract positive measures directly (they may be infinite).

Solution [Supplement]**.** *Existence (λ σ-finite positive).* Write $X = \bigsqcup_m Y_m$ with $\lambda(Y_m) < \infty$ and let $\lambda^{(m)}(E) = \lambda(E \cap Y_m)$, a finite positive measure. By the theorem, $\lambda^{(m)} = \lambda_a^{(m)} + \lambda_s^{(m)}$ with $d\lambda_a^{(m)} = h_m\, d\mu$, $h_m \geq 0$, and $\lambda_s^{(m)}$ concentrated on a set A_m with $\mu(A_m) = 0$. Replacing A_m by $A_m \cap Y_m$ and h_m by $h_m\chi_{Y_m}$ changes nothing (both measures vanish outside Y_m). Put

$$\lambda_a = \sum_m \lambda_a^{(m)}, \quad \lambda_s = \sum_m \lambda_s^{(m)}, \quad h = \sum_m h_m, \quad A = \bigcup_m A_m.$$

Then $\lambda = \lambda_a + \lambda_s$, $\lambda_a(E) = \int_E h\, d\mu$ (monotone convergence), $\lambda_a \ll \mu$, $\mu(A) = 0$ and $\lambda_s(E) = \lambda_s(E \cap A)$, so $\lambda_s \perp \mu$.

Uniqueness. Suppose also $\lambda = \lambda'_a + \lambda'_s$, $d\lambda'_a = h'\, d\mu$, λ'_s concentrated on A' with $\mu(A') = 0$. Let $N = A \cup A'$, so $\mu(N) = 0$. On N^c both singular parts vanish, and on N both absolutely continuous parts vanish; hence for every E

$$\lambda_s(E) = \lambda(E \cap N) = \lambda'_s(E), \qquad \lambda_a(E) = \lambda(E \setminus N) = \lambda'_a(E).$$

No subtraction was used. For h: on each $Y_m \cap X_n$ (with $\mu(X_n) < \infty$) the measure λ_a is finite, so $\int_E (h - h')\, d\mu = 0$ for all $E \subset Y_m \cap X_n$ with h, h' integrable there, giving $h = h'$ a.e. on each piece, hence a.e.

Signed case. If $\lambda = \rho_1 - \rho_2$ with ρ_i σ-finite positive measures (at least one finite, so λ is defined), decompose each $\rho_i = \rho_{i,a} + \rho_{i,s}$ as above and set $\lambda_a = \rho_{1,a} - \rho_{2,a}$, $\lambda_s = \rho_{1,s} - \rho_{2,s}$, $h = h_1 - h_2$. These are well defined (the finite ρ_i has finite parts), and linear combinations preserve $\ll \mu$ and $\perp \mu$. Uniqueness follows exactly as above using the null set N: $\lambda_s(E) = \lambda(E \cap N)$ and $\lambda_a(E) = \lambda(E \setminus N)$ for *any* decomposition. □

When λ_a and μ satisfy (9) we write

$$d\lambda_a = h\, d\mu, \qquad h = \Big[\frac{d\lambda_a}{d\mu}\Big].$$

Exercise 7.2.2

If $d\lambda = g\,d\mu$, show that $\lambda(\{x : g(x) = 0\}) = 0$.

Solution [Supplement]. Let $Z = \{x : g(x) = 0\}$. Then $\lambda(Z) = \int_Z g\,d\mu = \int_Z 0\,d\mu = 0$. (More generally $\lambda(E) = 0$ and even $|\lambda|(E) = 0$ for every measurable $E \subset Z$.) □

7.2.2 Polar decomposition

The Lebesgue–Radon–Nikodym theorem has many applications in measure theory; we look at a few.

Theorem 7.2.2 (polar decomposition)

For every complex measure λ on a set X there is a unique $h \in L^1(X, |\lambda|)$ with

$$d\lambda = h\,d|\lambda|, \qquad |h| = 1.$$

Proof. Since $\lambda \ll |\lambda|$, there is a unique $h \in L^1(|\lambda|)$ with $d\lambda = h\,d|\lambda|$ (Theorem 7.2.1, the singular part being 0). If $|\lambda|(E) > 0$, then

$$\Big|\frac{1}{|\lambda|(E)}\int_E h\,d|\lambda|\Big| = \frac{|\lambda(E)|}{|\lambda|(E)} \le 1,$$

so by Fact B.27, $|h| \le 1$ almost everywhere. On the other hand, for $r > 0$ put $A_r = \{x \in X : |h(x)| < r\}$. If $A_r = \bigsqcup_{n=1}^{\infty} E_n$, then

$$\sum_{n=1}^{\infty}|\lambda(E_n)| = \sum_{n=1}^{\infty}\Big|\int_{E_n} h\,d|\lambda|\Big| \le \sum_{n=1}^{\infty} r|\lambda|(E_n) = r|\lambda|(A_r),$$

so $|\lambda|(A_r) \le r|\lambda|(A_r)$, and for $r < 1$ this forces $|\lambda|(A_r) = 0$. Hence $|h| \ge 1$ almost everywhere, and we may redefine h on a null set so that $|h| = 1$. ■

7.2.3 Integration with respect to a complex measure

The polar decomposition suggests how to define integration with respect to a complex measure. Given $d\lambda = h\,d|\lambda|$, for every $f \in L^1(X, |\lambda|)$ define

$$\int_X f\,d\lambda = \int_X fh\,d|\lambda|.$$

If we decompose λ into positive measures as in (6), then for every $f \in L^1(X, |\lambda|)$

$$\int_X f\,d\lambda = \Big(\int_X f\,d\lambda_{1+} - \int_X f\,d\lambda_{1-}\Big) + i\Big(\int_X f\,d\lambda_{2+} - \int_X f\,d\lambda_{2-}\Big). \tag{14}$$

If f is a characteristic function this is just the definition of $d\lambda = h\,d|\lambda|$, so it holds for every simple function. If $f \in L^1(X, |\lambda|)$ and $f \ge 0$, Fact B.5 gives an increasing sequence of simple functions $\langle s_n\rangle$ converging pointwise to f. Since f belongs to $L^1(X, |\lambda|)$ and to $L^1(X, \lambda_{i\pm})$ $(i = 1, 2$; note $\lambda_{i\pm} \le |\lambda|)$, the Lebesgue convergence theorem gives (14), and therefore (14) holds for every complex $f \in L^1(X, |\lambda|)$.

7.2.4 The dual of $L^p(\mu)$

Finally, another application of Radon–Nikodym: the dual of $L^p(\mu)$ is $L^q(\mu)$ $(1 \le p < \infty)$; again σ-finiteness is needed. Recall from Example 2 of §6.1 that

$$g \mapsto \varphi_g : L^q(\mu) \to L^p(\mu)^*$$

is an isometry; it remains to show it is surjective. (For $p = 1$, $q = \infty$, showing this map is an isometry used σ-finiteness of μ.) Let $\varphi \in L^p(\mu)^*$; we find $g \in L^q(\mu)$ with $\varphi = \varphi_g$.

Finite μ. For $E \in \mathcal{S}$ put

$$\lambda(E) = \varphi(\chi_E).$$

Since $\chi_{E_1 \sqcup E_2} = \chi_{E_1} + \chi_{E_2}$, $\lambda(E_1 \sqcup E_2) = \lambda(E_1) + \lambda(E_2)$. If $E = \bigsqcup_{n=1}^{\infty} E_n$ and $A_n = E_1 \sqcup \cdots \sqcup E_n$, then

$$\lim_{n\to\infty} \|\chi_E - \chi_{A_n}\|_p = \lim_{n\to\infty} \left[\mu(E \setminus A_n)\right]^{1/p} = 0$$

(here $p < \infty$ and $\mu(E) < \infty$), and since $\varphi \in L^p(\mu)^*$,

$$\lambda(E) = \lim_{n\to\infty} \lambda(A_n) = \lim_{n\to\infty} \sum_{k=1}^{n} \lambda(E_k).$$

So λ is a complex measure. Clearly $\lambda \ll \mu$ (if $\mu(E) = 0$ then $\chi_E = 0$ in L^p), so taking the Radon–Nikodym derivative $g \in L^1(\mu)$,

$$\varphi(\chi_E) = \int_E g \, d\mu = \int_X \chi_E g \, d\mu = \varphi_g(\chi_E), \qquad E \in \mathcal{S}.$$

Hence $\varphi(f) = \varphi_g(f)$ for every simple f. The simple functions are dense in $L^\infty(\mu)$, and φ, φ_g are continuous on $L^\infty(\mu)$ (for finite μ, $\|f\|_p \le \mu(X)^{1/p}\|f\|_\infty$; and $g \in L^1$), so $\varphi(f) = \varphi_g(f)$ for every $f \in L^\infty(\mu)$. It remains to show $g \in L^q(\mu)$ and $\varphi = \varphi_g$ on all of $L^p(\mu)$.

Exercise 7.2.3

For a finite positive measure μ, show that the vector space of simple functions is dense in $L^p(\mu)$ for $1 \le p \le \infty$. [Hint: treat $1 \le p < \infty$ and $p = \infty$ separately; Fact B.20.]

Solution [Supplement]. $p = \infty$. Let $f \in L^\infty(\mu)$; changing f on a null set, $|f| \le \|f\|_\infty = M$ everywhere. Cover the disc $\{|z| \le M\}$ by finitely many disjoint Borel sets $B_1, \dots, B_N$ of diameter $< \epsilon$, pick $z_k \in B_k$, and let $s = \sum_k z_k \chi_{f^{-1}(B_k)}$. Then $|f - s| < \epsilon$ everywhere, so $\|f - s\|_\infty \le \epsilon$. (No finiteness of μ is needed here.) $1 \le p < \infty$. By splitting into real and imaginary, positive and negative parts, assume $f \ge 0$. By Fact B.5 there are simple $0 \le s_n \nearrow f$. Then $|f - s_n|^p \le f^p \in L^1$ and $|f - s_n|^p \to 0$ pointwise, so $\|f - s_n\|_p \to 0$ by the Lebesgue convergence theorem. (Each $s_n \le f$ is in L^p; finiteness of μ ensures every simple function, including χ_X, is in L^p.) □

$p = 1$, $q = \infty$. For every $E \in \mathcal{S}$,

$$\left| \int_E g \, d\mu \right| = |\varphi(\chi_E)| \le \|\varphi\| \|\chi_E\|_1 = \|\varphi\| \mu(E),$$

so by Fact B.27, $\|g\|_\infty \le \|\varphi\|$.

$1 < p < \infty$. The basic method is the same as for ℓ^p. For $n = 1, 2, \dots$ put $E_n = \{x \in X : |g(x)| \le n\}$ and $f_n = \chi_{E_n} |g|^{q-1} \cdot \operatorname{sgn} g$ (with $\operatorname{sgn} g = \overline{g}/|g|$ where $g \ne 0$, so that $f_n g = \chi_{E_n} |g|^q$). Then $f_n \in L^\infty(\mu)$ and

$$\int_{E_n} |g|^q = \int_X f_n g = \varphi_g(f_n) = \varphi(f_n) \le \|\varphi\| \|f_n\|_p = \|\varphi\| \Big(\int_{E_n} |g|^q \Big)^{1/p}.$$

Dividing both sides by the last factor and applying the monotone convergence theorem,

$$\|g\|_q = \lim_{n\to\infty} \Big(\int_X \chi_{E_n} |g|^q \Big)^{1/q} \le \|\varphi\|.$$

Now that $g \in L^q(\mu)$, φ_g is continuous on $L^p(\mu)$; since simple functions are dense in $L^p(\mu)$, $\varphi(f) = \varphi_g(f)$ for every $f \in L^p(\mu)$. So $L^p(\mu)^* = L^q(\mu)$ for finite μ and $1 \le p < \infty$.

σ-finite μ. Extending to σ-finite μ is similar to the argument in the Lebesgue–Radon–Nikodym theorem. Let μ_0 and $k \in L^1(\mu)$ be defined by (12), (13), so that $d\mu_0 = k\,d\mu$. Then

$$K : f \mapsto k^{1/p} f : L^p(\mu_0) \to L^p(\mu)$$

is a surjective isometry, so $\varphi \circ K$ is a bounded linear functional on $L^p(\mu_0)$. By the finite case there is $g \in L^q(\mu_0)$ with

$$(\varphi \circ K)(f) = \int_X fg\,d\mu_0, \qquad f \in L^p(\mu_0).$$

Rewriting,

$$\varphi(Kf) = \int_X fgk\,d\mu = \int_X k^{1/p} f g k^{1/q}\,d\mu = \int_X (Kf)\cdot(gk^{1/q})\,d\mu$$

(with $1/\infty = 0$). Hence $gk^{1/q}$ is the function we were looking for.

Exercise 7.2.4

In the argument above, show that K is a surjective isometry from $L^p(\mu_0)$ onto $L^p(\mu)$, and that $gk^{1/q} \in L^q(\mu)$.

Solution [Supplement]. $k > 0$ everywhere (each x lies in some X_n and $\alpha_n > 0$). *Isometry:* $\int_X |k^{1/p} f|^p\,d\mu = \int_X |f|^p k\,d\mu = \int_X |f|^p\,d\mu_0$, using $\int F\,d\mu_0 = \int Fk\,d\mu$ for measurable $F \ge 0$. *Surjective:* for $u \in L^p(\mu)$, $f = k^{-1/p}u$ satisfies $\int |f|^p\,d\mu_0 = \int |u|^p k^{-1} k\,d\mu = \|u\|_p^p < \infty$ and $Kf = u$. Also μ_0 and μ have the same null sets ($k > 0$), so a.e. statements agree. *$gk^{1/q} \in L^q(\mu)$:* for $1 < q < \infty$, $\int |gk^{1/q}|^q\,d\mu = \int |g|^q k\,d\mu = \int |g|^q\,d\mu_0 < \infty$. For $q = \infty$ ($p = 1$), $k^{1/q} = 1$ and $\|g\|_{L^\infty(\mu)} = \|g\|_{L^\infty(\mu_0)} < \infty$ because the null sets agree. □

Summarizing:

Theorem 7.2.3

For a σ-finite positive measure μ on a σ-algebra of X, $L^p(\mu)^* = L^q(\mu)$, where p, q are conjugate exponents and $1 \le p < \infty$.

Exercise 7.2.5

Let $X = \{a, b\}$ with $\mu(\{a\}) = 1$, $\mu(\{b\}) = \infty$. Is the dual of $L^1(\mu)$ equal to $L^\infty(\mu)$?

Solution [Supplement]. A measurable f is in $L^1(\mu)$ iff $|f(a)| + \infty \cdot |f(b)| < \infty$, i.e. iff $f(b) = 0$. So $L^1(\mu) \cong \mathbb{C}$ via $f \mapsto f(a)$, with $\|f\|_1 = |f(a)|$, and $L^1(\mu)^* \cong \mathbb{C}$. On the other hand $\{b\}$ is not a null set, so $L^\infty(\mu) \cong \mathbb{C}^2$ with the max norm. The map $g \mapsto \varphi_g$, $\varphi_g(f) = g(a)f(a)$, is surjective onto $L^1(\mu)^*$ but not injective: $g = \chi_{\{b\}}$ gives $\varphi_g = 0$ though $\|g\|_\infty = 1$. So the natural map is not an isometric isomorphism, and $L^1(\mu)^* \ne L^\infty(\mu)$; this shows σ-finiteness cannot be dropped in Theorem 7.2.3 for $p = 1$. □

Corollary 7.2.4

Let μ be a σ-finite positive measure on X and $1 \le p \le \infty$. If a measurable function g on X satisfies

$$h \in L^q(\mu) \implies \left| \int_X gh\,d\mu \right| \le M\|h\|_q \qquad (p, q \text{ conjugate}),$$

then $g \in L^p(\mu)$ and $\|g\|_p \le M$.

Proof. The hypothesis says $\varphi_g \in L^q(\mu)^*$ and $\|\varphi_g\| \le M$. So if $1 < p \le \infty$ (i.e. $1 \le q < \infty$), Theorem 7.2.3 applies and the isometry gives $\|g\|_p = \|\varphi_g\| \le M$. For $p = 1$, the hypothesis gives

$$\int_X |g|\,d\mu = \int_X g \cdot (\operatorname{sgn} g)\,d\mu \le M\|\operatorname{sgn} g\|_\infty \le M,$$

where $\operatorname{sgn} g = \overline{g}/|g|$ on $\{g \neq 0\}$ and 0 elsewhere. ∎

Corollary 7.2.5

For every $f \in L^1(\mathbb{R})$ and $g \in L^p(\mathbb{R})$ $(1 \le p \le \infty)$, $f * g \in L^p(\mathbb{R})$ and $\|f * g\|_p \le \|f\|_1\|g\|_p$.

Proof. For every $h \in L^q(\mathbb{R})$, writing $g_y(x) = g(x - y)$,

$$\begin{aligned}\Big|\int_{\mathbb{R}} (f * g)(x)h(x)\,dx\Big| &= \Big|\int_{\mathbb{R}}\int_{\mathbb{R}} f(y)g(x-y)\,dy\,h(x)\,dx\Big| \\ &\le \int_{\mathbb{R}} \Big|\int_{\mathbb{R}} g_y(x)h(x)\,dx\Big|\,|f(y)|\,dy \\ &\le \|g_y\|_p\|h\|_q\|f\|_1 = \|f\|_1\|g\|_p\|h\|_q\end{aligned}$$

(Fubini, Hölder, translation invariance). Corollary 7.2.4 gives the result. ∎

(To be careful, apply the computation first to $|f|, |g|, |h|$: Tonelli gives $\iint |f(y)g(x-y)h(x)|\,dy\,dx \le \|f\|_1\|g\|_p\|h\|_q < \infty$, which justifies Fubini and shows $(f * g)(x)$ is defined a.e.)

7.3 Borel Measures on the Real Line

Motivation. On $\mathbb{R}$ every finite Borel measure comes from an increasing function, as in the Riemann–Stieltjes integral. The decomposition of the measure into discrete, singular continuous and absolutely continuous parts corresponds to a decomposition of that function, and its absolutely continuous part is recovered by differentiation.

7.3.1 Regular Borel measures and their distribution functions

In this section we find all complex measures defined on the Borel sets of the real line. As in §4.1, $\mathcal{B}$ denotes the smallest σ-algebra containing all open subsets of $\mathbb{R}$; its members are the Borel sets. A complex measure on the Borel sets is a **complex Borel measure**. Every complex measure is a linear combination of positive measures, so we focus on positive Borel measures. Here we consider positive measures whose value on every bounded interval is finite; such a Borel measure is called a **regular positive Borel measure**.

For a regular positive Borel measure λ on the real line define $\alpha_\lambda : \mathbb{R} \to \mathbb{R}$ by

$$\alpha_\lambda(x) = \begin{cases} \lambda((0, x]), & x > 0, \\ 0, & x = 0, \\ -\lambda((x, 0]), & x < 0. \end{cases}$$

Then α_λ is clearly nondecreasing. It is also right-continuous. First let $x > 0$ and take any decreasing sequence $\langle h_n \rangle$ in $(0, x)$ converging to 0. Then

$$\alpha_\lambda(x + h_n) - \alpha_\lambda(x) = \lambda((0, x + h_n]) - \lambda((0, x]) = \lambda((x, x + h_n]),$$

and since $\bigcap_{n=1}^\infty (x, x + h_n] = \emptyset$ and $\lambda((x, x + h_1]) < \infty$, continuity from above gives

$$\lim_{n\to\infty} \{\alpha_\lambda(x + h_n) - \alpha_\lambda(x)\} = 0.$$

Thus $\lim_{h\to 0^+}\alpha_\lambda(x+h)=\alpha_\lambda(x)$. The cases $x=0$ and $x<0$ are handled the same way, so α_λ is right-continuous.

Exercise 7.3.1

For a regular positive Borel measure λ on the real line, show that $\alpha_\lambda:\mathbb{R}\to\mathbb{R}$ is a nondecreasing right-continuous function.

Solution [Supplement]**.** In all cases $\alpha_\lambda(b)-\alpha_\lambda(a)=\lambda((a,b])$ for $a<b$: e.g. for $a<0<b$, $\alpha_\lambda(b)-\alpha_\lambda(a)=\lambda((0,b])+\lambda((a,0])=\lambda((a,b])$; the cases $0\le a<b$ and $a<b\le 0$ are similar. Hence $\alpha_\lambda(b)-\alpha_\lambda(a)\ge 0$ (nondecreasing). For right-continuity at any x and $h_n\searrow 0$: $\alpha_\lambda(x+h_n)-\alpha_\lambda(x)=\lambda((x,x+h_n])\to\lambda(\emptyset)=0$ by continuity from above, valid since $\lambda((x,x+h_1])<\infty$ by regularity. As α_λ is monotone, convergence along every decreasing sequence gives $\lim_{h\to 0^+}\alpha_\lambda(x+h)=\alpha_\lambda(x)$. □

7.3.2 Riemann–Stieltjes integrals

For a bounded real function $f:[a,b]\to\mathbb{R}$ consider the Riemann–Stieltjes integral $\int_a^b f\,d\alpha_\lambda$, which proceeds like the Riemann integral (Fact B.13). For a partition $P=\{x_0,x_1,\dots,x_n\}$ of $[a,b]$, define U_P and L_P as in Fact B.13. Then the Riemann–Stieltjes sums are

$$\begin{aligned}U_a^b(f,P,\alpha_\lambda)&=\sum_{i=1}^n M_i(\alpha_\lambda(x_i)-\alpha_\lambda(x_{i-1}))=\sum_{i=1}^n M_i\lambda((x_{i-1},x_i])\\&=\int_{[a,b]}U_P\,d\lambda-f(a)\lambda(\{a\})=\int_{(a,b]}U_P\,d\lambda,\end{aligned}$$

and similarly $L_a^b(f,P,\alpha_\lambda)=\int_{(a,b]}L_P\,d\lambda$. Hence for every partition $P\in P[a,b]$

$$L_a^b(f,P,\alpha_\lambda)\le\int_{(a,b]}f\,d\lambda\le U_a^b(f,P,\alpha_\lambda).$$

If $f:\mathbb{R}\to\mathbb{R}$ is continuous with support in a bounded interval (A,B), then f is Riemann–Stieltjes integrable on $[A,B]$ with respect to α_λ, so

$$\int_{\mathbb{R}}f\,d\lambda=\int_A^B f(x)\,d\alpha_\lambda(x).$$

So on $C_c(\mathbb{R})$, integration with respect to a regular positive Borel measure is just Riemann–Stieltjes integration. The Riemann–Stieltjes integral is close to the general measure-theoretic integral, but its integrability becomes a big problem when the integrand is not continuous. By contrast, integrals with respect to regular positive Borel measures or Lebesgue measure have difficulties in defining improper integrals.

7.3.3 Continuity of α_λ

We now apply the Lebesgue–Radon–Nikodym theorem to Borel measures on the real line. First we need to know when the nondecreasing function α_λ is continuous or absolutely continuous. Taking a decreasing sequence $h_n\to 0$, $\bigcap_{n=1}^\infty(x-h_n,x]=\{x\}$, so

$$\lambda(\{x\})=\lim_{n\to\infty}\lambda((x-h_n,x])=\lim_{n\to\infty}\{\alpha_\lambda(x)-\alpha_\lambda(x-h_n)\}.$$

So α_λ is left-continuous at x iff $\lambda(\{x\})=0$. A measure for which every point has measure 0 is a **continuous measure**. Since α_λ is always right-continuous:

Proposition 7.3.1

For a regular positive Borel measure λ on the real line, α_λ is continuous at x if and only if $\lambda(\{x\}) = 0$.

The next theorem explains why the term "absolutely continuous" is used for both measures and functions. To discuss absolute continuity between two measures they must be defined on the same σ-algebra; Lebesgue measure restricted to the Borel sets, $m|_{\mathcal{B}}$, is also written m.

Theorem 7.3.2

For a regular positive Borel measure λ on the real line, the following are equivalent:

(a) $\lambda \ll m$;

(b) α_λ is absolutely continuous on every bounded closed interval.

Proof. (a)⇒(b). If $\lambda \ll m$, by the Radon–Nikodym theorem there is a Borel function $h : \mathbb{R} \to [0, \infty]$ with

$$\lambda(E) = \int_E h(x)\,dx, \qquad E \in \mathcal{B}.$$

Fix a bounded closed interval I; we show α_λ is absolutely continuous on I. Since $\int_I h = \lambda(I) < \infty$, $h \in L^1(I, m)$. By Fact B.15, for every $\epsilon > 0$ there is $\delta > 0$ with

$$A \in \mathcal{B},\ m(A) < \delta \implies \lambda(A) < \epsilon.$$

This δ works in the definition of absolute continuity of functions. If $\{(a_i, b_i) : i = 1, \dots, n\}$ are disjoint intervals in I with $\sum_{i=1}^n (b_i - a_i) < \delta$, then $m(\bigcup_{i=1}^n (a_i, b_i]) < \delta$, so

$$\sum_{i=1}^n |\alpha_\lambda(b_i) - \alpha_\lambda(a_i)| = \sum_{i=1}^n \lambda((a_i, b_i]) = \lambda\Big(\bigcup_{i=1}^n (a_i, b_i]\Big) < \epsilon.$$

(b)⇒(a). Assume α_λ is absolutely continuous on every bounded closed interval I, let $\epsilon > 0$, and let $\delta_I > 0$ be the number from the definition of absolute continuity of $\alpha_\lambda : I \to \mathbb{R}$. If $m(A) = 0$ then $m(A \cap I) = 0$, so there is a sequence of disjoint open intervals $\langle (c_n, d_n) : n = 1, 2, \dots \rangle$ with

$$A \cap I \subset \bigcup_{n=1}^\infty (c_n, d_n), \qquad \sum_{n=1}^\infty (d_n - c_n) < \delta_I.$$

Writing (a_n, b_n) for the intersection of (c_n, d_n) with the interior $\operatorname{int} I$ of I,

$$A \cap \operatorname{int} I \subset \bigcup_{n=1}^\infty (a_n, b_n) \subset I, \qquad \sum_{n=1}^\infty (b_n - a_n) < \delta_I.$$

Hence $\sum_{n=1}^N [\alpha_\lambda(b_n) - \alpha_\lambda(a_n)] < \epsilon$ for every N, so $\sum_{n=1}^\infty [\alpha_\lambda(b_n) - \alpha_\lambda(a_n)] \le \epsilon$. Since α_λ is continuous (being absolutely continuous), $\alpha_\lambda(b_n) - \alpha_\lambda(a_n) = \lambda((a_n, b_n))$ (no point masses), and

$$\lambda(A \cap \operatorname{int} I) \le \lambda\Big(\bigcup_{n=1}^\infty (a_n, b_n)\Big) = \sum_{n=1}^\infty \lambda((a_n, b_n)) = \sum_{n=1}^\infty [\alpha_\lambda(b_n) - \alpha_\lambda(a_n)] \le \epsilon.$$

So $\lambda(A \cap \operatorname{int} I) = 0$ for every bounded open interval $\operatorname{int} I$, and therefore $\lambda(A) = 0$. ∎

7.3.4 Regularity

We now look for the condition on α_λ for $\lambda \perp m$. First some properties of regular positive Borel measures on the real line.

Theorem 7.3.3 (regularity)

A regular positive Borel measure λ on the real line satisfies, for every Borel set $E \in \mathcal{B}$,

$$\lambda(E) = \inf\{\lambda(U) : U \supset E,\ U \text{ open}\},$$
$$\lambda(E) = \sup\{\lambda(K) : K \subset E,\ K \text{ compact}\}.$$

Proof. Clearly every open set satisfies the first identity. Since every open set is the union of an increasing sequence of compact sets, the second identity also holds for open E. Let $\mathcal{C}$ be the collection of Borel sets satisfying both identities; it suffices to show $\mathcal{C}$ is a σ-algebra.

Countable unions. Take $\langle E_n\rangle$ in $\mathcal{C}$ and $E = \bigcup_n E_n$. First the case $\lambda(E) < \infty$: choose open $U_n \supset E_n$ with $\lambda(U_n) < \lambda(E_n) + \epsilon/2^n$ and let U be their union. Since $U \setminus E \subset \bigcup_n (U_n \setminus E_n)$,

$$\lambda(U) - \lambda(E) = \lambda(U \setminus E) \le \sum_{n=1}^{\infty} \lambda(U_n \setminus E_n) < \epsilon,$$

so E satisfies the first identity. If $\lambda(E) = \infty$ the first identity is trivial. For the second, for each j choose compact sets $\langle K_{ij} : i = 1, 2, \dots\rangle$ with

$$K_{ij} \subset E_j, \qquad \lim_{i\to\infty} \lambda(K_{ij}) = \lambda(E_j).$$

With $K_n = \bigcup_{i,j=1}^{n} K_{ij}$ we can show $\lambda(K_n) \to \lambda(E)$, so $E \in \mathcal{C}$ (Exercise 7.3.2).

Complements. Let $E \in \mathcal{C}$ and choose open $U_n \supset E$ decreasing with $\lambda(U_n) \to \lambda(E)$. With $L_n = [-n, n] \setminus U_n$ (compact), $L_n \subset E^c$ and $\lambda(L_n) \to \lambda(E^c)$ (Exercise 7.3.2). It remains to show the first identity for E^c; for this we first show

$$I \text{ a bounded closed interval},\ I \cap E \in \mathcal{C} \implies I \setminus E \in \mathcal{C}.$$

Choose an open $U \supset I \cap E$ with $\lambda(U) < \lambda(I \cap E) + \epsilon$. Then $K = I \setminus U$ is compact, $K \subset I \setminus E$, and since $\lambda(I) < \infty$,

$$\lambda(K) = \lambda(I) - \lambda(I \cap U) \ge \lambda(I) - \lambda(U) > \lambda(I) - \lambda(I \cap E) - \epsilon = \lambda(I \setminus E) - \epsilon,$$

so $I \setminus E$ satisfies the second identity. For the first, choose a bounded open interval $J \supset I$ with $\lambda(J) < \lambda(I) + \epsilon$, and a compact $K \subset I \cap E$ with $\lambda(K) > \lambda(I \cap E) - \epsilon$. Put $U = J \setminus K$; then U is open, $U \supset I \setminus E$, and

$$\lambda(U) = \lambda(J) - \lambda(K) < \lambda(I) + \epsilon - \lambda(I \cap E) + \epsilon = \lambda(I \setminus E) + 2\epsilon,$$

so $I \setminus E$ satisfies the first identity.

If $E \in \mathcal{C}$, then $E \cap [-n, n] \in \mathcal{C}$ for each n (one checks this as above), so $[-n, n] \setminus E \in \mathcal{C}$. Since $\mathcal{C}$ is already closed under countable unions, $E^c = \bigcup_n ([-n, n] \setminus E) \in \mathcal{C}$. ∎

Exercise 7.3.2

Show that every open set is the union of an increasing sequence of compact sets. In the proof of Theorem 7.3.3 show that $\lim_n \lambda(K_n) = \lambda(E)$ and $\lim_n \lambda(L_n) = \lambda(E^c)$.

Solution [Supplement]. *Open sets.* For open $U \subset \mathbb{R}$ ($U \neq \mathbb{R}$) let $C_n = \{x \in U : |x| \le n,\ d(x, U^c) \ge \frac{1}{n}\}$; for $U = \mathbb{R}$ let $C_n = [-n, n]$. Each C_n is closed and bounded, hence compact, $C_n \subset C_{n+1}$, and every $x \in U$ has $d(x, U^c) > 0$, so $x \in C_n$ for large n.

$\lambda(K_n) \to \lambda(E)$. $K_n = \bigcup_{i,j\le n} K_{ij}$ is compact, $K_n \subset K_{n+1}$ and $K_n \subset E$, so $L = \lim_n \lambda(K_n) \le \lambda(E)$ exists. If some $\lambda(E_j) = \infty$, then $L \ge \lambda(K_{ij}) \to \infty$ as $i \to \infty$, so $L = \lambda(E) = \infty$. Otherwise fix N and i. For $n \ge \max(i, N)$, $K_n \supset \bigcup_{j\le N} K_{ij}$, and

$$\lambda\Big(\bigcup_{j\le N} E_j\Big) - \lambda\Big(\bigcup_{j\le N} K_{ij}\Big) \le \sum_{j\le N} \lambda(E_j \setminus K_{ij}) = \sum_{j\le N} \big(\lambda(E_j) - \lambda(K_{ij})\big) \xrightarrow[i\to\infty]{} 0.$$

Hence $L \geq \lambda(E_1 \cup \cdots \cup E_N)$ for every N, and continuity from below gives $L \geq \lambda(E)$.

$\lambda(L_n) \to \lambda(E^c)$. Suppose $\lambda(E) < \infty$. $L_n = [-n, n] \setminus U_n$ is compact, increasing (as U_n decreases), and $L_n \subset E^c$. Since $([-n, n] \cap E^c) \setminus L_n = [-n, n] \cap U_n \cap E^c \subset U_n \setminus E$,

$$0 \leq \lambda([-n, n] \cap E^c) - \lambda(L_n) \leq \lambda(U_n \setminus E) = \lambda(U_n) - \lambda(E) \to 0,$$

while $\lambda([-n, n] \cap E^c) \to \lambda(E^c)$ by continuity from below. Hence $\lambda(L_n) \to \lambda(E^c)$. (If $\lambda(E) = \infty$, apply this to the sets $E \cap [-N, N]$, which have finite measure; this is exactly how the end of the proof reduces to $[-n, n] \setminus E$.) □

Corollary 7.3.4

If $\lambda \ll m$ and $\alpha'_\lambda = 0$ almost everywhere, then $\lambda = 0$.

Proof. By Theorem 7.3.2, α_λ is absolutely continuous, so Fact B.16 gives

$$\lambda((a, b)) = \alpha_\lambda(b) - \alpha_\lambda(a) = \int_a^b \alpha'_\lambda(x)\, dx = 0.$$

Hence $\lambda(U) = 0$ for every open U (a countable disjoint union of open intervals), and Theorem 7.3.3 gives $\lambda = 0$. ■

7.3.5 Singular measures

Theorem 7.3.5

For a regular positive Borel measure λ on the real line, the following are equivalent:

(a) $\lambda \perp m$;

(b) $\alpha'_\lambda = 0$ almost everywhere.

Proof. (b)⇒(a). Assume $\alpha'_\lambda = 0$ a.e. Write the Lebesgue decomposition $\lambda = \lambda_a + \lambda_s$; then $\alpha_\lambda = \alpha_{\lambda_a} + \alpha_{\lambda_s}$, both nondecreasing, so $0 \leq \alpha'_{\lambda_a} \leq \alpha'_\lambda = 0$ and hence $\alpha'_{\lambda_a} = 0$ almost everywhere. Since $\lambda_a \ll m$, Corollary 7.3.4 gives $\lambda_a = 0$, so $\lambda = \lambda_s$ is singular.

(a)⇒(b). Assume $\lambda \perp m$ and put

$$\nu(E) = \int_E \alpha'_\lambda\, dm, \qquad E \in \mathcal{B}. \tag{15}$$

Then obviously $\nu \ll m$. On the other hand, by Fact B.14,

$$\nu((a, b]) = \int_{(a,b]} \alpha'_\lambda\, dm \leq \alpha_\lambda(b) - \alpha_\lambda(a) = \lambda((a, b]).$$

Every open set is a countable disjoint union of intervals of the form $(a, b]$, so $\nu(U) \leq \lambda(U)$ for every open U. Since ν and λ are both regular positive Borel measures, Theorem 7.3.3 gives $\nu(E) \leq \lambda(E)$ for every Borel E, hence $\nu \ll \lambda$. Since also $\lambda \perp m$, we get $\nu \perp m$ (Exercise 7.3.3), and together with $\nu \ll m$ this forces $\nu = 0$. As $\alpha'_\lambda \geq 0$ almost everywhere, $\alpha'_\lambda = 0$ almost everywhere. ■

Exercise 7.3.3

Show that if $\nu \ll \lambda$ and $\lambda \perp m$, then $\nu \perp m$.

Solution [Supplement]**.** Choose disjoint A, B with $\lambda(E) = \lambda(E \cap A)$ and $m(E) = m(E \cap B)$ for all E. Then $\lambda(E \setminus A) = 0$, so $\nu(E \setminus A) = 0$ by $\nu \ll \lambda$, i.e. $\nu(E) = \nu(E \cap A)$. Together with $m(E) = m(E \cap B)$, the disjoint pair A, B shows $\nu \perp m$. □

7.3.6 The Radon–Nikodym derivative is α'_λ

For an arbitrary regular positive Borel measure λ with Lebesgue decomposition $\lambda = \lambda_a + \lambda_s$, define ν by (15). By Theorem 7.3.5,

$$\alpha'_\lambda = \alpha'_{\lambda_a} + \alpha'_{\lambda_s} = \alpha'_{\lambda_a}$$

almost everywhere, and by Fact B.16 (α_{λ_a} absolutely continuous by Theorem 7.3.2)

$$\nu((c,d)) = \int_{(c,d)} \alpha'_\lambda \, dm = \int_{(c,d)} \alpha'_{\lambda_a} \, dm = \alpha_{\lambda_a}(d) - \alpha_{\lambda_a}(c) = \lambda_a((c,d)).$$

Hence $\nu(U) = \lambda_a(U)$ for every open U, and since ν is also regular, $\nu = \lambda_a$ (Theorem 7.3.3). That is,

$$\lambda_a(E) = \int_E \alpha'_\lambda \, dm$$

for every Borel set $E \in \mathcal{B}$: $\alpha'_{\lambda_a} = \alpha'_\lambda$ is precisely the Radon–Nikodym derivative of λ_a with respect to m.

Corollary 7.3.6

If a regular positive Borel measure λ on the real line is absolutely continuous with respect to Lebesgue measure m, then $\left[\frac{d\lambda}{dm}\right] = \alpha'_\lambda$ almost everywhere.

7.3.7 Discrete, continuous singular, and absolutely continuous parts

For a regular positive Borel measure λ put

$$D = \{x \in \mathbb{R} : \lambda(\{x\}) > 0\},$$

and for $E \in \mathcal{B}$

$$\lambda_d(E) = \lambda(E \cap D), \qquad \lambda_c(E) = \lambda(E \setminus D).$$

For each n, $\lambda(D \cap [-n,n]) \le \lambda([-n,n]) < \infty$, so $D \cap [-n,n]$ is countable (for each k, at most $k\lambda([-n,n])$ points have mass $> 1/k$); hence D is countable, $D = \{x_n : n = 1,2,\dots\}$. Then for every $E \in \mathcal{B}$

$$\lambda_d(E) = \sum_{x_n \in E} \lambda(\{x_n\}) = \sum_{n=1}^{\infty} \lambda(\{x_n\}) \delta_{x_n}(E),$$

so λ_d is a sum of Dirac measures. The resulting nondecreasing function α_{λ_d} has a discontinuity at each point $x \in D$. On the other hand, by Proposition 7.3.1 the nondecreasing function α_{λ_c} is continuous. If $\lambda_c \perp m$, then by Theorem 7.3.5 its derivative is 0 almost everywhere. A typical example of a non-constant continuous function whose derivative vanishes almost everywhere is the Cantor function.

Summary

Every regular positive Borel measure λ on the real line is the sum of three measures:

$$\lambda = \lambda_a + \lambda_s + \lambda_d,$$

an absolutely continuous measure λ_a, a continuous singular measure λ_s, and a discrete (Dirac) measure λ_d. With the corresponding nondecreasing functions $\alpha_a, \alpha_s, \alpha_d$: α_a is absolutely continuous, α_s is a continuous function whose derivative is 0 almost everywhere, and α_d is a discontinuous (jump) function. If α is the nondecreasing function of λ, then $\alpha = \alpha_a + \alpha_s + \alpha_d$

and

$$\lambda_a(E) = \int_E \alpha_a' \, dm = \int_E \alpha' \, dm, \qquad E \in \mathcal{B}.$$

Given a regular positive Borel measure λ, there is a corresponding nondecreasing right-continuous function α, and the two linear functionals on $C_c(\mathbb{R})$

$$f \mapsto \int_{\mathbb{R}} f \, d\lambda, \qquad f \mapsto \int_{\operatorname{supp} f} f(x) \, d\alpha(x)$$

coincide; the left is the integral with respect to the Borel measure λ, the right the Riemann–Stieltjes integral with respect to the nondecreasing function α. In the next chapter we shall see that, conversely, every nondecreasing function gives rise to a corresponding regular positive Borel measure.